

数字溯源表（报告 + PPT）

720 报告中的数字 输出 635 · 推导 38 · 设定 47 · 未找到 0	332 PPT 中的数字（含演讲备注） 输出 275 · 推导 37 · 设定 20 · 未找到 0	0 找不到来源的数字	19 + 1 输出文件 + notebook 打印输出作为来源
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- **输出：**在 提交版/outputs/ 的 JSON / CSV 或 notebook 打印输出里找到同一数值（按报告的小数位四舍五入；“×100”表示文件存比例、文中写百分数）。同值字段有多个时，按上下文关键词选最贴切的一个。
- **推导：**由其他已溯源数字算出（差值、比值、nats → bits 换算等），推导式已写明。
- **设定：**代码里的参数或常量（如 10 个分箱、k = 5、depth 10），不是计算结果。
- **未计入：**小于 10 的整数设定（如 k = 5、3 个步骤）、年份、学号、表/图/节编号。报告数字取自 report.tex 正文与表格；PPT 数字取自 19 页幻灯片的文字和演讲备注（图片内的数字以生成图片的 notebook 结果为准）。

一、报告（A2 Report, LaTeX 版）

数字	上下文	来源（输出文件 → 字段 / 推导式）	类型
1 Introduction			
10	...hosts who meet a set of criteria. The criteria are at least 10 completed reservations (or 3 reservatio...	Airbnb 官方超级房东标准（参考文献 Airbnb, n.d.）	推导
100	...ompleted reservations (or 3 reservations totalling at least 100 nights), an overall rating of 4.8 or hi...	Airbnb 官方超级房东标准（参考文献 Airbnb, n.d.）	推导
4.8	...ations totalling at least 100 nights), an overall rating of 4.8 or higher, a 90% response rate to new m...	Airbnb 官方超级房东标准（参考文献 Airbnb, n.d.）	推导
90%	...at least 100 nights), an overall rating of 4.8 or higher, a 90% response rate to new messages within 24...	Airbnb 官方超级房东标准（参考文献 Airbnb, n.d.）	推导
24	...f 4.8 or higher, a 90% response rate to new messages within 24 hours, and a cancellation rate below 1%...	Airbnb 官方超级房东标准（参考文献 Airbnb, n.d.）	推导
1%	...new messages within 24 hours, and a cancellation rate below 1%, subject to specified exceptions (Airbn...	Airbnb 官方超级房东标准（参考文献 Airbnb, n.d.）	推导
25,728	... Airbnb Melbourne dataset (Inside Airbnb, 2026), containing 25,728 listings described by 90 columns; the d...	evidence_cluster.json → n_rows_full	输出
90	...side Airbnb, 2026), containing 25,728 listings described by 90 columns; the days_since_last_review ref...	listings.csv 原始列数（读入后 90 列）	推导
2.1 Data Preprocessing · Table 1			
4,477	...ing days_since_last_review from the raw last_review field 4,477 rows (17.4%) have no review date at all...	evidence_preprocess.json → zero_review_count	输出
17.4%	...ce_last_review from the raw last_review field 4,477 rows (17.4%) have no review date at all, and their ...	4,477 ÷ 25,728	推导
6.59%	...4%) have no review date at all, and their superhost rate is 6.59% Selected 2 Counting amenity_count...	evidence_preprocess.json → superhost_share_no_review (×100)	输出
44.08	...parsing the structured amenities field Superhosts average 44.08 amenities, non-superhosts 31.03 Selec...	evidence_preprocess.json → amenity_count_mean_superhost	输出
31.03	... field Superhosts average 44.08 amenities, non-superhosts 31.03 Selected 3 Imputing missing price ...	evidence_preprocess.json → amenity_count_mean_nonsuperhost	输出
6,553	...s via median imputation plus a missing indicator variable 6,553 rows (25.47%) have no price Selected ...	evidence_preprocess.json → price_missing_count	输出
25.47%	... imputation plus a missing indicator variable 6,553 rows (25.47%) have no price Selected 4 Imputin...	evidence_preprocess.json → price_missing_pct	输出
18.19%	...ng missing values in the bedrooms and beds fields Missing 18.19% for bedrooms and 27.44% for beds, and t...	evidence_preprocess.json → bedrooms_missing_pct	输出
27.44%	... bedrooms and beds fields Missing 18.19% for bedrooms and 27.44% for beds, and the missingness is condit...	evidence_preprocess.json → beds_missing_pct	输出
82	...ing rare categories for the property_type variable Of the 82 categories, 41 contain fewer than 10 li...	evidence_preprocess.json → property_type_category_count	输出
41	...ries for the property_type variable Of the 82 categories, 41 contain fewer than 10 listings, and the...	evidence_preprocess.json → property_type_under_10_count	输出
10	...type variable Of the 82 categories, 41 contain fewer than 10 listings, and these 41 together account...	代码中的设定值（常量 / 参数）	设定
41	...82 categories, 41 contain fewer than 10 listings, and these 41 together account for only 111 rows (0.4...	evidence_preprocess.json → property_type_under_10_count	输出
111	...er than 10 listings, and these 41 together account for only 111 rows (0.43%) Rejected 6 Splitting...	代码中的设定值（常量 / 参数）	设定

0.43%	... listings, and these 41 together account for only 111 rows (0.43%) Rejected 6 Splitting minimum_nig...	evidence_preprocess.json → property_type_rare_row_pct	输出
30	... 6 Splitting minimum_nights into short/long groups at a 30–night threshold Only 426 rows (1.66%)...	代码中的设定值 (常量 / 参数)	设定
426	...ights into short/long groups at a 30–night threshold Only 426 rows (1.66%) require 30 nights or more ...	evidence_preprocess.json → min_nights_ge30_count	输出
1.66%	... short/long groups at a 30–night threshold Only 426 rows (1.66%) require 30 nights or more Rejected ...	evidence_preprocess.json → min_nights_ge30_pct	输出
30	...ups at a 30–night threshold Only 426 rows (1.66%) require 30 nights or more Rejected ...	代码中的设定值 (常量 / 参数)	设定

2.1 Data Preprocessing

4,477	...from the raw last_review field was selected as dropping the 4,477 listings with no review date (17.4% of ...	evidence_preprocess.json → zero_review_count	输出
17.4%	...elected as dropping the 4,477 listings with no review date (17.4% of all rows) would raise the superhost ...	4,477 ÷ 25,728	推导
30.65%	...ate (17.4% of all rows) would raise the superhost rate from 30.65% to 35.72%. Furthermore, 295 superhost l...	evidence_cluster.json → superhost_rate_full (×100)	输出
35.72%	... of all rows) would raise the superhost rate from 30.65% to 35.72%. Furthermore, 295 superhost listings ar...	evidence_cluster.json → superhost_rate_reviewed (×100)	输出
295	...aise the superhost rate from 30.65% to 35.72%. Furthermore, 295 superhost listings are still in the no ...	evidence_preprocess.json → superhost_count_no_review	输出
99.85%	... missing at random (MAR, not missing completely at random). 99.85% of unavailable listings (availability_3...	evidence_preprocess.json → p_price_missing_given_unavailable (×100)	输出
2.22%	...able listings (availability_365 = 0) have no price, against 2.22% of available ones. As unavailable listi...	evidence_preprocess.json → p_price_missing_given_available (×100)	输出
8.13%	...st 2.22% of available ones. As unavailable listings have an 8.13% superhost rate against 37.68% for avail...	evidence_preprocess.json → superhost_share_unavailable (×100)	输出
37.68%	...s unavailable listings have an 8.13% superhost rate against 37.68% for available listings, deletion would ...	evidence_preprocess.json → superhost_share_available (×100)	输出
30.65%	...le listings, deletion would inflate the superhost rate from 30.65% to 38.06%....	evidence_cluster.json → superhost_rate_full (×100)	输出
38.06%	...s, deletion would inflate the superhost rate from 30.65% to 38.06%....	evidence_preprocess.json → superhost_share_if_price_missing_dropped (×100)	输出
18.19%	...ngness is structural by room type. Bedrooms are missing for 18.19% (beds 27.44%), but the data is split by...	evidence_preprocess.json → bedrooms_missing_pct	输出
27.44%	...uctural by room type. Bedrooms are missing for 18.19% (beds 27.44%), but the data is split by room type, w...	evidence_preprocess.json → beds_missing_pct	输出
2.47%	...t the data is split by room type, with entire homes missing 2.47%, private rooms 61.28%. Neither field en...	evidence_preprocess.json → bedrooms_missing_share_by_room_type.Entire home/apt (×100)	输出
61.28%	...y room type, with entire homes missing 2.47%, private rooms 61.28%. Neither field enters the models in any...	evidence_preprocess.json → bedrooms_missing_share_by_room_type.Private room (×100)	输出
41	...ging property_types was rejected because although there are 41 property types having fewer than 10 lis...	evidence_preprocess.json → property_type_under_10_count	输出
10	...ause although there are 41 property types having fewer than 10 listings (out of 82 total property type...	代码中的设定值 (常量 / 参数)	设定
82	...are 41 property types having fewer than 10 listings (out of 82 total property types), these 41 account...	evidence_preprocess.json → property_type_category_count	输出
41	...er than 10 listings (out of 82 total property types), these 41 account for only 111 rows (0.43%), whic...	evidence_preprocess.json → property_type_under_10_count	输出
111	...(out of 82 total property types), these 41 account for only 111 rows (0.43%), which is a trivial change...	evidence_preprocess.json → property_type_rare_row_count	输出
0.43%	... total property types), these 41 account for only 111 rows (0.43%), which is a trivial change to the data...	evidence_preprocess.json → property_type_rare_row_pct	输出
426	...nimum_nights into short/long stay bins was rejected as only 426 listings (1.66%) require a minimum stay...	evidence_preprocess.json → min_nights_ge30_count	输出
1.66%	...nto short/long stay bins was rejected as only 426 listings (1.66%) require a minimum stay of 30 nights or...	evidence_preprocess.json → min_nights_ge30_pct	输出
30	...cted as only 426 listings (1.66%) require a minimum stay of 30 nights or more. The long–stay group wou...	代码中的设定值 (常量 / 参数)	设定

2.2 Correlation Analysis

10	...y (y = 1 when host_is_superhost = 't'). Five variables give 10 unique pairs and 40 values....	代码中的设定值 (常量 / 参数)	设定
40	...s_superhost = 't'). Five variables give 10 unique pairs and 40 values....	代码中的设定值 (常量 / 参数)	设定
25.47%	...Price is excluded because 25.47% of prices are missing and 93.35% of tho...	evidence_preprocess.json → price_missing_pct	输出
93.35%	...Price is excluded because 25.47% of prices are missing and 93.35% of those missing rows are unavailable l...	evidence_preprocess.json → p_unavailable_given_price_missing (×100)	输出
11	...e, accommodates has the lowest mutual information among the 11 features (0.0065), so dist_cbd is used ...	evidence_model.json → rq[4].n_features	输出
0.0065	...es has the lowest mutual information among the 11 features (0.0065), so dist_cbd is used as the single fun...	evidence_model.json → feature_ranking.accommodates.filter_MI	输出
667	...ions. host_listings_count captures scale (median 2, maximum 667 platform-wide; the largest host in this...	evidence_cluster.json → max_before_log.host_listings_count	输出
319	...mum 667 platform-wide; the largest host in this dataset has 319 listings), availability_365 reflects ca...	evidence_preprocess.json → largest_host_listing_count_in_data	输出
21.8%	... listings), availability_365 reflects calendar positioning (21.8% of listings at 0 days, 1.56% at 365) an...	evidence_correlation.json → av365_share_0 (×100)	输出
1.56%	...reflects calendar positioning (21.8% of listings at 0 days, 1.56% at 365) and days_since_last_review meas...	evidence_correlation.json → av365_share_365 (×100)	输出
365	...calendar positioning (21.8% of listings at 0 days, 1.56% at 365) and days_since_last_review measures re...	evidence_cluster.json → max_before_log.availability_365	输出
84	...and days_since_last_review measures recent activity (median 84 days, maximum 4,593). The variables are...	evidence_preprocess.json → days_since_last_review_median	输出
4,593	...st_review measures recent activity (median 84 days, maximum 4,593). The variables are tilted 3:1 towards ...	evidence_cluster.json → max_before_log.days_since_last_review	输出
10	...variables are binned using equal-frequency bins. We found n=10 bins to be a good parameter: each bin h...	代码中的设定值 (常量 / 参数)	设定
2,125	...nd n=10 bins to be a good parameter: each bin holds roughly 2,125 rows, and the expected count in each ce...	21,251 ÷ 10 个分箱	推导
10	...oughly 2,125 rows, and the expected count in each cell of a 10×10 contingency table is around 212, lar...	代码中的设定值 (常量 / 参数)	设定
10	...hly 2,125 rows, and the expected count in each cell of a 10×10 contingency table is around 212, large ...	代码中的设定值 (常量 / 参数)	设定
212	...d count in each cell of a 10×10 contingency table is around 212, large enough to be reliable. Equal-wid...	21,251 ÷ (10 × 10) 个格子	推导
10	... enough to be reliable. Equal-width binning was rejected as 10 equal-width bins would concentrate 91.4...	代码中的设定值 (常量 / 参数)	设定
91.4%	...nning was rejected as 10 equal-width bins would concentrate 91.4% of host_listings_count rows in one bin ...	evidence_correlation.json → hlc_equal_width_max_bin.10 (×100)	输出
84.6%	...d concentrate 91.4% of host_listings_count rows in one bin (84.6% at 20 bins), which lowers MI. Correlati...	evidence_correlation.json → hlc_equal_width_max_bin.20 (×100)	输出
20	...rate 91.4% of host_listings_count rows in one bin (84.6% at 20 bins), which lowers MI. Correlations we...	代码中的设定值 (常量 / 参数)	设定
21,251	... bins), which lowers MI. Correlations were only computed on 21,251 listings with at least one review, beca...	evidence_correlation.json → n_rows	输出
4,477	...mputed on 21,251 listings with at least one review, because 4,477 zero-review listings carry the imputati...	evidence_preprocess.json → zero_review_count	输出
4,594	... 4,477 zero-review listings carry the imputation fill value 4,594 for days_since_last_review, which would...	evidence_preprocess.json → days_since_last_review_fill_value	输出

2.3 Supervised Learning Models and Evaluation

25,728	...Data: All 25,728 listings are used, with zero-review lis...	evidence_cluster.json → n_rows_full	输出
14,113	...ry listing held by the same host carries the same label. Of 14,113 hosts, 3,770 are superhosts (26.71%), w...	evidence_cluster.json → n_hosts_full	输出
3,770	...d by the same host carries the same label. Of 14,113 hosts, 3,770 are superhosts (26.71%), whereas superh...	evidence_preprocess.json → superhost_host_count	输出
26.71%	...ries the same label. Of 14,113 hosts, 3,770 are superhosts (26.71%), whereas superhost listings make up 30...	evidence_preprocess.json → superhost_host_share (×100)	输出
30.65%	...are superhosts (26.71%), whereas superhost listings make up 30.65% of all listings. Because the label belo...	evidence_cluster.json → superhost_rate_full (×100)	输出
11	...Target and features: The 11 features comprise three variables relat...	evidence_model.json → rq[4].n_features	输出

231	...inimum_nights, host_listings_count). room_type_3 merges the 231 shared-room and 52 hotel-room rows (283...	代码中的设定值 (常量 / 参数)	设定
52	...listings_count). room_type_3 merges the 231 shared-room and 52 hotel-room rows (283 in total, 1.1%) in...	代码中的设定值 (常量 / 参数)	设定
283	...m_type_3 merges the 231 shared-room and 52 hotel-room rows (283 in total, 1.1%) into Other, leaving thr...	evidence_preprocess.json → other_room_type_count	输出
1.1%	...s the 231 shared-room and 52 hotel-room rows (283 in total, 1.1%) into Other, leaving three categories t...	evidence_model.json → rq[6].cv_std (×100)	输出
0.682	...h price_num and accommodates have a Spearman correlation of 0.682. This classification does not drive the...	evidence_preprocess.json → price_accommodates_spearman	输出
0.1844	...oving price into the fundamentals group leaves its CV F1 at 0.1844 for the decision tree and 0.2647 for k-...	rq_experiments_extra.csv → A2_fundamentals_plus_price / cv_mean	输出
0.2647	... group leaves its CV F1 at 0.1844 for the decision tree and 0.2647 for k-NN, against 0.5577 and 0.5442 for...	rq_experiments_extra.csv → A2_fundamentals_plus_price / cv_mean	输出
0.5577	...t 0.1844 for the decision tree and 0.2647 for k-NN, against 0.5577 and 0.5442 for the operational group, s...	evidence_model.json → rq_extra[7].cv_mean	输出
0.5442	...r the decision tree and 0.2647 for k-NN, against 0.5577 and 0.5442 for the operational group, so the concl...	rq_experiments_extra.csv → B2_operational_minus_price / cv_mean	输出
80	...atifying by y, and take one fold as the test set, giving an 80/20 split. The 20,582 training rows cont...	划分比例: 5 折取 1 折作测试集 (设定)	推导
20,582	...d take one fold as the test set, giving an 80/20 split. The 20,582 training rows contain 6,308 superhost l...	evidence_model.json → split.train_rows	输出
6,308	...et, giving an 80/20 split. The 20,582 training rows contain 6,308 superhost listings and 14,274 non-super...	evidence_model.json → split_details.train_superhost_rows	输出
14,274	...e 20,582 training rows contain 6,308 superhost listings and 14,274 non-superhost listings from 11,292 host...	evidence_model.json → split_details.train_non_superhost_rows	输出
11,292	...8 superhost listings and 14,274 non-superhost listings from 11,292 hosts, while the 5,146 test rows contai...	evidence_model.json → split.train_hosts	输出
5,146	... 14,274 non-superhost listings from 11,292 hosts, while the 5,146 test rows contain 1,577 and 3,569 listi...	evidence_model.json → split.test_rows	输出
1,577	...stings from 11,292 hosts, while the 5,146 test rows contain 1,577 and 3,569 listings from 2,821 hosts, wi...	evidence_model.json → split_details.test_superhost_rows	输出
3,569	...m 11,292 hosts, while the 5,146 test rows contain 1,577 and 3,569 listings from 2,821 hosts, with no host...	evidence_model.json → split_details.test_non_superhost_rows	输出
2,821	...e the 5,146 test rows contain 1,577 and 3,569 listings from 2,821 hosts, with no host on both sides. The ...	evidence_model.json → split.test_hosts	输出
30.65%	... with no host on both sides. The superhost share is held at 30.65% on each side. Grouping is necessary bec...	evidence_preprocess.json → superhost_share_overall (×100)	输出
55.8%	...ping is necessary because the label belongs to the host and 55.8% of listings share a host with at least ...	evidence_preprocess.json → multi_listing_host_listing_share (×100)	输出
20,582	...Validation strategy: Hyperparameters are tuned on the 20,582 training rows using 5-fold, host-groupe...	evidence_model.json → split.train_rows	输出
4,116	...1 for the superhost class. Each validation fold holds about 4,116–4,117 rows, including 1,261–1,262 super...	evidence_model.json → split_details.cv_validation_folds[1].rows	输出
4,117	...the superhost class. Each validation fold holds about 4,116–4,117 rows, including 1,261–1,262 superhost l...	evidence_model.json → split_details.cv_validation_folds[0].rows	输出
1,261	...ach validation fold holds about 4,116–4,117 rows, including 1,261–1,262 superhost listings from 2,258–2,2...	evidence_model.json → split_details.cv_validation_folds[1].superhost_rows	输出
1,262	...lidation fold holds about 4,116–4,117 rows, including 1,261–1,262 superhost listings from 2,258–2,259 hos...	evidence_model.json → split_details.cv_validation_folds[0].superhost_rows	输出
2,258	...6–4,117 rows, including 1,261–1,262 superhost listings from 2,258–2,259 hosts. Ten folds were not used be...	evidence_model.json → split_details.cv_validation_folds[1].hosts	输出
2,259	...7 rows, including 1,261–1,262 superhost listings from 2,258–2,259 hosts. Ten folds were not used because ...	evidence_model.json → split_details.cv_validation_folds[0].hosts	输出
319	...sts. Ten folds were not used because the largest host holds 319 listings; that host would occupy 7.75% ...	evidence_model.json → split_details.largest_host_rows	输出
7.75%	...the largest host holds 319 listings; that host would occupy 7.75% of a validation fold under five folds a...	$319 \div (20,582 \div 5)$	推导
15.5%	...ould occupy 7.75% of a validation fold under five folds and 15.5% under ten, so a single host could domin...	$319 \div (20,582 \div 10)$	推导
11	...Hyperparameter grids: k-NN is trained with k \$ \{1, 5, 11, 21, 51, 101, 151, 201, 301\} and the de...	代码中的设定值 (常量 / 参数)	设定

21	...Hyperparameter grids: k–NN is trained with k \$ \$ \1, 5, 11, 21, 51, 101, 151, 201, 301\ and the decisi...	代码中的设定值 (常量 / 参数)	设定
51	...rparameter grids: k–NN is trained with k \$ \$ \1, 5, 11, 21, 51, 101, 151, 201, 301\ and the decision t...	代码中的设定值 (常量 / 参数)	设定
101	...ameter grids: k–NN is trained with k \$ \$ \1, 5, 11, 21, 51, 101, 151, 201, 301\ and the decision tree w...	代码中的设定值 (常量 / 参数)	设定
151	...r grids: k–NN is trained with k \$ \$ \1, 5, 11, 21, 51, 101, 151, 201, 301\ and the decision tree with m...	代码中的设定值 (常量 / 参数)	设定
201	...ds: k–NN is trained with k \$ \$ \1, 5, 11, 21, 51, 101, 151, 201, 301\ and the decision tree with max_de...	代码中的设定值 (常量 / 参数)	设定
301	...–NN is trained with k \$ \$ \1, 5, 11, 21, 51, 101, 151, 201, 301\ and the decision tree with max_depth \$...	代码中的设定值 (常量 / 参数)	设定
10	...and the decision tree with max_depth \$ \$ \2, 3, 4, 5, 6, 8, 10, 15, None\, where None means unlimited ...	evidence_model.json → best_param.DT	输出
15	...the decision tree with max_depth \$ \$ \2, 3, 4, 5, 6, 8, 10, 15, None\, where None means unlimited dept...	cv_results.csv → DT / param	输出
0.5018	...ed depth and shows the cost of memorising the training set (0.5018). Only odd values of k are tested so th...	cv_results.csv → DT / cv_mean	输出
0.694	...ch predicts “not a superhost” for every listing; it reaches 0.694 accuracy and an F1 score of exactly 0. ...	evidence_model.json → test.Baseline.accuracy	输出
1,000	...y and an F1 score of exactly 0. A percentile bootstrap with 1,000 resamples drawn with replacement gives ...	代码中的设定值 (常量 / 参数)	设定
95%	...otstrap with 1,000 resamples drawn with replacement gives a 95% confidence interval for test F1. The te...	代码中的设定值 (常量 / 参数)	设定
2,821	... 95% confidence interval for test F1. The test set contains 2,821 hosts and 5,146 rows, and listings of t...	evidence_model.json → split.test_hosts	输出
5,146	...interval for test F1. The test set contains 2,821 hosts and 5,146 rows, and listings of the same host sha...	evidence_model.json → split.test_rows	输出

2.4 Feature Selection

10	...od: Impurity–based importance is taken from the tuned depth–10 decision tree, measured by Gini impurit...	evidence_model.json → best_param.DT	输出
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2.5 Dimensionality Reduction & Clustering

21,251	...Candidate feature sets. Clustering runs on the 21,251 reviewed listings, since the 4,477 zero...	evidence_cluster.json → n_rows_reviewed	输出
4,477	... Clustering runs on the 21,251 reviewed listings, since the 4,477 zero–review listings share the imputed ...	evidence_preprocess.json → zero_review_count	输出
4,594	...ince the 4,477 zero–review listings share the imputed value 4,594, producing a separate cluster. Three ca...	evidence_preprocess.json → days_since_last_review_fill_value	输出
667	...ia StandardScaler. Without log transform a single host with 667 listings would dominate a component on ...	evidence_cluster.json → max_before_log.host_listings_count	输出

2.5 Dimensionality Reduction & Clustering · Table 2

5,000	...uster superhost rates the silhouette score is computed on a 5,000–row sample, and the smallest cluster is...	notebook 4.5 打印输出	输出
0.5331	...menity_count, days_since_last_review, host_listings_count 0.5331 0.2942 3,243 S2 Fundamentals on...	evidence_cluster.json → candidate_set_comparison.S1_operational_only.superhost_rate_spread	输出
0.2942	...unt, days_since_last_review, host_listings_count 0.5331 0.2942 3,243 S2 Fundamentals only dist...	evidence_cluster.json → candidate_set_comparison.S1_operational_only.silhouette_5000_sample	输出
3,243	..._since_last_review, host_listings_count 0.5331 0.2942 3,243 S2 Fundamentals only dist_cbd, ac...	evidence_cluster.json → candidate_set_comparison.S1_operational_only.smallest_cluster_size	输出
0.0754	... 3,243 S2 Fundamentals only dist_cbd, accommodates 0.0754 0.3997 780 S3 Mixed (both famili...	evidence_cluster.json → candidate_set_comparison.S2_fundamentals_onl y.superhost_rate_spread	输出
0.3997	... S2 Fundamentals only dist_cbd, accommodates 0.0754 0.3997 780 S3 Mixed (both families) ava...	evidence_cluster.json → candidate_set_comparison.S2_fundamentals_onl y.silhouette_5000_sample	输出
780	...damentals only dist_cbd, accommodates 0.0754 0.3997 780 S3 Mixed (both families) availabil...	evidence_cluster.json → candidate_set_comparison.S2_fundamentals_onl y.smallest_cluster_size	输出
0.4719	..._last_review, host_listings_count, dist_cbd, accommodates 0.4719 0.2232 2,365 ...	evidence_cluster.json → kmeans_superhost_rate_spread	输出
0.2232	...iew, host_listings_count, dist_cbd, accommodates 0.4719 0.2232 2,365 ...	evidence_cluster.json → kmeans_sweep[3].silhouette_5000_sample	输出

2,365	...listings_count, dist_cbd, accommodates 0.4719 0.2232 2,365 ...	evidence_cluster.json → kmeans_sizes.3	输出
2.5 Dimensionality Reduction & Clustering			
0.4719	...etaining most of S1's separation (a superhost-rate range of 0.4719 versus 0.5331). The very small range fo...	evidence_cluster.json → kmeans_superhost_rate_spread	输出
0.5331	...of S1's separation (a superhost-rate range of 0.4719 versus 0.5331). The very small range for S2 (0.0754) ...	evidence_cluster.json → candidate_set_comparison.S1_operational_only.superhost_rate_spread	输出
0.0754	...ange of 0.4719 versus 0.5331). The very small range for S2 (0.0754) is itself informative, because cluster...	evidence_cluster.json → candidate_set_comparison.S2_fundamentals_only.superhost_rate_spread	输出
16.0%	...he SSE curve: the SSE reduction when adding each cluster is 16.0%, 12.8%, 12.0% and 6.8% for k=2→3, 3→4, ...	evidence_cluster.json → kmeans_sweep[1].sse_drop_pct	输出
12.8%	...curve: the SSE reduction when adding each cluster is 16.0%, 12.8%, 12.0% and 6.8% for k=2→3, 3→4, 4→5 and...	evidence_cluster.json → kmeans_sweep[2].sse_drop_pct	输出
12.0%	...the SSE reduction when adding each cluster is 16.0%, 12.8%, 12.0% and 6.8% for k=2→3, 3→4, 4→5 and 5→6. T...	evidence_cluster.json → kmeans_sweep[3].sse_drop_pct	输出
6.8%	...duction when adding each cluster is 16.0%, 12.8%, 12.0% and 6.8% for k=2→3, 3→4, 4→5 and 5→6. The silhou...	evidence_cluster.json → kmeans_sweep[4].sse_drop_pct	输出
0.2232	..., 4→5 and 5→6. The silhouette score is also highest at k=5 (0.2232, versus 0.2227 at k=4 and 0.2177 at k=6...	evidence_cluster.json → kmeans_sweep[3].silhouette_5000_sample	输出
0.2227	...The silhouette score is also highest at k=5 (0.2232, versus 0.2227 at k=4 and 0.2177 at k=6). For hierarch...	evidence_cluster.json → kmeans_sweep[2].silhouette_5000_sample	输出
0.2177	...re is also highest at k=5 (0.2232, versus 0.2227 at k=4 and 0.2177 at k=6). For hierarchical clustering, W...	evidence_cluster.json → kmeans_sweep[4].silhouette_5000_sample	输出
21,247	...k=5, into one dominant cluster: single linkage gives sizes [21,247, 1, 1, 1, 1], average linkage [19,482, ...	evidence_cluster.json → linkage_comparison_sizes.single[0]	输出
19,482	... linkage gives sizes [21,247, 1, 1, 1, 1], average linkage [19,482, 1,401, 341, 26, 1], and complete linka...	evidence_cluster.json → linkage_comparison_sizes.average[0]	输出
1,401	... gives sizes [21,247, 1, 1, 1, 1], average linkage [19,482, 1,401, 341, 26, 1], and complete linkage a la...	evidence_cluster.json → linkage_comparison_sizes.average[1]	输出
341	...sizes [21,247, 1, 1, 1, 1], average linkage [19,482, 1,401, 341, 26, 1], and complete linkage a largest...	evidence_cluster.json → linkage_comparison_sizes.average[2]	输出
26	... [21,247, 1, 1, 1, 1], average linkage [19,482, 1,401, 341, 26, 1], and complete linkage a largest clu...	evidence_cluster.json → linkage_comparison_sizes.average[3]	输出
12,473	...401, 341, 26, 1], and complete linkage a largest cluster of 12,473. Ward instead merges the two clusters t...	evidence_cluster.json → linkage_comparison_sizes.complete[0]	输出
3.1 Data Preprocessing · Table 3			
4,477	...ince_last_review from last_review last_review missing for 4,477 rows, could not derive missing values ...	evidence_preprocess.json → zero_review_count	输出
4,594	... derive missing values days_since_last_review filled with 4,594 days for missing last_review rows. 0 mi...	evidence_preprocess.json → days_since_last_review_fill_value	输出
84	...ys for missing last_review rows. 0 missing rows with median 84 days. Counting amenity_count by parsi...	evidence_preprocess.json → days_since_last_review_median	输出
3,096	...particular listing. A naive comma count would be wrong on 3,096 rows (12.03%), over-counting by up to 2...	evidence_preprocess.json → amenity_naive_diff_count	输出
12.03%	...isting. A naive comma count would be wrong on 3,096 rows (12.03%), over-counting by up to 21 amenities f...	evidence_preprocess.json → amenity_naive_diff_pct	输出
21	...uld be wrong on 3,096 rows (12.03%), over-counting by up to 21 amenities for a single listing. Imputi...	evidence_preprocess.json → amenity_naive_max_diff	输出
25.47%	...s via median imputation plus a missing indicator variable 25.47% of prices missing; the missingness depe...	evidence_preprocess.json → price_missing_pct	输出
0%	...e missingness depends on availability_365 (MAR, not MCAR) 0% prices missing, imputed within the pipe...	evidence_cluster.json → skew_before_log.availability_365 (数值为 0, 多个字段同值)	输出
245.00	...puted within the pipeline using the training-set median of \$245.00. price_missing marks every imputed row....	evidence_model.json → split_details.price_median_train	输出
3.2 Correlation Analysis · Table 4			
21,251	...d mutual information for each pair of the five variables (n=21,251 listings with at least one review). Row...	evidence_correlation.json → n_rows	输出
-0.438	...) NMI Bins days_since_last_review availability_365 -0.438 -0.395 0.2639 0.0856 10x8 day...	correlation_matrix.csv → days_since_last_review ~ availability_365 / pearson	输出

-0.395	... Bins days_since_last_review availability_365 -0.438 -0.395 0.2639 0.0856 10x8 days_since_l...	correlation_matrix.csv → days_since_last_review ~ availability_365 / spearman	输出
0.2639	...days_since_last_review availability_365 -0.438 -0.395 0.2639 0.0856 10x8 days_since_last_revie...	correlation_matrix.csv → days_since_last_review ~ availability_365 / mi_bits	输出
0.0856	...e_last_review availability_365 -0.438 -0.395 0.2639 0.0856 10x8 days_since_last_review y -0....	evidence_correlation.json → mi_bin_sensitivity.10.nmi	输出
10	...view availability_365 -0.438 -0.395 0.2639 0.0856 10x8 days_since_last_review y -0.337 ...	代码中的设定值 (常量 / 参数)	设定
-0.337	...0.395 0.2639 0.0856 10x8 days_since_last_review y -0.337 -0.443 0.1672 0.0785 10x2 avai...	correlation_matrix.csv → days_since_last_review ~ y / pearson	输出
-0.443	...2639 0.0856 10x8 days_since_last_review y -0.337 -0.443 0.1672 0.0785 10x2 availability_...	correlation_matrix.csv → days_since_last_review ~ y / spearman	输出
0.1672	...0856 10x8 days_since_last_review y -0.337 -0.443 0.1672 0.0785 10x2 availability_365 y 0...	correlation_matrix.csv → days_since_last_review ~ y / mi_bits	输出
0.0785	...0x8 days_since_last_review y -0.337 -0.443 0.1672 0.0785 10x2 availability_365 y 0.146 0....	correlation_matrix.csv → days_since_last_review ~ y / nmi	输出
10	...s_since_last_review y -0.337 -0.443 0.1672 0.0785 10x2 availability_365 y 0.146 0.155 ...	代码中的设定值 (常量 / 参数)	设定
0.146	...337 -0.443 0.1672 0.0785 10x2 availability_365 y 0.146 0.155 0.0613 0.0324 8x2 host_...	correlation_matrix.csv → availability_365 ~ y / pearson	输出
0.155	...443 0.1672 0.0785 10x2 availability_365 y 0.146 0.155 0.0613 0.0324 8x2 host_listings...	correlation_matrix.csv → availability_365 ~ y / spearman	输出
0.0613	...1672 0.0785 10x2 availability_365 y 0.146 0.155 0.0613 0.0324 8x2 host_listings_count di...	correlation_matrix.csv → availability_365 ~ y / mi_bits	输出
0.0324	...0785 10x2 availability_365 y 0.146 0.155 0.0613 0.0324 8x2 host_listings_count dist_cbd ...	correlation_matrix.csv → availability_365 ~ y / nmi	输出
-0.131	...55 0.0613 0.0324 8x2 host_listings_count dist_cbd -0.131 -0.225 0.0837 0.0285 7x10 host...	correlation_matrix.csv → host_listings_count ~ dist_cbd / pearson	输出
-0.225	...13 0.0324 8x2 host_listings_count dist_cbd -0.131 -0.225 0.0837 0.0285 7x10 host_listings...	correlation_matrix.csv → host_listings_count ~ dist_cbd / spearman	输出
0.0837	...24 8x2 host_listings_count dist_cbd -0.131 -0.225 0.0837 0.0285 7x10 host_listings_count da...	correlation_matrix.csv → host_listings_count ~ dist_cbd / mi_bits	输出
0.0285	... host_listings_count dist_cbd -0.131 -0.225 0.0837 0.0285 7x10 host_listings_count days_since...	correlation_matrix.csv → host_listings_count ~ dist_cbd / nmi	输出
10	...ings_count dist_cbd -0.131 -0.225 0.0837 0.0285 7x10 host_listings_count days_since_last_re...	代码中的设定值 (常量 / 参数)	设定
-0.113	...0.0285 7x10 host_listings_count days_since_last_review -0.113 -0.194 0.0806 0.0274 7x10 hos...	correlation_matrix.csv → host_listings_count ~ days_since_last_review / pearson	输出
-0.194	...7x10 host_listings_count days_since_last_review -0.113 -0.194 0.0806 0.0274 7x10 host_listing...	correlation_matrix.csv → host_listings_count ~ days_since_last_review / spearman	输出
0.0806	...t_listings_count days_since_last_review -0.113 -0.194 0.0806 0.0274 7x10 host_listings_count a...	correlation_matrix.csv → host_listings_count ~ days_since_last_review / mi_bits	输出
0.0274	...s_count days_since_last_review -0.113 -0.194 0.0806 0.0274 7x10 host_listings_count availabili...	correlation_matrix.csv → host_listings_count ~ days_since_last_review / nmi	输出
10	...s_since_last_review -0.113 -0.194 0.0806 0.0274 7x10 host_listings_count availability_365 ...	代码中的设定值 (常量 / 参数)	设定
0.059	...06 0.0274 7x10 host_listings_count availability_365 0.059 0.225 0.0621 0.0230 7x8 availa...	correlation_matrix.csv → host_listings_count ~ availability_365 / pearson	输出
0.225	...274 7x10 host_listings_count availability_365 0.059 0.225 0.0621 0.0230 7x8 availability_3...	correlation_matrix.csv → host_listings_count ~ availability_365 / spearman	输出
0.0621	...10 host_listings_count availability_365 0.059 0.225 0.0621 0.0230 7x8 availability_365 dist_c...	correlation_matrix.csv → host_listings_count ~ availability_365 / mi_bits	输出
0.0230	..._listings_count availability_365 0.059 0.225 0.0621 0.0230 7x8 availability_365 dist_cbd 0.17...	correlation_matrix.csv → host_listings_count ~ availability_365 / nmi	输出
0.170	... 0.225 0.0621 0.0230 7x8 availability_365 dist_cbd 0.170 0.124 0.0455 0.0147 8x10 days...	correlation_matrix.csv → availability_365 ~ dist_cbd / pearson	输出
0.124	... 0.0621 0.0230 7x8 availability_365 dist_cbd 0.170 0.124 0.0455 0.0147 8x10 days_since_l...	correlation_matrix.csv → availability_365 ~ dist_cbd / spearman	输出
0.0455	... 0.0230 7x8 availability_365 dist_cbd 0.170 0.124 0.0455 0.0147 8x10 days_since_last_revie...	correlation_matrix.csv → availability_365 ~ dist_cbd / mi_bits	输出
0.0147	... 7x8 availability_365 dist_cbd 0.170 0.124 0.0455 0.0147 8x10 days_since_last_review dist_cb...	correlation_matrix.csv → availability_365 ~ dist_cbd / nmi	输出

10	...lability_365 dist_cbd 0.170 0.124 0.0455 0.0147 8×10 days_since_last_review dist_cbd −0...	代码中的设定值 (常量 / 参数)	设定
−0.081	... 0.0455 0.0147 8×10 days_since_last_review dist_cbd −0.081 0.061 0.0458 0.0138 10×10 host...	correlation_matrix.csv → days_since_last_review ~ dist_cbd / pearson	输出
0.061	... 0.0147 8×10 days_since_last_review dist_cbd −0.081 0.061 0.0458 0.0138 10×10 host_listing...	correlation_matrix.csv → days_since_last_review ~ dist_cbd / spearman	输出
0.0458	... 8×10 days_since_last_review dist_cbd −0.081 0.061 0.0458 0.0138 10×10 host_listings_count y...	correlation_matrix.csv → days_since_last_review ~ dist_cbd / mi_bits	输出
0.0138	...days_since_last_review dist_cbd −0.081 0.061 0.0458 0.0138 10×10 host_listings_count y −0.069...	correlation_matrix.csv → days_since_last_review ~ dist_cbd / nmi	输出
10	...e_last_review dist_cbd −0.081 0.061 0.0458 0.0138 10×10 host_listings_count y −0.069 0...	代码中的设定值 (常量 / 参数)	设定
10	...ast_review dist_cbd −0.081 0.061 0.0458 0.0138 10×10 host_listings_count y −0.069 0.070...	代码中的设定值 (常量 / 参数)	设定
−0.069	... 0.061 0.0458 0.0138 10×10 host_listings_count y −0.069 0.070 0.0169 0.0097 7×2 dist_...	correlation_matrix.csv → host_listings_count ~ y / pearson	输出
0.070	... 0.0458 0.0138 10×10 host_listings_count y −0.069 0.070 0.0169 0.0097 7×2 dist_cbd y ...	correlation_matrix.csv → host_listings_count ~ y / spearman	输出
0.0169	... 0.0138 10×10 host_listings_count y −0.069 0.070 0.0169 0.0097 7×2 dist_cbd y 0.076 0...	correlation_matrix.csv → host_listings_count ~ y / mi_bits	输出
0.0097	... 10×10 host_listings_count y −0.069 0.070 0.0169 0.0097 7×2 dist_cbd y 0.076 0.009 0...	correlation_matrix.csv → host_listings_count ~ y / nmi	输出
0.076	...t y −0.069 0.070 0.0169 0.0097 7×2 dist_cbd y 0.076 0.009 0.0101 0.0047 10×2 ...	correlation_matrix.csv → dist_cbd ~ y / pearson	输出
0.009	...069 0.070 0.0169 0.0097 7×2 dist_cbd y 0.076 0.009 0.0101 0.0047 10×2 ...	correlation_matrix.csv → dist_cbd ~ y / spearman	输出
0.0101	...070 0.0169 0.0097 7×2 dist_cbd y 0.076 0.009 0.0101 0.0047 10×2 ...	correlation_matrix.csv → dist_cbd ~ y / mi_bits	输出
0.0047	...0169 0.0097 7×2 dist_cbd y 0.076 0.009 0.0101 0.0047 10×2 ...	correlation_matrix.csv → dist_cbd ~ y / nmi	输出
10	...0097 7×2 dist_cbd y 0.076 0.009 0.0101 0.0047 10×2 ...	代码中的设定值 (常量 / 参数)	设定

3.2 Correlation Analysis

−0.069	...pearman disagree on host_listings_count with the target, at −0.069 and +0.070, because the superhost rate ...	correlation_matrix.csv → host_listings_count ~ y / pearson	输出
+0.070	...agree on host_listings_count with the target, at −0.069 and +0.070, because the superhost rate does not mo...	correlation_matrix.csv → host_listings_count ~ y / spearman	输出

3.2 Correlation Analysis · Figure 1

21,251	...Superhost rate by host_listings_count band (n = 21,251 reviewed listings). The red line marks ...	evidence_cluster.json → n_rows_reviewed	输出
35.7%	...listings). The red line marks the overall superhost rate of 35.7%. ...	evidence_cluster.json → superhost_rate_reviewed (×100)	输出

3.3 Supervised Learning Models and Evaluation · Table 5

20,582	...re at every hyperparameter value tried (five grouped folds, 20,582 training listings). Selected configurat...	evidence_model.json → split.train_rows	输出
11	...of F1 across the five folds. > *9> X k-NN: k 1 5 11 21 51 101 151 201 301 CV F...	代码中的设定值 (常量 / 参数)	设定
21	... across the five folds. > *9> X k-NN: k 1 5 11 21 51 101 151 201 301 CV F1 (me...	代码中的设定值 (常量 / 参数)	设定
51	...ss the five folds. > *9> X k-NN: k 1 5 11 21 51 101 151 201 301 CV F1 (mean) ...	代码中的设定值 (常量 / 参数)	设定
101	...e five folds. > *9> X k-NN: k 1 5 11 21 51 101 151 201 301 CV F1 (mean) 0.480...	cv_results.csv → KNN / param	输出
151	... folds. > *9> X k-NN: k 1 5 11 21 51 101 151 201 301 CV F1 (mean) 0.4802 0...	cv_results.csv → KNN / param	输出
201	... > *9> X k-NN: k 1 5 11 21 51 101 151 201 301 CV F1 (mean) 0.4802 0.5020 ...	cv_results.csv → KNN / param	输出
301	... *9> X k-NN: k 1 5 11 21 51 101 151 201 301 CV F1 (mean) 0.4802 0.5020 0.507...	cv_results.csv → KNN / param	输出
0.4802	... 5 11 21 51 101 151 201 301 CV F1 (mean) 0.4802 0.5020 0.5071 0.5190 0.5299 0...	cv_results.csv → KNN / cv_mean	输出
0.5020	... 21 51 101 151 201 301 CV F1 (mean) 0.4802 0.5020 0.5071 0.5190 0.5299 0.5370 0...	cv_results.csv → KNN / cv_mean	输出

0.5071	...1 101 151 201 301 CV F1 (mean) 0.4802 0.5020 0.5071 0.5190 0.5299 0.5370 0.5312 0...	cv_results.csv → KNN / cv_mean	输出
0.5190	... 151 201 301 CV F1 (mean) 0.4802 0.5020 0.5071 0.5190 0.5299 0.5370 0.5312 0.5309 0...	cv_results.csv → KNN / cv_mean	输出
0.5299	...1 301 CV F1 (mean) 0.4802 0.5020 0.5071 0.5190 0.5299 0.5370 0.5312 0.5309 0.5249 ±...	cv_results.csv → KNN / cv_mean	输出
0.5370	...CV F1 (mean) 0.4802 0.5020 0.5071 0.5190 0.5299 0.5370 0.5312 0.5309 0.5249 ± SD 0.0...	evidence_model.json → rq[4].cv_mean	输出
0.5312	...an) 0.4802 0.5020 0.5071 0.5190 0.5299 0.5370 0.5312 0.5309 0.5249 ± SD 0.0175 0.0...	cv_results.csv → KNN / cv_mean	输出
0.5309	...802 0.5020 0.5071 0.5190 0.5299 0.5370 0.5312 0.5309 0.5249 ± SD 0.0175 0.0159 0.0...	cv_results.csv → KNN / cv_mean	输出
0.5249	...020 0.5071 0.5190 0.5299 0.5370 0.5312 0.5309 0.5249 ± SD 0.0175 0.0159 0.0176 0.0...	cv_results.csv → KNN / cv_mean	输出
0.0175	...5190 0.5299 0.5370 0.5312 0.5309 0.5249 ± SD 0.0175 0.0159 0.0176 0.0112 0.0121 0...	cv_results.csv → KNN / cv_std	输出
0.0159	...5299 0.5370 0.5312 0.5309 0.5249 ± SD 0.0175 0.0159 0.0176 0.0112 0.0121 0.0133 0...	evidence_model.json → rq_extra[3].cv_std	输出
0.0176	...5370 0.5312 0.5309 0.5249 ± SD 0.0175 0.0159 0.0176 0.0112 0.0121 0.0133 0.0147 0...	cv_results.csv → KNN / cv_std	输出
0.0112	...5312 0.5309 0.5249 ± SD 0.0175 0.0159 0.0176 0.0112 0.0121 0.0133 0.0147 0.0175 0...	cv_results.csv → KNN / cv_std	输出
0.0121	...5309 0.5249 ± SD 0.0175 0.0159 0.0176 0.0112 0.0121 0.0133 0.0147 0.0175 0.0226 ...	cv_results.csv → KNN / cv_std	输出
0.0133	...5249 ± SD 0.0175 0.0159 0.0176 0.0112 0.0121 0.0133 0.0147 0.0175 0.0226 Tree: max...	evidence_model.json → rq[4].cv_std	输出
0.0147	... SD 0.0175 0.0159 0.0176 0.0112 0.0121 0.0133 0.0147 0.0175 0.0226 Tree: max_depth ...	cv_results.csv → KNN / cv_std	输出
0.0175	...175 0.0159 0.0176 0.0112 0.0121 0.0133 0.0147 0.0175 0.0226 Tree: max_depth 2 3 4...	cv_results.csv → KNN / cv_std	输出
0.0226	...159 0.0176 0.0112 0.0121 0.0133 0.0147 0.0175 0.0226 Tree: max_depth 2 3 4 5 6 ...	cv_results.csv → KNN / cv_std	输出
10	...0175 0.0226 Tree: max_depth 2 3 4 5 6 8 10 15 None CV F1 (mean) 0.5158 0....	cv_results.csv → DT / param	输出
15	... 0.0226 Tree: max_depth 2 3 4 5 6 8 10 15 None CV F1 (mean) 0.5158 0.5611 ...	cv_results.csv → DT / param	输出
0.5158	...th 2 3 4 5 6 8 10 15 None CV F1 (mean) 0.5158 0.5611 0.5712 0.5384 0.5512 0...	cv_results.csv → DT / cv_mean	输出
0.5611	...3 4 5 6 8 10 15 None CV F1 (mean) 0.5158 0.5611 0.5712 0.5384 0.5512 0.5622 0...	cv_results.csv → DT / cv_mean	输出
0.5712	... 6 8 10 15 None CV F1 (mean) 0.5158 0.5611 0.5712 0.5384 0.5512 0.5622 0.5745 0...	cv_results.csv → DT / cv_mean	输出
0.5384	... 10 15 None CV F1 (mean) 0.5158 0.5611 0.5712 0.5384 0.5512 0.5622 0.5745 0.5363 0...	cv_results.csv → DT / cv_mean	输出
0.5512	... None CV F1 (mean) 0.5158 0.5611 0.5712 0.5384 0.5512 0.5622 0.5745 0.5363 0.5018 ±...	cv_results.csv → DT / cv_mean	输出
0.5622	...CV F1 (mean) 0.5158 0.5611 0.5712 0.5384 0.5512 0.5622 0.5745 0.5363 0.5018 ± SD 0.0...	cv_results.csv → DT / cv_mean	输出
0.5745	...an) 0.5158 0.5611 0.5712 0.5384 0.5512 0.5622 0.5745 0.5363 0.5018 ± SD 0.0503 0.0...	evidence_model.json → rq[5].cv_mean	输出
0.5363	...158 0.5611 0.5712 0.5384 0.5512 0.5622 0.5745 0.5363 0.5018 ± SD 0.0503 0.0431 0.0...	cv_results.csv → DT / cv_mean	输出
0.5018	...611 0.5712 0.5384 0.5512 0.5622 0.5745 0.5363 0.5018 ± SD 0.0503 0.0431 0.0177 0.0...	cv_results.csv → DT / cv_mean	输出
0.0503	...5384 0.5512 0.5622 0.5745 0.5363 0.5018 ± SD 0.0503 0.0431 0.0177 0.0424 0.0240 0...	cv_results.csv → DT / cv_std	输出
0.0431	...5512 0.5622 0.5745 0.5363 0.5018 ± SD 0.0503 0.0431 0.0177 0.0424 0.0240 0.0123 0...	cv_results.csv → DT / cv_std	输出
0.0177	...5622 0.5745 0.5363 0.5018 ± SD 0.0503 0.0431 0.0177 0.0424 0.0240 0.0123 0.0130 0...	cv_results.csv → DT / cv_std	输出
0.0424	...5745 0.5363 0.5018 ± SD 0.0503 0.0431 0.0177 0.0424 0.0240 0.0123 0.0130 0.0171 0...	cv_results.csv → DT / cv_std	输出
0.0240	...5363 0.5018 ± SD 0.0503 0.0431 0.0177 0.0424 0.0240 0.0123 0.0130 0.0171 0.0072 ...	cv_results.csv → DT / cv_std	输出

0.0123	...5018 ± SD 0.0503 0.0431 0.0177 0.0424 0.0240 0.0123 0.0130 0.0171 0.0072 ...	cv_results.csv → DT / cv_std	输出
0.0130	... SD 0.0503 0.0431 0.0177 0.0424 0.0240 0.0123 0.0130 0.0171 0.0072 ...	evidence_model.json → rq[5].cv_std	输出
0.0171	...503 0.0431 0.0177 0.0424 0.0240 0.0123 0.0130 0.0171 0.0072 ...	cv_results.csv → DT / cv_std	输出
0.0072	...431 0.0177 0.0424 0.0240 0.0123 0.0130 0.0171 0.0072 ...	cv_results.csv → DT / cv_std	输出

3.3 Supervised Learning Models and Evaluation · Table 6

101	...1 Effect k-NN Number of neighbours (k) k = 5 k = 101 0.5020 0.5370 +0.0350 Decision ...	cv_results.csv → KNN / param	输出
0.5020	...fect k-NN Number of neighbours (k) k = 5 k = 101 0.5020 0.5370 +0.0350 Decision Tree Ma...	cv_results.csv → KNN / cv_mean	输出
0.5370	...-NN Number of neighbours (k) k = 5 k = 101 0.5020 0.5370 +0.0350 Decision Tree Maximum dep...	evidence_model.json → rq[4].cv_mean	输出
+0.0350	...ber of neighbours (k) k = 5 k = 101 0.5020 0.5370 +0.0350 Decision Tree Maximum depth of tree...	0.5370 - 0.5020	推导
10	...depth of tree (max_depth) max_depth = None max_depth = 10 0.5018 0.5745 +0.0727 ...	evidence_model.json → best_param.DT	输出
0.5018	... of tree (max_depth) max_depth = None max_depth = 10 0.5018 0.5745 +0.0727 ...	cv_results.csv → DT / cv_mean	输出
0.5745	...(max_depth) max_depth = None max_depth = 10 0.5018 0.5745 +0.0727 ...	cv_results.csv → DT / cv_mean	输出
+0.0727	...h) max_depth = None max_depth = 10 0.5018 0.5745 +0.0727 ...	0.5745 - 0.5018	推导

3.3 Supervised Learning Models and Evaluation · Table 7

5,146	...Test-set performance (n= 5,146), with precision, recall, accuracy and ...	evidence_model.json → split.test_rows	输出
0.6935	... Recall Accuracy F1 Score Baseline Non-superhost 0.6935 1.0000 0.6935 0.8190 Superhost ...	evidence_model.json → test.Baseline.class_0.precision	输出
1.0000	... Accuracy F1 Score Baseline Non-superhost 0.6935 1.0000 0.6935 0.8190 Superhost 0.0000 ...	evidence_model.json → hard_case.comparison.has_review.non_superhost_median	输出
0.6935	... F1 Score Baseline Non-superhost 0.6935 1.0000 0.6935 0.8190 Superhost 0.0000 0.0000 ...	evidence_model.json → test.Baseline.accuracy	输出
0.8190	...re Baseline Non-superhost 0.6935 1.0000 0.6935 0.8190 Superhost 0.0000 0.0000 0.000...	evidence_model.json → test.Baseline.class_0.f1-score	输出
0.0000	...superhost 0.6935 1.0000 0.6935 0.8190 Superhost 0.0000 0.0000 0.0000 k-NN (k=101) Non...	evidence_model.json → test.Baseline.class_1.f1-score (数值为 0, 多个字段同值)	输出
0.0000	... 0.6935 1.0000 0.6935 0.8190 Superhost 0.0000 0.0000 0.0000 k-NN (k=101) Non-superhos...	evidence_model.json → test_extra.Baseline.predicted_superhost (数值为 0, 多个字段同值)	输出
0.0000	... 1.0000 0.6935 0.8190 Superhost 0.0000 0.0000 0.0000 k-NN (k=101) Non-superhost 0.8106 ...	evidence_model.json → hard_case.tree_path[10].superhost_share (数值为 0, 多个字段同值)	输出
101	... 0.8190 Superhost 0.0000 0.0000 0.0000 k-NN (k=101) Non-superhost 0.8106 0.8322 0....	代码中的设定值 (常量 / 参数)	设定
0.8106	... 0.0000 0.0000 0.0000 k-NN (k=101) Non-superhost 0.8106 0.8322 0.7487 0.8212 Superhost ...	evidence_model.json → test.KNN.class_0.precision	输出
0.8322	... 0.0000 0.0000 k-NN (k=101) Non-superhost 0.8106 0.8322 0.7487 0.8212 Superhost 0.5958 ...	evidence_model.json → test.KNN.class_0.recall	输出
0.7487	... 0.0000 k-NN (k=101) Non-superhost 0.8106 0.8322 0.7487 0.8212 Superhost 0.5958 0.5599 ...	evidence_model.json → test.KNN.accuracy	输出
0.8212	... k-NN (k=101) Non-superhost 0.8106 0.8322 0.7487 0.8212 Superhost 0.5958 0.5599 0.577...	evidence_model.json → test.KNN.class_0.f1-score	输出
0.5958	...superhost 0.8106 0.8322 0.7487 0.8212 Superhost 0.5958 0.5599 0.5773 Decision Tree (dep...	evidence_model.json → test.KNN.class_1.precision	输出
0.5599	... 0.8106 0.8322 0.7487 0.8212 Superhost 0.5958 0.5599 0.5773 Decision Tree (depth=10) ...	evidence_model.json → test.KNN.class_1.recall	输出
0.5773	... 0.8322 0.7487 0.8212 Superhost 0.5958 0.5599 0.5773 Decision Tree (depth=10) Non-superho...	evidence_model.json → test.KNN.f1	输出
10	...uperhost 0.5958 0.5599 0.5773 Decision Tree (depth=10) Non-superhost 0.8445 0.8067 0....	evidence_model.json → best_param.DT	输出

0.8445	...5599 0.5773 Decision Tree (depth=10) Non-superhost 0.8445 0.8067 0.7629 0.8252 Superhost ...	<code>evidence_model.json → test.DT.class_0.precision</code>	输出
0.8067	...0.5773 Decision Tree (depth=10) Non-superhost 0.8445 0.8067 0.7629 0.8252 Superhost 0.6028 ...	<code>evidence_model.json → test.DT.class_0.recall</code>	输出
0.7629	...ecision Tree (depth=10) Non-superhost 0.8445 0.8067 0.7629 0.8252 Superhost 0.6028 0.6639 ...	<code>evidence_model.json → test.DT.accuracy</code>	输出
0.8252	...ree (depth=10) Non-superhost 0.8445 0.8067 0.7629 0.8252 Superhost 0.6028 0.6639 0.631...	<code>evidence_model.json → test.DT.class_0.f1-score</code>	输出
0.6028	...superhost 0.8445 0.8067 0.7629 0.8252 Superhost 0.6028 0.6639 0.6319 ...	<code>evidence_model.json → test.DT.class_1.precision</code>	输出
0.6639	... 0.8445 0.8067 0.7629 0.8252 Superhost 0.6028 0.6639 0.6319 ...	<code>evidence_model.json → test.DT.class_1.recall</code>	输出
0.6319	... 0.8067 0.7629 0.8252 Superhost 0.6028 0.6639 0.6319 ...	<code>evidence_model.json → test.DT.f1</code>	输出

3.3 Supervised Learning Models and Evaluation · Table 8

95%	...95% percentile-bootstrap intervals (1,000 p...	代码中的设定值 (常量 / 参数)	设定
1,000	...95% percentile-bootstrap intervals (1,000 paired host resamples) for test-set sup...	<code>notebook 3.4</code> 打印输出	输出
95%	... B and F are decision-tree results. L> > Model/Pair 95% CI Point estimate k-NN [0.5423, ...	代码中的设定值 (常量 / 参数)	设定
0.5423	... L> > Model/Pair 95% CI Point estimate k-NN [0.5423 , 0.6155] 0.5773 Decision Tree [0....	<code>evidence_model.json → boot_KNN_vs_DT.a[0]</code>	输出
0.6155	... Model/Pair 95% CI Point estimate k-NN [0.5423, 0.6155] 0.5773 Decision Tree [0.5940, 0....	<code>evidence_model.json → boot_KNN_vs_DT.a[1]</code>	输出
0.5773	...Pair 95% CI Point estimate k-NN [0.5423, 0.6155] 0.5773 Decision Tree [0.5940, 0.6757] 0....	<code>evidence_model.json → test.KNN.f1</code>	输出
0.5940	...mate k-NN [0.5423, 0.6155] 0.5773 Decision Tree [0.5940 , 0.6757] 0.6319 DT — k-NN [0.0130,...	<code>evidence_model.json → boot_KNN_vs_DT.b[0]</code>	输出
0.6757	...k-NN [0.5423, 0.6155] 0.5773 Decision Tree [0.5940, 0.6757] 0.6319 DT — k-NN [0.0130, 0.0986]...	<code>evidence_model.json → boot_KNN_vs_DT.b[1]</code>	输出
0.6319	...5423, 0.6155] 0.5773 Decision Tree [0.5940, 0.6757] 0.6319 DT — k-NN [0.0130, 0.0986] 0.0546 ...	<code>evidence_model.json → test.DT.f1</code>	输出
0.0130	...3 Decision Tree [0.5940, 0.6757] 0.6319 DT — k-NN [0.0130 , 0.0986] 0.0546 A (fundamentals) ...	<code>evidence_model.json → boot_KNN_vs_DT.b_minus_a[0]</code>	输出
0.0986	...sion Tree [0.5940, 0.6757] 0.6319 DT — k-NN [0.0130, 0.0986] 0.0546 A (fundamentals) [0.1535,...	<code>evidence_model.json → boot_KNN_vs_DT.b_minus_a[1]</code>	输出
0.0546	... [0.5940, 0.6757] 0.6319 DT — k-NN [0.0130, 0.0986] 0.0546 A (fundamentals) [0.1535, 0.2384] ...	<code>0.6319 - 0.5773</code>	推导
0.1535	...DT — k-NN [0.0130, 0.0986] 0.0546 A (fundamentals) [0.1535 , 0.2384] 0.1955 B (operational) [0...	<code>evidence_model.json → boot_A_vs_B.a[0]</code>	输出
0.2384	...N [0.0130, 0.0986] 0.0546 A (fundamentals) [0.1535, 0.2384] 0.1955 B (operational) [0.5981, 0...	<code>evidence_model.json → boot_A_vs_B.a[1]</code>	输出
0.1955	...0, 0.0986] 0.0546 A (fundamentals) [0.1535, 0.2384] 0.1955 B (operational) [0.5981, 0.6844] 0...	<code>rq_experiments.csv → A_fundamentals / test_f1</code>	输出
0.5981	...ndamentals) [0.1535, 0.2384] 0.1955 B (operational) [0.5981 , 0.6844] 0.6389 B — A [0.3813, 0...	<code>evidence_model.json → boot_A_vs_B.b[0]</code>	输出
0.6844	...ls) [0.1535, 0.2384] 0.1955 B (operational) [0.5981, 0.6844] 0.6389 B — A [0.3813, 0.5046] ...	<code>evidence_model.json → boot_A_vs_B.b[1]</code>	输出
0.6389	...535, 0.2384] 0.1955 B (operational) [0.5981, 0.6844] 0.6389 B — A [0.3813, 0.5046] 0.4434 F ...	<code>rq_experiments.csv → B_operational / test_f1</code>	输出
0.3813	...955 B (operational) [0.5981, 0.6844] 0.6389 B — A [0.3813 , 0.5046] 0.4434 F (operational, no r...	<code>evidence_model.json → boot_A_vs_B.b_minus_a[0]</code>	输出
0.5046	...operational) [0.5981, 0.6844] 0.6389 B — A [0.3813, 0.5046] 0.4434 F (operational, no review pa...	<code>evidence_model.json → boot_A_vs_B.b_minus_a[1]</code>	输出
0.4434	...l) [0.5981, 0.6844] 0.6389 B — A [0.3813, 0.5046] 0.4434 F (operational, no review pair) [0.4...	<code>0.6389 - 0.1955</code>	推导
0.4923	...3813, 0.5046] 0.4434 F (operational, no review pair) [0.4923 , 0.6313] 0.5662 F — A [0.2819, 0...	<code>evidence_model.json → boot_A_vs_F.b[0]</code>	输出
0.6313	...5046] 0.4434 F (operational, no review pair) [0.4923, 0.6313] 0.5662 F — A [0.2819, 0.4536] ...	<code>evidence_model.json → boot_A_vs_F.b[1]</code>	输出
0.5662	...4434 F (operational, no review pair) [0.4923, 0.6313] 0.5662 F — A [0.2819, 0.4536] 0.3707 ...	<code>rq_experiments.csv → F_operational_no_review / test_f1</code>	输出

0.2819	...nal, no review pair) [0.4923, 0.6313] 0.5662 F — A [0.2819, 0.4536] 0.3707 ...	evidence_model.json → boot_A_vs_F.b_minus_a[0]	输出
0.4536	...review pair) [0.4923, 0.6313] 0.5662 F — A [0.2819, 0.4536] 0.3707 ...	evidence_model.json → boot_A_vs_F.b_minus_a[1]	输出
0.3707	...r) [0.4923, 0.6313] 0.5662 F — A [0.2819, 0.4536] 0.3707 ...	0.5662 — 0.1955	推导
3.4 Feature Selection			
35667203	...ost confident miss is listing id 6812677, belonging to host 35667203, who has only this one listing; it is a...	evidence_model.json → hard_case.host_id	输出
0.000	...he tree classified as a non-superhost with a probability of 0.000. It is one of 530 false negatives....	evidence_model.json → hard_case.proba_superhost (数值为 0, 多个字段同值)	输出
530	...s a non-superhost with a probability of 0.000. It is one of 530 false negatives....	evidence_model.json → test_extra.DT.false_negative	输出
3.4 Feature Selection · Table 9			
11	...Full ranking of the 11 features by mutual information and deci...	evidence_model.json → feature_ranking.accommodates.filter_rank	输出
0.1273	... permutation k-NN permutation days_since_last_review 0.1273 1 0.5005 1 0.2269 0.0747 ho...	evidence_model.json → feature_ranking.days_since_last_review.filter_MI	输出
0.5005	... k-NN permutation days_since_last_review 0.1273 1 0.5005 1 0.2269 0.0747 host_listings_c...	evidence_model.json → feature_ranking.days_since_last_review.DT_builtin	输出
0.2269	...ation days_since_last_review 0.1273 1 0.5005 1 0.2269 0.0747 host_listings_count 0.1087...	evidence_model.json → feature_ranking.days_since_last_review.DT_perm	输出
0.0747	...days_since_last_review 0.1273 1 0.5005 1 0.2269 0.0747 host_listings_count 0.1087 2 0....	evidence_model.json → feature_ranking.days_since_last_review.KNN_perm	输出
0.1087	... 1 0.5005 1 0.2269 0.0747 host_listings_count 0.1087 2 0.2698 2 0.0195 0.0037 ava...	evidence_model.json → feature_ranking.host_listings_count.filter_MI	输出
0.2698	... 1 0.2269 0.0747 host_listings_count 0.1087 2 0.2698 2 0.0195 0.0037 availability_365...	evidence_model.json → feature_ranking.host_listings_count.DT_builtin	输出
0.0195	... 0.0747 host_listings_count 0.1087 2 0.2698 2 0.0195 0.0037 availability_365 0.0759 3...	evidence_model.json → feature_ranking.host_listings_count.DT_perm	输出
0.0037	... host_listings_count 0.1087 2 0.2698 2 0.0195 0.0037 availability_365 0.0759 3 0.0356...	evidence_model.json → feature_ranking.host_listings_count.KNN_perm	输出
0.0759	...1087 2 0.2698 2 0.0195 0.0037 availability_365 0.0759 3 0.0356 4 0.0131 0.0169 am...	evidence_model.json → feature_ranking.availability_365.filter_MI	输出
0.0356	...2698 2 0.0195 0.0037 availability_365 0.0759 3 0.0356 4 0.0131 0.0169 amenity_count ...	evidence_model.json → feature_ranking.availability_365.DT_builtin	输出
0.0131	...0195 0.0037 availability_365 0.0759 3 0.0356 4 0.0131 0.0169 amenity_count 0.0749 4 ...	evidence_model.json → feature_ranking.availability_365.DT_perm	输出
0.0169	...0037 availability_365 0.0759 3 0.0356 4 0.0131 0.0169 amenity_count 0.0749 4 0.1023 ...	evidence_model.json → feature_ranking.availability_365.KNN_perm	输出
0.0749	...0.0759 3 0.0356 4 0.0131 0.0169 amenity_count 0.0749 4 0.1023 3 0.0810 0.1433 pri...	evidence_model.json → feature_ranking.amenity_count.filter_MI	输出
0.1023	...0.0356 4 0.0131 0.0169 amenity_count 0.0749 4 0.1023 3 0.0810 0.1433 price_num 0.06...	evidence_model.json → feature_ranking.amenity_count.DT_builtin	输出
0.0810	...0.0131 0.0169 amenity_count 0.0749 4 0.1023 3 0.0810 0.1433 price_num 0.0657 5 0.02...	evidence_model.json → feature_ranking.amenity_count.DT_perm	输出
0.1433	...0.0169 amenity_count 0.0749 4 0.1023 3 0.0810 0.1433 price_num 0.0657 5 0.0277 6 ...	evidence_model.json → feature_ranking.amenity_count.KNN_perm	输出
0.0657	...nt 0.0749 4 0.1023 3 0.0810 0.1433 price_num 0.0657 5 0.0277 6 0.0109 0.0039 pr...	evidence_model.json → feature_ranking.price_num.filter_MI	输出
0.0277	... 4 0.1023 3 0.0810 0.1433 price_num 0.0657 5 0.0277 6 0.0109 0.0039 price_missing ...	evidence_model.json → feature_ranking.price_num.DT_builtin	输出
0.0109	... 3 0.0810 0.1433 price_num 0.0657 5 0.0277 6 0.0109 0.0039 price_missing 0.0451 6 ...	evidence_model.json → feature_ranking.price_num.DT_perm	输出
0.0039	...10 0.1433 price_num 0.0657 5 0.0277 6 0.0109 0.0039 price_missing 0.0451 6 0.0024 ...	evidence_model.json → feature_ranking.price_num.KNN_perm	输出
0.0451	...0.0657 5 0.0277 6 0.0109 0.0039 price_missing 0.0451 6 0.0024 9 0.0219 0.0621 has...	evidence_model.json → feature_ranking.price_missing.filter_MI	输出

0.0024	...0.0277 6 0.0109 0.0039 price_missing 0.0451 6 0.0024 9 0.0219 0.0621 has_review 0.0...	evidence_model.json → feature_ranking.price_missing.DT_builtin	输出
0.0219	...0.0109 0.0039 price_missing 0.0451 6 0.0024 9 0.0219 0.0621 has_review 0.0344 7 0.0...	evidence_model.json → feature_ranking.price_missing.DT_perm	输出
0.0621	...0.0039 price_missing 0.0451 6 0.0024 9 0.0219 0.0621 has_review 0.0344 7 0.0000 11 ...	evidence_model.json → feature_ranking.price_missing.KNN_perm	输出
0.0344	...g 0.0451 6 0.0024 9 0.0219 0.0621 has_review 0.0344 7 0.0000 11 0.0000 0.0073 d...	evidence_model.json → feature_ranking.has_review.filter_MI	输出
0.0000	...6 0.0024 9 0.0219 0.0621 has_review 0.0344 7 0.0000 11 0.0000 0.0073 dist_cbd 0.0...	evidence_model.json → feature_ranking.has_review.DT_perm (数值为 0, 多个字段同值)	输出
11	...4 9 0.0219 0.0621 has_review 0.0344 7 0.0000 11 0.0000 0.0073 dist_cbd 0.0340 ...	evidence_model.json → feature_ranking.has_review.embedded_rank	输出
0.0000	... 0.0219 0.0621 has_review 0.0344 7 0.0000 11 0.0000 0.0073 dist_cbd 0.0340 8 0.02...	evidence_model.json → feature_ranking.has_review.DT_perm (数值为 0, 多个字段同值)	输出
0.0073	... 0.0621 has_review 0.0344 7 0.0000 11 0.0000 0.0073 dist_cbd 0.0340 8 0.0278 5 ...	evidence_model.json → feature_ranking.has_review.KNN_perm	输出
0.0340	...w 0.0344 7 0.0000 11 0.0000 0.0073 dist_cbd 0.0340 8 0.0278 5 0.0054 0.0324 min...	evidence_model.json → feature_ranking.dist_cbd.filter_MI	输出
0.0278	...7 0.0000 11 0.0000 0.0073 dist_cbd 0.0340 8 0.0278 5 0.0054 0.0324 minimum_nights ...	evidence_model.json → feature_ranking.dist_cbd.DT_builtin	输出
0.0054	...11 0.0000 0.0073 dist_cbd 0.0340 8 0.0278 5 0.0054 0.0324 minimum_nights 0.0097 9 ...	evidence_model.json → feature_ranking.dist_cbd.DT_perm	输出
0.0324	...00 0.0073 dist_cbd 0.0340 8 0.0278 5 0.0054 0.0324 minimum_nights 0.0097 9 0.0157 ...	evidence_model.json → feature_ranking.dist_cbd.KNN_perm	输出
0.0097	...0.0340 8 0.0278 5 0.0054 0.0324 minimum_nights 0.0097 9 0.0157 8 0.0014 0.0006 ro...	evidence_model.json → feature_ranking.minimum_nights.filter_MI	输出
0.0157	...0.0278 5 0.0054 0.0324 minimum_nights 0.0097 9 0.0157 8 0.0014 0.0006 room_type_3 0...	evidence_model.json → feature_ranking.minimum_nights.DT_builtin	输出
0.0014	...0.0054 0.0324 minimum_nights 0.0097 9 0.0157 8 0.0014 0.0006 room_type_3 0.0089 10 ...	evidence_model.json → feature_ranking.minimum_nights.DT_perm	输出
0.0006	...0.0324 minimum_nights 0.0097 9 0.0157 8 0.0014 0.0006 room_type_3 0.0089 10 0.0001 ...	evidence_model.json → feature_ranking.minimum_nights.KNN_perm	输出
0.0089	... 0.0097 9 0.0157 8 0.0014 0.0006 room_type_3 0.0089 10 0.0001 10 0.0000 0.0168 a...	evidence_model.json → feature_ranking.room_type_3.filter_MI	输出
10	... 9 0.0157 8 0.0014 0.0006 room_type_3 0.0089 10 0.0001 10 0.0000 0.0168 accomm...	evidence_model.json → feature_ranking.room_type_3.filter_rank	输出
0.0001	... 0.0157 8 0.0014 0.0006 room_type_3 0.0089 10 0.0001 10 0.0000 0.0168 accommodates ...	evidence_model.json → feature_ranking.room_type_3.DT_builtin	输出
10	... 8 0.0014 0.0006 room_type_3 0.0089 10 0.0001 10 0.0000 0.0168 accommodates 0.006...	evidence_model.json → feature_ranking.room_type_3.filter_rank	输出
0.0000	...0.0014 0.0006 room_type_3 0.0089 10 0.0001 10 0.0000 0.0168 accommodates 0.0065 11 ...	evidence_model.json → hard_case.tree_path[10].superhost_share (数值为 0, 多个字段同值)	输出
0.0168	...0.0006 room_type_3 0.0089 10 0.0001 10 0.0000 0.0168 accommodates 0.0065 11 0.0181 ...	evidence_model.json → feature_ranking.room_type_3.KNN_perm	输出
0.0065	...0.0089 10 0.0001 10 0.0000 0.0168 accommodates 0.0065 11 0.0181 7 0.0062 -0.0064 ...	evidence_model.json → feature_ranking.accommodates.filter_MI	输出
11	...10 0.0001 10 0.0000 0.0168 accommodates 0.0065 11 0.0181 7 0.0062 -0.0064 ...	evidence_model.json → feature_ranking.accommodates.filter_rank	输出
0.0181	...0.0001 10 0.0000 0.0168 accommodates 0.0065 11 0.0181 7 0.0062 -0.0064 ...	evidence_model.json → feature_ranking.accommodates.DT_builtin	输出
0.0062	... 0.0000 0.0168 accommodates 0.0065 11 0.0181 7 0.0062 -0.0064 ...	evidence_model.json → feature_ranking.accommodates.DT_perm	输出
-0.0064	... 0.0168 accommodates 0.0065 11 0.0181 7 0.0062 -0.0064 ...	evidence_model.json → feature_ranking.accommodates.KNN_perm	输出
3.5 Dimensionality Reduction & Clustering			
69.4%	...PCA. The first three components carry 69.4% of the variance between them, and Table...	pca_table.csv → PC3 / cumulative_variance (×100)	输出
3.5 Dimensionality Reduction & Clustering · Table 10			
0.3211	...nent EVR Cumulative Rank 1 Rank 2 Rank 3 PC1 0.3211 0.3211 days_since_last_review +0.56...	pca_table.csv → PC1 / explained_variance	输出

0.3211	...R Cumulative Rank 1 Rank 2 Rank 3 PC1 0.3211 0.3211 days_since_last_review +0.563 ameni...	pca_table.csv → PC1 / explained_variance	输出
+0.563	... Rank 3 PC1 0.3211 0.3211 days_since_last_review + 0.563 amenity_count −0.504 availability_3...	evidence_cluster.json → pca_top_loadings.PC1[0][1]	输出
−0.504	...11 0.3211 days_since_last_review +0.563 amenity_count − 0.504 availability_365 −0.447 PC2 0.204...	evidence_cluster.json → pca_top_loadings.PC1[1][1]	输出
−0.447	...ast_review +0.563 amenity_count −0.504 availability_365 − 0.447 PC2 0.2045 0.5255 dist_cbd +0.7...	evidence_cluster.json → pca_top_loadings.PC1[2][1]	输出
0.2045	...63 amenity_count −0.504 availability_365 −0.447 PC2 0.2045 0.5255 dist_cbd +0.786 host_listi...	pca_table.csv → PC2 / explained_variance	输出
0.5255	...ity_count −0.504 availability_365 −0.447 PC2 0.2045 0.5255 dist_cbd +0.786 host_listings_count...	pca_table.csv → PC2 / cumulative_variance	输出
+0.786	... availability_365 −0.447 PC2 0.2045 0.5255 dist_cbd + 0.786 host_listings_count −0.556 accommod...	evidence_cluster.json → pca_top_loadings.PC2[0][1]	输出
−0.556	...2 0.2045 0.5255 dist_cbd +0.786 host_listings_count − 0.556 accommodates +0.208 PC3 0.1681 0...	evidence_cluster.json → pca_top_loadings.PC2[1][1]	输出
+0.208	...dist_cbd +0.786 host_listings_count −0.556 accommodates + 0.208 PC3 0.1681 0.6936 accommodates +...	evidence_cluster.json → pca_top_loadings.PC2[2][1]	输出
0.1681	...6 host_listings_count −0.556 accommodates +0.208 PC3 0.1681 0.6936 accommodates +0.671 availa...	pca_table.csv → PC3 / explained_variance	输出
0.6936	...listings_count −0.556 accommodates +0.208 PC3 0.1681 0.6936 accommodates +0.671 availability_36...	pca_table.csv → PC3 / cumulative_variance	输出
+0.671	... accommodates +0.208 PC3 0.1681 0.6936 accommodates + 0.671 availability_365 −0.591 amenity_cou...	evidence_cluster.json → pca_top_loadings.PC3[0][1]	输出
−0.591	... 0.1681 0.6936 accommodates +0.671 availability_365 − 0.591 amenity_count +0.355 ...	evidence_cluster.json → pca_top_loadings.PC3[1][1]	输出
+0.355	...commodates +0.671 availability_365 −0.591 amenity_count + 0.355 ...	evidence_cluster.json → pca_top_loadings.PC3[2][1]	输出

3.5 Dimensionality Reduction & Clustering · Figure 2

21,251	...PCA scatter of the 21,251 reviewed listings on the first two comp...	evidence_cluster.json → n_rows_reviewed	输出
5,000	...us, with K-Means centroids (k=5) marked. A random sample of 5,000 listings is shown. Centroids numbered 1...	notebook 4.5 打印输出	输出

3.5 Dimensionality Reduction & Clustering · Table 11

4,541	... days_since_last_review host_listings_count K1 4,541 21.4% 35.3% 1.42 4 188 41 ...	evidence_cluster.json → cluster_profiles[0].listings	输出
21.4%	...since_last_review host_listings_count K1 4,541 21.4% 35.3% 1.42 4 188 41 50 50...	evidence_cluster.json → cluster_profiles[0].share (×100)	输出
35.3%	...ast_review host_listings_count K1 4,541 21.4% 35.3% 1.42 4 188 41 50 50 K2 ...	evidence_cluster.json → cluster_profiles[0].superhost_rate (×100)	输出
1.42	...iew host_listings_count K1 4,541 21.4% 35.3% 1.42 4 188 41 50 50 K2 5,340 ...	evidence_cluster.json → cluster_profiles[0].dist_cbd	输出
188	...listings_count K1 4,541 21.4% 35.3% 1.42 4 188 41 50 50 K2 5,340 25.1% 5...	evidence_cluster.json → cluster_profiles[0].availability_365	输出
41	...ngs_count K1 4,541 21.4% 35.3% 1.42 4 188 41 50 50 K2 5,340 25.1% 53.9% ...	evidence_preprocess.json → property_type_under_10_count	输出
50	...count K1 4,541 21.4% 35.3% 1.42 4 188 41 50 50 K2 5,340 25.1% 53.9% 2.2...	evidence_cluster.json → cluster_profiles[0].host_listings_count	输出
50	... K1 4,541 21.4% 35.3% 1.42 4 188 41 50 50 K2 5,340 25.1% 53.9% 2.25 4...	evidence_cluster.json → cluster_profiles[0].host_listings_count	输出
5,340	...,541 21.4% 35.3% 1.42 4 188 41 50 50 K2 5,340 25.1% 53.9% 2.25 4 157 47 ...	evidence_cluster.json → cluster_profiles[1].listings	输出
25.1%	...1.4% 35.3% 1.42 4 188 41 50 50 K2 5,340 25.1% 53.9% 2.25 4 157 47 40 2 ...	evidence_cluster.json → cluster_profiles[1].share (×100)	输出
53.9%	...5.3% 1.42 4 188 41 50 50 K2 5,340 25.1% 53.9% 2.25 4 157 47 40 2 K3 4,...	evidence_cluster.json → cluster_profiles[1].superhost_rate (×100)	输出
2.25	...42 4 188 41 50 50 K2 5,340 25.1% 53.9% 2.25 4 157 47 40 2 K3 4,689 2...	evidence_cluster.json → cluster_profiles[1].dist_cbd	输出
157	...88 41 50 50 K2 5,340 25.1% 53.9% 2.25 4 157 47 40 2 K3 4,689 22.1% 6.8...	evidence_cluster.json → cluster_profiles[1].availability_365	输出
47	...1 50 50 K2 5,340 25.1% 53.9% 2.25 4 157 47 40 2 K3 4,689 22.1% 6.8% 4...	evidence_cluster.json → cluster_profiles[1].amenity_count	输出

40	...0 50 K2 5,340 25.1% 53.9% 2.25 4 157 47 40 2 K3 4,689 22.1% 6.8% 4.54 ...	代码中的设定值 (常量 / 参数)	设定
4,689	... 5,340 25.1% 53.9% 2.25 4 157 47 40 2 K3 4,689 22.1% 6.8% 4.54 2 0 23 1,...	evidence_cluster.json → kmeans_sizes.2	输出
22.1%	... 25.1% 53.9% 2.25 4 157 47 40 2 K3 4,689 22.1% 6.8% 4.54 2 0 23 1,883 1 ...	evidence_cluster.json → cluster_profiles[2].share (×100)	输出
6.8%	... 53.9% 2.25 4 157 47 40 2 K3 4,689 22.1% 6.8% 4.54 2 0 23 1,883 1 K4 ...	evidence_cluster.json → kmeans_sweep[4].sse_drop_pct	输出
4.54	... 2.25 4 157 47 40 2 K3 4,689 22.1% 6.8% 4.54 2 0 23 1,883 1 K4 2,365 ...	evidence_cluster.json → cluster_profiles[2].dist_cbd	输出
23	...7 47 40 2 K3 4,689 22.1% 6.8% 4.54 2 0 23 1,883 1 K4 2,365 11.1% 43.7...	evidence_cluster.json → cluster_profiles[2].amenity_count	输出
1,883	...7 40 2 K3 4,689 22.1% 6.8% 4.54 2 0 23 1,883 1 K4 2,365 11.1% 43.7% 20.5...	evidence_cluster.json → cluster_profiles[2].days_since_last_review	输出
2,365	...4,689 22.1% 6.8% 4.54 2 0 23 1,883 1 K4 2,365 11.1% 43.7% 20.56 8 218 51 ...	evidence_cluster.json → kmeans_sizes.3	输出
11.1%	...22.1% 6.8% 4.54 2 0 23 1,883 1 K4 2,365 11.1% 43.7% 20.56 8 218 51 58 3...	evidence_cluster.json → cluster_profiles[3].share (×100)	输出
43.7%	...6.8% 4.54 2 0 23 1,883 1 K4 2,365 11.1% 43.7% 20.56 8 218 51 58 3 K5 4...	evidence_cluster.json → cluster_profiles[3].superhost_rate (×100)	输出
20.56	...54 2 0 23 1,883 1 K4 2,365 11.1% 43.7% 20.56 8 218 51 58 3 K5 4,316 2...	evidence_cluster.json → cluster_profiles[3].dist_cbd	输出
218	... 23 1,883 1 K4 2,365 11.1% 43.7% 20.56 8 218 51 58 3 K5 4,316 20.3% 40....	evidence_cluster.json → cluster_profiles[3].availability_365	输出
51	... 1,883 1 K4 2,365 11.1% 43.7% 20.56 8 218 51 58 3 K5 4,316 20.3% 40.7% ...	代码中的设定值 (常量 / 参数)	设定
58	...3 1 K4 2,365 11.1% 43.7% 20.56 8 218 51 58 3 K5 4,316 20.3% 40.7% 17.82...	evidence_cluster.json → cluster_profiles[3].days_since_last_review	输出
4,316	...2,365 11.1% 43.7% 20.56 8 218 51 58 3 K5 4,316 20.3% 40.7% 17.82 2 310 33 ...	evidence_cluster.json → kmeans_sizes.4	输出
20.3%	...11.1% 43.7% 20.56 8 218 51 58 3 K5 4,316 20.3% 40.7% 17.82 2 310 33 76 2...	evidence_model.json → rq_extra[5].test_f1 (×100)	输出
40.7%	...43.7% 20.56 8 218 51 58 3 K5 4,316 20.3% 40.7% 17.82 2 310 33 76 2 W1 ...	evidence_cluster.json → cluster_profiles[4].superhost_rate (×100)	输出
17.82	...20.56 8 218 51 58 3 K5 4,316 20.3% 40.7% 17.82 2 310 33 76 2 W1 8,256 ...	evidence_cluster.json → cluster_profiles[4].dist_cbd	输出
310	...218 51 58 3 K5 4,316 20.3% 40.7% 17.82 2 310 33 76 2 W1 8,256 38.9% 49...	evidence_cluster.json → cluster_profiles[4].availability_365	输出
33	...51 58 3 K5 4,316 20.3% 40.7% 17.82 2 310 33 76 2 W1 8,256 38.9% 49.9% ...	evidence_cluster.json → cluster_profiles[4].amenity_count	输出
76	...58 3 K5 4,316 20.3% 40.7% 17.82 2 310 33 76 2 W1 8,256 38.9% 49.9% 5.36...	evidence_cluster.json → cluster_profiles[4].days_since_last_review	输出
8,256	...,316 20.3% 40.7% 17.82 2 310 33 76 2 W1 8,256 38.9% 49.9% 5.36 4 174 47 ...	evidence_cluster.json → hierarchical_sizes.0	输出
38.9%	...0.3% 40.7% 17.82 2 310 33 76 2 W1 8,256 38.9% 49.9% 5.36 4 174 47 56 1 ...	evidence_cluster.json → cluster_profiles[5].share (×100)	输出
49.9%	...0.7% 17.82 2 310 33 76 2 W1 8,256 38.9% 49.9% 5.36 4 174 47 56 1 W2 3,...	evidence_cluster.json → cluster_profiles[5].superhost_rate (×100)	输出
5.36	...7.82 2 310 33 76 2 W1 8,256 38.9% 49.9% 5.36 4 174 47 56 1 W2 3,829 1...	evidence_cluster.json → cluster_profiles[5].dist_cbd	输出
174	...310 33 76 2 W1 8,256 38.9% 49.9% 5.36 4 174 47 56 1 W2 3,829 18.0% 30...	evidence_cluster.json → cluster_profiles[5].availability_365	输出
47	...33 76 2 W1 8,256 38.9% 49.9% 5.36 4 174 47 56 1 W2 3,829 18.0% 30.1% ...	evidence_cluster.json → cluster_profiles[1].amenity_count	输出
56	...76 2 W1 8,256 38.9% 49.9% 5.36 4 174 47 56 1 W2 3,829 18.0% 30.1% 7.32 ...	evidence_cluster.json → cluster_profiles[5].days_since_last_review	输出
3,829	... 8,256 38.9% 49.9% 5.36 4 174 47 56 1 W2 3,829 18.0% 30.1% 7.32 2 267 22 ...	evidence_cluster.json → hierarchical_sizes.1	输出
18.0%	... 38.9% 49.9% 5.36 4 174 47 56 1 W2 3,829 18.0% 30.1% 7.32 2 267 22 100 3...	evidence_model.json → hard_case.tree_path[3].value	输出
30.1%	... 49.9% 5.36 4 174 47 56 1 W2 3,829 18.0% 30.1% 7.32 2 267 22 100 3 W3 ...	evidence_cluster.json → cluster_profiles[6].superhost_rate (×100)	输出
7.32	... 5.36 4 174 47 56 1 W2 3,829 18.0% 30.1% 7.32 2 267 22 100 3 W3 4,716 ...	evidence_cluster.json → cluster_profiles[6].dist_cbd	输出

267	... 174 47 56 1 W2 3,829 18.0% 30.1% 7.32 2 267 22 100 3 W3 4,716 22.2% 3...	evidence_cluster.json → cluster_profiles[6].availability_365	输出
22	... 47 56 1 W2 3,829 18.0% 30.1% 7.32 2 267 22 100 3 W3 4,716 22.2% 37.5% ...	evidence_cluster.json → cluster_profiles[6].amenity_count	输出
100	... 56 1 W2 3,829 18.0% 30.1% 7.32 2 267 22 100 3 W3 4,716 22.2% 37.5% 1.79...	代码中的设定值 (常量 / 参数)	设定
4,716	...,829 18.0% 30.1% 7.32 2 267 22 100 3 W3 4,716 22.2% 37.5% 1.79 4 175 42 ...	evidence_cluster.json → hierarchical_sizes.2	输出
22.2%	...8.0% 30.1% 7.32 2 267 22 100 3 W3 4,716 22.2% 37.5% 1.79 4 175 42 55 45...	evidence_cluster.json → minimum_nights_skew	输出
37.5%	...0.1% 7.32 2 267 22 100 3 W3 4,716 22.2% 37.5% 1.79 4 175 42 55 45 W4 3...	evidence_cluster.json → cluster_profiles[7].superhost_rate (×100)	输出
1.79	...32 2 267 22 100 3 W3 4,716 22.2% 37.5% 1.79 4 175 42 55 45 W4 3,249 ...	evidence_cluster.json → cluster_profiles[7].dist_cbd	输出
175	...67 22 100 3 W3 4,716 22.2% 37.5% 1.79 4 175 42 55 45 W4 3,249 15.3% 2....	evidence_cluster.json → cluster_profiles[7].availability_365	输出
42	...2 100 3 W3 4,716 22.2% 37.5% 1.79 4 175 42 55 45 W4 3,249 15.3% 2.9% ...	evidence_cluster.json → cluster_profiles[7].amenity_count	输出
55	...00 3 W3 4,716 22.2% 37.5% 1.79 4 175 42 55 45 W4 3,249 15.3% 2.9% 4.34 ...	evidence_cluster.json → cluster_profiles[7].days_since_last_review	输出
45	...3 W3 4,716 22.2% 37.5% 1.79 4 175 42 55 45 W4 3,249 15.3% 2.9% 4.34 2 ...	evidence_cluster.json → cluster_profiles[7].host_listings_count	输出
3,249	...4,716 22.2% 37.5% 1.79 4 175 42 55 45 W4 3,249 15.3% 2.9% 4.34 2 0 20 2,...	evidence_cluster.json → hierarchical_sizes.3	输出
15.3%	...22.2% 37.5% 1.79 4 175 42 55 45 W4 3,249 15.3% 2.9% 4.34 2 0 20 2,364 1 ...	evidence_model.json → boot_A_vs_B.a[0] (×100)	输出
2.9%	...37.5% 1.79 4 175 42 55 45 W4 3,249 15.3% 2.9% 4.34 2 0 20 2,364 1 W5 ...	evidence_cluster.json → cluster_profiles[8].superhost_rate (×100)	输出
4.34	... 1.79 4 175 42 55 45 W4 3,249 15.3% 2.9% 4.34 2 0 20 2,364 1 W5 1,201 ...	evidence_cluster.json → cluster_profiles[8].dist_cbd	输出
20	... 42 55 45 W4 3,249 15.3% 2.9% 4.34 2 0 20 2,364 1 W5 1,201 5.7% 37.8%...	代码中的设定值 (常量 / 参数)	设定
2,364	... 55 45 W4 3,249 15.3% 2.9% 4.34 2 0 20 2,364 1 W5 1,201 5.7% 37.8% 17.55...	evidence_cluster.json → cluster_profiles[8].days_since_last_review	输出
1,201	...3,249 15.3% 2.9% 4.34 2 0 20 2,364 1 W5 1,201 5.7% 37.8% 17.55 10 221 49 ...	evidence_cluster.json → hierarchical_sizes.4	输出
5.7%	...15.3% 2.9% 4.34 2 0 20 2,364 1 W5 1,201 5.7% 37.8% 17.55 10 221 49 49 ...	evidence_cluster.json → pc_vs_superhost.PC1.nmi_10_bins (×100)	输出
37.8%	... 2.9% 4.34 2 0 20 2,364 1 W5 1,201 5.7% 37.8% 17.55 10 221 49 49 12 ...	evidence_cluster.json → cluster_profiles[9].superhost_rate (×100)	输出
17.55	...4.34 2 0 20 2,364 1 W5 1,201 5.7% 37.8% 17.55 10 221 49 49 12 ...	evidence_cluster.json → cluster_profiles[9].dist_cbd	输出
10	... 0 20 2,364 1 W5 1,201 5.7% 37.8% 17.55 10 221 49 49 12 ...	代码中的设定值 (常量 / 参数)	设定
221	... 20 2,364 1 W5 1,201 5.7% 37.8% 17.55 10 221 49 49 12 ...	evidence_cluster.json → cluster_profiles[9].availability_365	输出
49	... 2,364 1 W5 1,201 5.7% 37.8% 17.55 10 221 49 49 12 ...	evidence_cluster.json → cluster_profiles[9].amenity_count	输出
49	...4 1 W5 1,201 5.7% 37.8% 17.55 10 221 49 49 12 ...	evidence_cluster.json → cluster_profiles[9].amenity_count	输出
12	... W5 1,201 5.7% 37.8% 17.55 10 221 49 49 12 ...	$0.0896 \div 0.0799 - 1$ (NMI 增幅)	推导

4.1 Data Preprocessing			
0.5745	...eatures (experiment D) moves the decision tree's CV F1 from 0.5745 to 0.5262 (test F1 0.6319 to 0.5667) an...	cv_results.csv → DT / cv_mean	输出
0.5262	...xperiment D) moves the decision tree's CV F1 from 0.5745 to 0.5262 (test F1 0.6319 to 0.5667) and k-NN fro...	rq_experiments.csv → D_all_no_review / cv_mean	输出
0.6319	...es the decision tree's CV F1 from 0.5745 to 0.5262 (test F1 0.6319 to 0.5667) and k-NN from 0.5370 to 0.50...	evidence_model.json → test.DT.f1	输出
0.5667	...ision tree's CV F1 from 0.5745 to 0.5262 (test F1 0.6319 to 0.5667) and k-NN from 0.5370 to 0.5032. Removi...	evidence_model.json → rq[7].test_f1	输出
0.5370	...m 0.5745 to 0.5262 (test F1 0.6319 to 0.5667) and k-NN from 0.5370 to 0.5032. Removing amenity_count (expe...	evidence_model.json → rq[4].cv_mean	输出

0.5032	...o 0.5262 (test F1 0.6319 to 0.5667) and k–NN from 0.5370 to 0.5032. Removing amenity_count (experiment G) ...	evidence_model.json → rq[6].cv_mean	输出
0.5370	...5032. Removing amenity_count (experiment G) moves k–NN from 0.5370 to 0.4747 (test 0.5773 to 0.5196), whil...	evidence_model.json → rq[4].cv_mean	输出
0.4747	...ving amenity_count (experiment G) moves k–NN from 0.5370 to 0.4747 (test 0.5773 to 0.5196), while the deci...	evidence_model.json → rq_extra[0].cv_mean	输出
0.5773	...count (experiment G) moves k–NN from 0.5370 to 0.4747 (test 0.5773 to 0.5196), while the decision tree is ...	evidence_model.json → test.KNN.f1	输出
0.5196	...eriment G) moves k–NN from 0.5370 to 0.4747 (test 0.5773 to 0.5196), while the decision tree is almost unc...	evidence_model.json → rq_extra[0].test_f1	输出
0.5747	... to 0.5196), while the decision tree is almost unchanged at 0.5747. Removing the price pair (experiment H)...	evidence_model.json → rq_extra[1].cv_mean	输出
0.5220	...5747. Removing the price pair (experiment H) moves k–NN to 0.5220 and the decision tree slightly up to 0....	rq_experiments_extra.csv → H_all_no_price / cv_mean	输出
0.5791	...) moves k–NN to 0.5220 and the decision tree slightly up to 0.5791. The decision tree therefore relies alm...	rq_experiments_extra.csv → H_all_no_price / cv_mean	输出
4.2 Correlation Analysis			
−0.438	...r longest pair, days_since_last_review with availability_365 (−0.438 vs −0.395), but they point in opposite ...	correlation_matrix.csv → days_since_last_review ~ availability_365 / pearson	输出
−0.395	...ir, days_since_last_review with availability_365 (−0.438 vs −0.395), but they point in opposite directions...	correlation_matrix.csv → days_since_last_review ~ availability_365 / spearman	输出
8,103	...h ranks: most listings fall in the 0–5 host–listings range (8,103, 2,560 and 2,832 listings across its th...	evidence_correlation.json → hlc_band.n.(−0.001, 1.0]	输出
2,560	...: most listings fall in the 0–5 host–listings range (8,103, 2,560 and 2,832 listings across its three ban...	evidence_correlation.json → hlc_band.n.(1.0, 2.0]	输出
2,832	...tings fall in the 0–5 host–listings range (8,103, 2,560 and 2,832 listings across its three bands), where...	evidence_correlation.json → hlc_band.n.(2.0, 5.0]	输出
28.5%	...cross its three bands), where the superhost rate rises from 28.5% to 46.1%, so the rank correlation is po...	evidence_correlation.json → hlc_band.rate.(−0.001, 1.0] (×100)	输出
46.1%	... three bands), where the superhost rate rises from 28.5% to 46.1%, so the rank correlation is positive at...	evidence_correlation.json → hlc_band.rate.(2.0, 5.0] (×100)	输出
+0.070	...from 28.5% to 46.1%, so the rank correlation is positive at +0.070. Pearson uses the raw values: 1,259 lis...	correlation_matrix.csv → host_listings_count ~ y / spearman	输出
1,259	...elation is positive at +0.070. Pearson uses the raw values: 1,259 listings belong to hosts with more than...	evidence_correlation.json → hlc_band.n.(100.0, 700.0]	输出
100	...e raw values: 1,259 listings belong to hosts with more than 100 listings (up to the platform–wide maxim...	代码中的设定值 (常量 / 参数)	设定
667	... more than 100 listings (up to the platform–wide maximum of 667), and their superhost rate is only 16.9...	evidence_preprocess.json → host_listings_count_max	输出
16.9%	...form–wide maximum of 667), and their superhost rate is only 16.9%, which pulls the coefficient to −0.069....	evidence_correlation.json → hlc_band.rate.(100.0, 700.0] (×100)	输出
−0.069	...uperhost rate is only 16.9%, which pulls the coefficient to −0.069. The two coefficients therefore describ...	evidence_model.json → test_extra.DT.gain_over_baseline.accuracy	输出
0.0097	...I do not assume monotonicity, yet NMI for this pair is only 0.0097....	correlation_matrix.csv → host_listings_count ~ y / nmi	输出
−0.081	..._last_review and dist_cbd the same pattern appears (Pearson −0.081, Spearman +0.061). When distance is spl...	correlation_matrix.csv → days_since_last_review ~ dist_cbd / pearson	输出
+0.061	...dist_cbd the same pattern appears (Pearson −0.081, Spearman +0.061). When distance is split into twenty eq...	correlation_matrix.csv → days_since_last_review ~ dist_cbd / spearman	输出
41	...equal–frequency bands, the median days_since_last_review is 41–60 days at 0.6–1.4 km, rises to 111 day...	evidence_correlation.json → dist_band_trend.3.dsl_median	输出
60	...al–frequency bands, the median days_since_last_review is 41–60 days at 0.6–1.4 km, rises to 111 days a...	evidence_correlation.json → dist_band_trend.0.dsl_median	输出
0.6	...y bands, the median days_since_last_review is 41–60 days at 0.6–1.4 km, rises to 111 days at 1.7 km, pe...	evidence_correlation.json → dist_band_trend.0.dist_km	输出
1.4	...nds, the median days_since_last_review is 41–60 days at 0.6–1.4 km, rises to 111 days at 1.7 km, peaks ...	correlation_matrix.csv → days_since_last_review ~ dist_cbd / nmi (×100)	输出
111	...ays_since_last_review is 41–60 days at 0.6–1.4 km, rises to 111 days at 1.7 km, peaks at 122 days near ...	evidence_correlation.json → dist_band_trend.5.dsl_median	输出
1.7	...st_review is 41–60 days at 0.6–1.4 km, rises to 111 days at 1.7 km, peaks at 122 days near 5.0 km, and ...	evidence_correlation.json → dist_band_trend.5.dist_km	输出

122	...0 days at 0.6–1.4 km, rises to 111 days at 1.7 km, peaks at 122 days near 5.0 km, and then falls to 69 ...	evidence_correlation.json → dist_band_trend.10.dsl_median	输出
5.0	...1.4 km, rises to 111 days at 1.7 km, peaks at 122 days near 5.0 km, and then falls to 69 days at 34.6 k...	evidence_correlation.json → dist_band_trend.10.dist_km	输出
69	...at 1.7 km, peaks at 122 days near 5.0 km, and then falls to 69 days at 34.6 km and 47 days at 50.6 km....	1 - 0.3065 ≈ 0.694 (基线准确率)	推导
34.6	...peaks at 122 days near 5.0 km, and then falls to 69 days at 34.6 km and 47 days at 50.6 km. Most listing...	evidence_correlation.json → dist_band_trend.18.dist_km	输出
47	... days near 5.0 km, and then falls to 69 days at 34.6 km and 47 days at 50.6 km. Most listings lie betw...	evidence_cluster.json → cluster_profiles[1].amenity_count	输出
50.6	...5.0 km, and then falls to 69 days at 34.6 km and 47 days at 50.6 km. Most listings lie between the centr...	evidence_correlation.json → dist_band_trend.19.dist_km	输出
0.0785	..., days_since_last_review has the strongest association (NMI 0.0785), followed by availability_365 (0.0324)...	correlation_matrix.csv → days_since_last_review ~ y / nmi	输出
0.0324	...est association (NMI 0.0785), followed by availability_365 (0.0324) and host_listings_count (0.0097). dist...	correlation_matrix.csv → availability_365 ~ y / nmi	输出
0.0097	...lowed by availability_365 (0.0324) and host_listings_count (0.0097). dist_cbd comes last (0.0047), so the ...	correlation_matrix.csv → host_listings_count ~ y / nmi	输出
0.0047	...324) and host_listings_count (0.0097). dist_cbd comes last (0.0047), so the only listing fundamentals feat...	correlation_matrix.csv → dist_cbd ~ y / nmi	输出
100	...note that these are associations only. Hosts with more than 100 listings may be running business or apa...	代码中的设定值 (常量 / 参数)	设定
-0.8155	...ince its Spearman correlation with number_of_reviews_ltm is -0.8155. This motivates experiment F, which rem...	evidence_preprocess.json → days_since_last_review_vs_reviews_ltm_spearman	输出
0.2819	...ted operational features; the test-set interval for F - A, [0.2819, 0.4536], remains clearly positive....	evidence_model.json → boot_A_vs_F.b_minus_a[0]	输出
0.4536	...ational features; the test-set interval for F - A, [0.2819, 0.4536], remains clearly positive....	evidence_model.json → boot_A_vs_F.b_minus_a[1]	输出
4.3 Supervised Learning Models and Evaluation			
0.632	...e official criteria excluded, the tree reaches a test F1 of 0.632 on the superhost class and k-NN 0.577, ...	evidence_model.json → test.DT.class_1.f1-score	输出
0.577	... reaches a test F1 of 0.632 on the superhost class and k-NN 0.577, against a majority-class baseline of e...	evidence_model.json → test.KNN.class_1.f1-score	输出
0.694	...e baseline model that ignores every feature already reaches 0.694 accuracy, which is just the non-superho...	evidence_model.json → test.Baseline.accuracy	输出
0.3065	...curacy, which is just the non-superhost share restated (1 - 0.3065 = 0.6935), so accuracy cannot separate ...	evidence_preprocess.json → superhost_share_overall	输出
0.6935	...hich is just the non-superhost share restated (1 - 0.3065 = 0.6935), so accuracy cannot separate a useful ...	evidence_model.json → test.Baseline.accuracy	输出
0.6319	...el from the baseline. Relative to the baseline, F1 rises by 0.6319 for the tree and 0.5773 for k-NN, accur...	evidence_model.json → test_extra.DT.gain_over_baseline.f1	输出
0.5773	...lative to the baseline, F1 rises by 0.6319 for the tree and 0.5773 for k-NN, accuracy by 0.0694 (0.7629 ve...	evidence_model.json → test_extra.KNN.gain_over_baseline.f1	输出
0.0694	...ses by 0.6319 for the tree and 0.5773 for k-NN, accuracy by 0.0694 (0.7629 versus 0.6935) and 0.0552, and ...	evidence_model.json → test_extra.DT.gain_over_baseline.accuracy	输出
0.7629	...6319 for the tree and 0.5773 for k-NN, accuracy by 0.0694 (0.7629 versus 0.6935) and 0.0552, and macro-F1...	evidence_model.json → test.DT.accuracy	输出
0.6935	...tree and 0.5773 for k-NN, accuracy by 0.0694 (0.7629 versus 0.6935) and 0.0552, and macro-F1 by 0.3190 and...	evidence_model.json → test.Baseline.accuracy	输出
0.0552	...773 for k-NN, accuracy by 0.0694 (0.7629 versus 0.6935) and 0.0552, and macro-F1 by 0.3190 and 0.2898 (0.7...	evidence_model.json → test_extra.KNN.gain_over_baseline.accuracy	输出
0.3190	...y 0.0694 (0.7629 versus 0.6935) and 0.0552, and macro-F1 by 0.3190 and 0.2898 (0.7285 and 0.6993 against 0...	evidence_model.json → test_extra.DT.gain_over_baseline.macro_f1	输出
0.2898	...7629 versus 0.6935) and 0.0552, and macro-F1 by 0.3190 and 0.2898 (0.7285 and 0.6993 against 0.4095)....	evidence_model.json → test_extra.KNN.gain_over_baseline.macro_f1	输出
0.7285	...rsus 0.6935) and 0.0552, and macro-F1 by 0.3190 and 0.2898 (0.7285 and 0.6993 against 0.4095)....	evidence_model.json → test_extra.DT.macro_f1	输出
0.6993	...) and 0.0552, and macro-F1 by 0.3190 and 0.2898 (0.7285 and 0.6993 against 0.4095)....	evidence_model.json → test_extra.KNN.macro_f1	输出
0.4095	...nd macro-F1 by 0.3190 and 0.2898 (0.7285 and 0.6993 against 0.4095)....	evidence_model.json → test_extra.Baseline.macro_f1	输出
0.0130	... of the two. Its paired bootstrap interval against k-NN is [0.0130, 0.0986], which does not include zero, ...	evidence_model.json → boot_KNN_vs_DT.b_minus_a[0]	输出

0.0986	...two. Its paired bootstrap interval against k-NN is [0.0130, 0.0986], which does not include zero, so the d...	evidence_model.json → boot_KNN_vs_DT.b_minus_a[1]	输出
1,737	...amplifying noise. The tree labels more listings as superhosts, 1,737 in total against 1,482 for k-NN, which ...	evidence_model.json → test_extra.DT.predicted_superhost	输出
1,482	... labels more listings as superhosts, 1,737 in total against 1,482 for k-NN, which is a trade-off: of these...	evidence_model.json → test_extra.KNN.predicted_superhost	输出
1,047	...tal against 1,482 for k-NN, which is a trade-off: of these, 1,047 and 883 are true positives, 690 and 599...	evidence_model.json → test_extra.DT.true_positive	输出
883	...t 1,482 for k-NN, which is a trade-off: of these, 1,047 and 883 are true positives, 690 and 599 are fal...	evidence_model.json → test_extra.KNN.true_positive	输出
690	...is a trade-off: of these, 1,047 and 883 are true positives, 690 and 599 are false positives...	evidence_model.json → test_extra.DT.false_positive	输出
599	...de-off: of these, 1,047 and 883 are true positives, 690 and 599 are false positives...	evidence_model.json → test_extra.KNN.false_positive	输出
0.2269	...mutation importance is highest for days_since_last_review (0.2269), followed by amenity_count (0.0810); f...	evidence_model.json → permutation_importance.DT.days_since_last_re view	输出
0.0810	...days_since_last_review (0.2269), followed by amenity_count (0.0810); for k-NN it is highest for amenity_co...	evidence_model.json → permutation_importance.DT.amenity_count	输出
0.1433	...y_count (0.0810); for k-NN it is highest for amenity_count (0.1433), followed by days_since_last_review (0...	evidence_model.json → feature_ranking.amenity_count.KNN_perm	输出
0.0747	...amenity_count (0.1433), followed by days_since_last_review (0.0747) and price_missing (0.0621). host_listi...	evidence_model.json → feature_ranking.days_since_last_review.KNN_p erm	输出
0.0621	...lowed by days_since_last_review (0.0747) and price_missing (0.0621). host_listings_count shows a striking ...	evidence_model.json → feature_ranking.price_missing.KNN_perm	输出
0.2698	...ount shows a striking contrast, with impurity importance of 0.2698 (second highest) but permutation import...	evidence_model.json → feature_ranking.host_listings_count.DT_built in	输出
0.0195	... 0.2698 (second highest) but permutation importance of only 0.0195 for the tree and 0.0037 for k-NN, meani...	evidence_model.json → permutation_importance.DT.host_listings_coun t	输出
0.0037	... but permutation importance of only 0.0195 for the tree and 0.0037 for k-NN, meaning the tree often splits...	evidence_model.json → permutation_importance.KNN.host_listings_cou nt	输出
10	...imits. Two further points qualify the results. First, depth 10 gives a CV F1 of 0.5745 ± 0.0130 and de...	cv_results.csv → DT / param	输出
0.5745	...oints qualify the results. First, depth 10 gives a CV F1 of 0.5745 ± 0.0130 and depth 4 gives 0.5712 ± 0.0...	cv_results.csv → DT / cv_mean	输出
0.0130	...lify the results. First, depth 10 gives a CV F1 of 0.5745 ± 0.0130 and depth 4 gives 0.5712 ± 0.0177, a di...	cv_results.csv → DT / cv_std	输出
0.5712	...depth 10 gives a CV F1 of 0.5745 ± 0.0130 and depth 4 gives 0.5712 ± 0.0177, a difference of only 0.0033 t...	cv_results.csv → DT / cv_mean	输出
0.0177	...gives a CV F1 of 0.5745 ± 0.0130 and depth 4 gives 0.5712 ± 0.0177, a difference of only 0.0033 that is sm...	cv_results.csv → DT / cv_std	输出
0.0033	...130 and depth 4 gives 0.5712 ± 0.0177, a difference of only 0.0033 that is smaller than the standard devia...	0.5745 - 0.5712 (cv_results.csv: depth 10 与 depth 4)	推导
10	...tion, so the two depths cannot be reliably separated. Depth 10 is retained because the selection rule,...	evidence_model.json → best_param.DT	输出
0.7188	...e decision tree shows mild overfitting, with training F1 of 0.7188 against test F1 of 0.6319 (a gap of 0.0...	evidence_model.json → test_extra.DT.train_f1	输出
0.6319	... overfitting, with training F1 of 0.7188 against test F1 of 0.6319 (a gap of 0.087), whereas k-NN has a sm...	evidence_model.json → test.DT.f1	输出
0.087	...h training F1 of 0.7188 against test F1 of 0.6319 (a gap of 0.087), whereas k-NN has a smaller gap, 0.602...	0.7188 - 0.6319	推导
0.6023	...of 0.6319 (a gap of 0.087), whereas k-NN has a smaller gap, 0.6023 against 0.5773, because k=101 applies s...	evidence_model.json → test_extra.KNN.train_f1	输出
0.5773	...p of 0.087), whereas k-NN has a smaller gap, 0.6023 against 0.5773, because k=101 applies strong smoothing...	evidence_model.json → test.KNN.f1	输出
101	...as k-NN has a smaller gap, 0.6023 against 0.5773, because k=101 applies strong smoothing...	代码中的设定值 (常量 / 参数)	设定
0.166	...tion most directly. On cross-validation, fundamentals reach 0.166 F1 and operational features 0.548; on t...	rq_experiments.csv → A_fundamentals / test_f1	输出
0.548	...ation, fundamentals reach 0.166 F1 and operational features 0.548; on the test set the corresponding valu...	rq_experiments.csv → B_operational / cv_mean	输出

0.1955	...eatures 0.548; on the test set the corresponding values are 0.1955 and 0.6389, and the 95% interval for th...	evidence_model.json → rq[1].test_f1	输出
0.6389	...48; on the test set the corresponding values are 0.1955 and 0.6389, and the 95% interval for the test-set ...	rq_experiments.csv → B_operational / test_f1	输出
95%	...set the corresponding values are 0.1955 and 0.6389, and the 95% interval for the test-set difference of...	代码中的设定值 (常量 / 参数)	设定
0.4434	...0.6389, and the 95% interval for the test-set difference of 0.4434 is [0.3813, 0.5046], which does not inc...	0.6389 - 0.1955	推导
0.3813	... the 95% interval for the test-set difference of 0.4434 is [0.3813, 0.5046], which does not include zero...	evidence_model.json → boot_A_vs_B.b_minus_a[0]	输出
0.5046	... interval for the test-set difference of 0.4434 is [0.3813, 0.5046], which does not include zero...	evidence_model.json → boot_A_vs_B.b_minus_a[1]	输出
4.4 Feature Selection			
0.0759	...atures: availability_365 ranks third by mutual information (0.0759) but fourth by impurity (0.0356), while...	evidence_model.json → feature_ranking.availability_365.filter_MI	输出
0.0356	...hird by mutual information (0.0759) but fourth by impurity (0.0356), while amenity_count ranks fourth by m...	evidence_model.json → feature_ranking.availability_365.DT_builtin	输出
0.0749	...6), while amenity_count ranks fourth by mutual information (0.0749) but third by impurity (0.1023). Mutual...	evidence_model.json → feature_ranking.amenity_count.filter_MI	输出
0.1023	...ourth by mutual information (0.0749) but third by impurity (0.1023). Mutual information scores each featur...	evidence_model.json → feature_ranking.amenity_count.DT_builtin	输出
0.0356	...ncy there is little left for it to explain, and it drops to 0.0356. amenity_count overlaps with recency fa...	evidence_model.json → feature_ranking.availability_365.DT_builtin	输出
0.1023	... amenity_count overlaps with recency far less, so it keeps 0.1023. Experiment E supports this reading: re...	evidence_model.json → feature_ranking.amenity_count.DT_builtin	输出
0.5846	...ng: removing availability_365 leaves CV F1 slightly higher (0.5846 vs 0.5745) and test F1 essentially unch...	evidence_model.json → rq[9].cv_mean	输出
0.5745	...ng availability_365 leaves CV F1 slightly higher (0.5846 vs 0.5745) and test F1 essentially unchanged (0.6...	evidence_model.json → rq[5].cv_mean	输出
0.6303	...igher (0.5846 vs 0.5745) and test F1 essentially unchanged (0.6303 vs 0.6319)....	evidence_model.json → rq[9].test_f1	输出
0.6319	...846 vs 0.5745) and test F1 essentially unchanged (0.6303 vs 0.6319)....	evidence_model.json → test.DT.f1	输出
0.000	... classed as a non-superhost with a predicted probability of 0.000. On the features the model can see it l...	evidence_model.json → hard_case.proba_superhost (数值为 0, 多个字段同值)	输出
18	...00. On the features the model can see it looks less active: 18 amenities against medians of 45 for sup...	evidence_model.json → hard_case.tree_path[3].value	输出
45	...n see it looks less active: 18 amenities against medians of 45 for superhosts and 32 for non-superhost...	evidence_model.json → hard_case.comparison.amenity_count.superhost_median	输出
32	...tive: 18 amenities against medians of 45 for superhosts and 32 for non-superhosts, availability of 116...	evidence_model.json → hard_case.comparison.amenity_count.non_superhost_median	输出
116	...5 for superhosts and 32 for non-superhosts, availability of 116 which is below even the non-superhost m...	evidence_model.json → hard_case.comparison.availability_365.this_listing	输出
124	...lity of 116 which is below even the non-superhost median of 124, a host with a single listing, and 156 ...	evidence_model.json → hard_case.comparison.availability_365.non_superhost_median	输出
156	...-superhost median of 124, a host with a single listing, and 156 days since the last review against a su...	evidence_model.json → hard_case.comparison.days_since_last_review.this_listing	输出
37	...56 days since the last review against a superhost median of 37. The tree path explains the result: the...	evidence_model.json → hard_case.comparison.days_since_last_review.superhost_median	输出
147.5	...ains the result: the root split is days_since_last_review ≤ 147.5, and its value of 156 falls only 8.5 da...	evidence_model.json → hard_case.tree_path[0].threshold	输出
156	...t split is days_since_last_review ≤ 147.5, and its value of 156 falls only 8.5 days onto the inactive s...	evidence_model.json → hard_case.tree_path[0].value	输出
8.5	..._since_last_review ≤ 147.5, and its value of 156 falls only 8.5 days onto the inactive side; it then me...	156 - 147.5	推导
1.5	...onto the inactive side; it then meets host_listings_count ≤ 1.5 and amenity_count ≤ 24.5 (it has 18), e...	evidence_model.json → rq_extra[6].cv_std (×100)	输出
24.5	...it then meets host_listings_count ≤ 1.5 and amenity_count ≤ 24.5 (it has 18), ending in a leaf of 86 tra...	evidence_model.json → hard_case.tree_path[9].threshold	输出

18	... host_listings_count ≤ 1.5 and amenity_count ≤ 24.5 (it has 18), ending in a leaf of 86 training listi...	evidence_model.json → hard_case.comparison.amenity_count.this_listing	输出
86	...5 and amenity_count ≤ 24.5 (it has 18), ending in a leaf of 86 training listings that are all non-supe...	evidence_model.json → hard_case.tree_path[10].training_listings	输出
0.000	...tings that are all non-superhosts, hence the probability of 0.000. k-NN also classifies it incorrectly, w...	evidence_model.json → hard_case.proba_superhost (数值为 0, 多个字段同值)	输出
0.267	... k-NN also classifies it incorrectly, with a probability of 0.267. On the two official criteria observabl...	evidence_model.json → hard_case.knn_proba_superhost	输出
24	...wo official criteria observable in the data, it meets both: 24 reviews in the last twelve months (at l...	evidence_model.json → hard_case.number_of_reviews_ltm	输出
10	... meets both: 24 reviews in the last twelve months (at least 10 required) and a rating of 4.8 (at least...	Airbnb 官方超级房东标准 (参考文献 Airbnb, n.d.)	推导
4.8	...e last twelve months (at least 10 required) and a rating of 4.8 (at least 4.8 required), but these two ...	Airbnb 官方超级房东标准 (参考文献 Airbnb, n.d.)	推导
4.8	...months (at least 10 required) and a rating of 4.8 (at least 4.8 required), but these two features are e...	Airbnb 官方超级房东标准 (参考文献 Airbnb, n.d.)	推导
54.6	...ded from the models. Its days_since_last_review sits at the 54.6th percentile among all listings yet the...	evidence_model.json → false_negative_context.hard_case_dsl_percentile_among_all_listings	输出
90.8	...ew sits at the 54.6th percentile among all listings yet the 90.8th percentile among reviewed superhosts,...	evidence_model.json → false_negative_context.hard_case_dsl_percentile_among_reviewed_superhosts	输出
0.0344	... coincide with a low embedded ranking. has_review has MI of 0.0344 (rank 7) but impurity of 0.0000 (rank 1...	evidence_model.json → feature_ranking.has_review.filter_MI	输出
0.0000	...nking. has_review has MI of 0.0344 (rank 7) but impurity of 0.0000 (rank 11), because has_review = 0 is ex...	evidence_model.json → feature_ranking.has_review.DT_perm (数值为 0, 多个字段同值)	输出
11	...view has MI of 0.0344 (rank 7) but impurity of 0.0000 (rank 11), because has_review = 0 is exactly equ...	evidence_model.json → feature_ranking.has_review.embedded_rank	输出
4,594	...view = 0 is exactly equivalent to days_since_last_review = 4,594, and once the tree splits on recency it...	evidence_preprocess.json → days_since_last_review_fill_value	输出
93.35%	...rice_missing ranks sixth by MI but ninth by impurity, since 93.35% of listings with a missing price also h...	evidence_preprocess.json → p_unavailable_given_price_missing (×100)	输出
4.5 Dimensionality Reduction & Clustering			
32.1%	...PCA. PC1 carries 32.1% of the variance, and its three largest ...	pca_table.csv → PC1 / explained_variance (×100)	输出
+0.563	...rgest loadings are all operational: days_since_last_review (+0.563), amenity_count (−0.504) and availabili...	evidence_cluster.json → pca_top_loadings.PC1[0][1]	输出
−0.504	...perational: days_since_last_review (+0.563), amenity_count (−0.504) and availability_365 (−0.447). PC2 car...	evidence_cluster.json → pca_top_loadings.PC1[1][1]	输出
−0.447	...view (+0.563), amenity_count (−0.504) and availability_365 (−0.447). PC2 carries 20.45% and compares dist_...	evidence_cluster.json → pca_top_loadings.PC1[2][1]	输出
20.45%	...y_count (−0.504) and availability_365 (−0.447). PC2 carries 20.45% and compares dist_cbd (+0.786) against ...	pca_table.csv → PC2 / explained_variance (×100)	输出
+0.786	...ity_365 (−0.447). PC2 carries 20.45% and compares dist_cbd (+0.786) against host_listings_count (−0.556), ...	evidence_cluster.json → pca_top_loadings.PC2[0][1]	输出
−0.556	...and compares dist_cbd (+0.786) against host_listings_count (−0.556), which separates distance from the CBD...	evidence_cluster.json → pca_top_loadings.PC2[1][1]	输出
16.8%	... the CBD from the size of the host's portfolio. PC3 carries 16.8% and compares accommodates (+0.671) agai...	pca_table.csv → PC3 / explained_variance (×100)	输出
+0.671	...t's portfolio. PC3 carries 16.8% and compares accommodates (+0.671) against availability_365 (−0.591). The...	evidence_cluster.json → pca_top_loadings.PC3[0][1]	输出
−0.591	...nd compares accommodates (+0.671) against availability_365 (−0.591). The components also differ in how the...	evidence_cluster.json → pca_top_loadings.PC3[1][1]	输出
−0.354	...iffer in how they relate to the target: PC1 has correlation −0.354 and NMI 0.057 with y, whereas PC2 has c...	evidence_cluster.json → pc_vs_superhost.PC1.point_biserial_r	输出
0.057	...ey relate to the target: PC1 has correlation −0.354 and NMI 0.057 with y, whereas PC2 has correlation +0...	evidence_cluster.json → pc_vs_superhost.PC1.nmi_10_bins	输出
+0.050	...on −0.354 and NMI 0.057 with y, whereas PC2 has correlation +0.050 and NMI 0.0036. PC2 explains 20.45% of ...	evidence_cluster.json → pc_vs_superhost.PC2.point_biserial_r	输出
0.0036	...MI 0.057 with y, whereas PC2 has correlation +0.050 and NMI 0.0036. PC2 explains 20.45% of the variance al...	evidence_cluster.json → pc_vs_superhost.PC2.nmi_10_bins	输出

20.45%	...eas PC2 has correlation +0.050 and NMI 0.0036. PC2 explains 20.45% of the variance along the geographic an...	pca_table.csv → PC2 / explained_variance (×100)	输出
22.1%	...ates two clusters at the extremes rather than one. K3 holds 22.1% of listings, with a median of 1,883 day...	evidence_cluster.json → cluster_profiles[2].share (×100)	输出
1,883	...ther than one. K3 holds 22.1% of listings, with a median of 1,883 days since the last review, zero availa...	evidence_cluster.json → cluster_profiles[2].days_since_last_review	输出
23	...ast review, zero availability across the year, one listing, 23 amenities and a superhost rate of 6.8%;...	evidence_cluster.json → cluster_profiles[2].amenity_count	输出
6.8%	...the year, one listing, 23 amenities and a superhost rate of 6.8%; K2 sits at the opposite end, with 40 d...	evidence_cluster.json → cluster_profiles[2].superhost_rate (×100)	输出
40	...a superhost rate of 6.8%; K2 sits at the opposite end, with 40 days since the last review, 47 amenitie...	evidence_cluster.json → cluster_profiles[1].days_since_last_review	输出
47	...ts at the opposite end, with 40 days since the last review, 47 amenities and a rate of 53.9%. K1 (35.3...	evidence_cluster.json → cluster_profiles[1].amenity_count	输出
53.9%	...h 40 days since the last review, 47 amenities and a rate of 53.9%. K1 (35.3%) and K5 (40.7%) stay close t...	evidence_cluster.json → cluster_profiles[1].superhost_rate (×100)	输出
35.3%	...ince the last review, 47 amenities and a rate of 53.9%. K1 (35.3%) and K5 (40.7%) stay close to the 35.72...	evidence_cluster.json → cluster_profiles[0].superhost_rate (×100)	输出
40.7%	...eview, 47 amenities and a rate of 53.9%. K1 (35.3%) and K5 (40.7%) stay close to the 35.72% base rate and...	evidence_cluster.json → cluster_profiles[4].superhost_rate (×100)	输出
35.72%	... rate of 53.9%. K1 (35.3%) and K5 (40.7%) stay close to the 35.72% base rate and K4 (43.7%) sits slightly ...	evidence_cluster.json → superhost_rate_reviewed (×100)	输出
43.7%	...) and K5 (40.7%) stay close to the 35.72% base rate and K4 (43.7%) sits slightly above it, and the three ...	evidence_cluster.json → cluster_profiles[3].superhost_rate (×100)	输出
50	...and size: K1 is a group of large operators with a median of 50 listings 1.4 km from the CBD, K4 is lar...	evidence_cluster.json → cluster_profiles[0].host_listings_count	输出
1.4	... is a group of large operators with a median of 50 listings 1.4 km from the CBD, K4 is larger homes at ...	evidence_correlation.json → dist_band_trend.4.dist_km	输出
20.6	...n of 50 listings 1.4 km from the CBD, K4 is larger homes at 20.6 km, and K5 is small listings at 17.8 km...	evidence_cluster.json → cluster_profiles[3].dist_cbd	输出
17.8	... K4 is larger homes at 20.6 km, and K5 is small listings at 17.8 km with 310 available days...	evidence_cluster.json → cluster_profiles[4].dist_cbd	输出
310	... homes at 20.6 km, and K5 is small listings at 17.8 km with 310 available days....	evidence_cluster.json → cluster_profiles[4].availability_365	输出
0.166	...es with the supervised result that fundamentals alone reach 0.166 CV F1 against 0.548 for operational fea...	rq_experiments.csv → A_fundamentals / test_f1	输出
0.548	...ed result that fundamentals alone reach 0.166 CV F1 against 0.548 for operational features....	rq_experiments.csv → B_operational / cv_mean	输出
0.377	...cture with different boundaries. The adjusted Rand index is 0.377 and the majority overlap is 68.4%. The ...	evidence_cluster.json → kmeans_vs_hierarchical_ari	输出
68.4%	...he adjusted Rand index is 0.377 and the majority overlap is 68.4%. The dormant cluster is stable under bo...	evidence_cluster.json → kmeans_vs_hierarchical_majority_overlap (×100)	输出
3,224	...is 68.4%. The dormant cluster is stable under both methods: 3,224 of the 4,689 listings in K3 fall into W...	evidence_cluster.json → crosstab.2.3	输出
4,689	... dormant cluster is stable under both methods: 3,224 of the 4,689 listings in K3 fall into W4, and 99.2% ...	evidence_cluster.json → cluster_profiles[2].listings	输出
99.2%	...ethods: 3,224 of the 4,689 listings in K3 fall into W4, and 99.2% of W4's 3,249 listings come from K3. Th...	3,224 + 3,249 (evidence_cluster.json → crosstab)	推导
3,249	...of the 4,689 listings in K3 fall into W4, and 99.2% of W4's 3,249 listings come from K3. The large-operat...	evidence_cluster.json → cluster_profiles[8].listings	输出
81.4%	 The large-operator cluster also corresponds closely, with 81.4% of K1 (3,696 listings) falling into W3,...	3,696 + 4,541 (K1 规模, Table 11)	推导
3,696	...perator cluster also corresponds closely, with 81.4% of K1 (3,696 listings) falling into W3, while 82.6% ...	evidence_cluster.json → crosstab.0.2	输出
82.6%	...y, with 81.4% of K1 (3,696 listings) falling into W3, while 82.6% of K2 (4,410 listings) falls into W1. T...	4,410 + 5,340 (evidence_cluster.json → crosstab)	推导
4,410	... of K1 (3,696 listings) falling into W3, while 82.6% of K2 (4,410 listings) falls into W1. The Ward profi...	evidence_cluster.json → crosstab.1.0	输出
8,256	...into W1. The Ward profiles complete the interpretation: W1 (8,256 listings, 49.9% superhosts, 47 amenitie...	evidence_cluster.json → cluster_profiles[5].listings	输出
49.9%	...d profiles complete the interpretation: W1 (8,256 listings, 49.9% superhosts, 47 amenities, 56 days since...	evidence_cluster.json → cluster_profiles[5].superhost_rate (×100)	输出

47	...e the interpretation: W1 (8,256 listings, 49.9% superhosts, 47 amenities, 56 days since review), W2 (3...	evidence_cluster.json → cluster_profiles[1].amenity_count	输出
56	...tation: W1 (8,256 listings, 49.9% superhosts, 47 amenities, 56 days since review), W2 (3,829, 30.1%, 2...	evidence_cluster.json → cluster_profiles[5].days_since_last_review	输出
3,829	... 49.9% superhosts, 47 amenities, 56 days since review), W2 (3,829, 30.1%, 22 amenities, 267 available day...	evidence_cluster.json → cluster_profiles[6].listings	输出
30.1%	...superhosts, 47 amenities, 56 days since review), W2 (3,829, 30.1%, 22 amenities, 267 available days), W3 ...	evidence_cluster.json → cluster_profiles[6].superhost_rate (×100)	输出
22	...sts, 47 amenities, 56 days since review), W2 (3,829, 30.1%, 22 amenities, 267 available days), W3 (4,7...	evidence_cluster.json → cluster_profiles[6].amenity_count	输出
267	...ies, 56 days since review), W2 (3,829, 30.1%, 22 amenities, 267 available days), W3 (4,716, 37.5%, 45 h...	evidence_cluster.json → cluster_profiles[6].availability_365	输出
4,716	...), W2 (3,829, 30.1%, 22 amenities, 267 available days), W3 (4,716, 37.5%, 45 host listings, 1.79 km from ...	evidence_cluster.json → cluster_profiles[7].listings	输出
37.5%	...3,829, 30.1%, 22 amenities, 267 available days), W3 (4,716, 37.5%, 45 host listings, 1.79 km from the CBD...	evidence_cluster.json → cluster_profiles[7].superhost_rate (×100)	输出
45	...30.1%, 22 amenities, 267 available days), W3 (4,716, 37.5%, 45 host listings, 1.79 km from the CBD), W...	evidence_cluster.json → cluster_profiles[7].host_listings_count	输出
1.79	...s, 267 available days), W3 (4,716, 37.5%, 45 host listings, 1.79 km from the CBD), W4 (3,249, 2.9%, 2,36...	evidence_cluster.json → cluster_profiles[7].dist_cbd	输出
3,249	...(4,716, 37.5%, 45 host listings, 1.79 km from the CBD), W4 (3,249, 2.9%, 2,364 days since review, zero av...	evidence_cluster.json → cluster_profiles[8].listings	输出
2.9%	... 37.5%, 45 host listings, 1.79 km from the CBD), W4 (3,249, 2.9%, 2,364 days since review, zero availabl...	correlation_matrix.csv → host_listings_count ~ dist_cbd / nmi (×100)	输出
2,364	..., 45 host listings, 1.79 km from the CBD), W4 (3,249, 2.9%, 2,364 days since review, zero available days)...	evidence_cluster.json → cluster_profiles[8].days_since_last_review	输出
1,201	...2.9%, 2,364 days since review, zero available days) and W5 (1,201, 37.8%, accommodates 10, 17.55 km from ...	evidence_cluster.json → hierarchical_sizes.4	输出
37.8%	...,364 days since review, zero available days) and W5 (1,201, 37.8%, accommodates 10, 17.55 km from the CBD...	evidence_cluster.json → cluster_profiles[9].superhost_rate (×100)	输出
10	...ew, zero available days) and W5 (1,201, 37.8%, accommodates 10, 17.55 km from the CBD)....	evidence_cluster.json → cluster_profiles[9].accommodates	输出
17.55	...zero available days) and W5 (1,201, 37.8%, accommodates 10, 17.55 km from the CBD)....	evidence_cluster.json → cluster_profiles[9].dist_cbd	输出
5 Limitations and Improvement Opportunities			
245.00	...sing — Price imputation assigns the training-set median of \$245.00 to 6,553 missing rows, and 93.35% of th...	evidence_model.json → split_details.price_median_train	输出
6,553	...ce imputation assigns the training-set median of \$245.00 to 6,553 missing rows, and 93.35% of those are u...	evidence_preprocess.json → price_missing_count	输出
93.35%	...e training-set median of \$245.00 to 6,553 missing rows, and 93.35% of those are unavailable listings (avai...	evidence_preprocess.json → p_unavailable_given_price_missing (×100)	输出
4,594	...he same amount. The imputed days_since_last_review value of 4,594 becomes an outlier once standardised fo...	evidence_preprocess.json → days_since_last_review_fill_value	输出
21,251	...Correlation — Observations are not independent. The 21,251 reviewed listings come from 11,474 host...	evidence_cluster.json → n_rows_reviewed	输出
11,474	...are not independent. The 21,251 reviewed listings come from 11,474 hosts, and a host's status is the same ...	evidence_cluster.json → n_hosts_reviewed	输出
0.1695	...days_since_last_review with availability_365, MI rises from 0.1695 at five bins to 0.3605 at twenty, an in...	evidence_correlation.json → mi_bin_sensitivity.5.mi ÷ ln 2 (nats → bits)	推导
0.3605	...with availability_365, MI rises from 0.1695 at five bins to 0.3605 at twenty, an increase of 113%, while N...	evidence_correlation.json → mi_bin_sensitivity.20.mi ÷ ln 2 (nats → bits)	推导
113%	...rom 0.1695 at five bins to 0.3605 at twenty, an increase of 113%, while NMI moves only from 0.0799 to 0...	0.3605 ÷ 0.1695 - 1	推导
0.0799	...5 at twenty, an increase of 113%, while NMI moves only from 0.0799 to 0.0896, or 12%. Comparisons between ...	evidence_correlation.json → mi_bin_sensitivity.5.nmi	输出
0.0896	...y, an increase of 113%, while NMI moves only from 0.0799 to 0.0896, or 12%. Comparisons between pairs ther...	evidence_correlation.json → mi_bin_sensitivity.20.nmi	输出
12%	...ase of 113%, while NMI moves only from 0.0799 to 0.0896, or 12%. Comparisons between pairs therefore us...	0.0896 ÷ 0.0799 - 1 (NMI 增幅)	推导
0.664	...lanced. Recall on the superhost class is the weak point, at 0.664 for the tree and 0.560 for k-NN....	evidence_model.json → test.DT.class_1.recall	输出
0.560	...uperhost class is the weak point, at 0.664 for the tree and 0.560 for k-NN....	evidence_model.json → test.KNN.class_1.recall	输出

10	...ax_depth was tuned. For decision trees, max_depth is set to 10 and min_samples_leaf is left at its def...	evidence_model.json → best_param.DT	输出
14,113	...: modelling at the host level, with one row for each of the 14,113 hosts, would remove within-host depende...	evidence_cluster.json → n_hosts_full	输出
0.0278	... For instance, dist_cbd ranks 5th by embedded importance at 0.0278 while the filter places it 8th at MI 0....	evidence_model.json → feature_ranking.dist_cbd.DT_builtin	输出
0.0340	...d importance at 0.0278 while the filter places it 8th at MI 0.0340....	evidence_model.json → feature_ranking.dist_cbd.filter_MI	输出
0.0010	...ry close. availability_365 and amenity_count differ by only 0.0010 on mutual information (0.0759 versus 0....	0.0759 – 0.0749	推导
0.0759	... amenity_count differ by only 0.0010 on mutual information (0.0759 versus 0.0749); MI is produced by a nea...	evidence_model.json → feature_ranking.availability_365.filter_MI	输出
0.0749	... differ by only 0.0010 on mutual information (0.0759 versus 0.0749); MI is produced by a nearest-neighbour...	evidence_model.json → feature_ranking.amenity_count.filter_MI	输出
69.4%	...e variance is unexplained. The first three components carry 69.4%, leaving 30.6% that none of the interpr...	pca_table.csv → PC3 / cumulative_variance (×100)	输出
30.6%	...nexplained. The first three components carry 69.4%, leaving 30.6% that none of the interpretations above ...	evidence_model.json → split.test_pos_rate (×100)	输出
32.1%	...he interpretations above covers, and PC1 alone accounts for 32.1%....	pca_table.csv → PC1 / explained_variance (×100)	输出
667	... long-tailed variables such as host_listings_count (maximum 667) and days_since_last_review (maximum 4,...	evidence_cluster.json → max_before_log.host_listings_count	输出
4,593	...ngs_count (maximum 667) and days_since_last_review (maximum 4,593) would dominate the leading components,...	evidence_cluster.json → max_before_log.days_since_last_review	输出
72,991.5	... no minimum to look for, and the values for k=4 and k=5 are 72,991.5 and 64,211.4, a difference of about 12%...	evidence_cluster.json → kmeans_sweep[2].sse	输出
64,211.4	...o look for, and the values for k=4 and k=5 are 72,991.5 and 64,211.4, a difference of about 12%. The silhoue...	evidence_cluster.json → chosen_k_sse	输出
12%	...=4 and k=5 are 72,991.5 and 64,211.4, a difference of about 12%. The silhouette scores for k=4 and k=5 ...	0.0896 ÷ 0.0799 – 1 (NMI 增幅)	推导
0.0005	...t 12%. The silhouette scores for k=4 and k=5 differ by only 0.0005 (0.2227 versus 0.2232), so the advantag...	0.2232 – 0.2227 (kmeans_sweep 的 silhouette)	推导
0.2227	...he silhouette scores for k=4 and k=5 differ by only 0.0005 (0.2227 versus 0.2232), so the advantage of k=5...	evidence_cluster.json → kmeans_sweep[2].silhouette_5000_sample	输出
0.2232	...scores for k=4 and k=5 differ by only 0.0005 (0.2227 versus 0.2232), so the advantage of k=5 is small. A g...	evidence_cluster.json → kmeans_sweep[3].silhouette_5000_sample	输出
6 Conclusion			
0.166	...e comparison: a decision tree on fundamentals alone reaches 0.166 CV F1 against 0.548 for operational fea...	rq_experiments.csv → A_fundamentals / test_f1	输出
0.548	...sion tree on fundamentals alone reaches 0.166 CV F1 against 0.548 for operational features, while on the ...	rq_experiments.csv → B_operational / cv_mean	输出
0.196	... operational features, while on the test set the values are 0.196 and 0.639, with the interval for the di...	evidence_model.json → rq[1].test_f1	输出
0.639	...al features, while on the test set the values are 0.196 and 0.639, with the interval for the difference [...	rq_experiments.csv → B_operational / test_f1	输出
0.3813	... are 0.196 and 0.639, with the interval for the difference [0.3813, 0.5046] excluding zero; all 11 feature...	evidence_model.json → boot_A_vs_B.b_minus_a[0]	输出
0.5046	...96 and 0.639, with the interval for the difference [0.3813, 0.5046] excluding zero; all 11 features reach ...	evidence_model.json → boot_A_vs_B.b_minus_a[1]	输出
11	...val for the difference [0.3813, 0.5046] excluding zero; all 11 features reach 0.5745 CV and 0.6319 tes...	fs_validation.csv → C_all / n_features	输出
0.5745	...ence [0.3813, 0.5046] excluding zero; all 11 features reach 0.5745 CV and 0.6319 test F1 against a zero ba...	fs_validation.csv → C_all / cv_mean	输出
0.6319	...0.5046] excluding zero; all 11 features reach 0.5745 CV and 0.6319 test F1 against a zero baseline. PCA an...	fs_validation.csv → C_all / test_f1	输出
32.1%	...clustering support this conclusion, with PC1 accounting for 32.1% of the variance and K-Means isolating 4...	pca_table.csv → PC1 / explained_variance (×100)	输出
4,689	... accounting for 32.1% of the variance and K-Means isolating 4,689 low-activity listings at a 6.8% superho...	evidence_cluster.json → cluster_profiles[2].listings	输出
6.8%	...ance and K-Means isolating 4,689 low-activity listings at a 6.8% superhost rate. The status can be infer...	evidence_cluster.json → cluster_profiles[2].superhost_rate (×100)	输出
0.632	...t rate. The status can be inferred only in part: test F1 is 0.632 and recall 0.664, finding 1,047 of 1,57...	evidence_model.json → test.DT.f1	输出

0.664	...s can be inferred only in part: test F1 is 0.632 and recall 0.664, finding 1,047 of 1,577 superhost listi...	evidence_model.json → test.DT.class_1.recall	输出
1,047	...ed only in part: test F1 is 0.632 and recall 0.664, finding 1,047 of 1,577 superhost listings while missi...	evidence_model.json → test_extra.DT.true_positive	输出
1,577	...n part: test F1 is 0.632 and recall 0.664, finding 1,047 of 1,577 superhost listings while missing 530, a...	evidence_model.json → split_details.test_superhost_rows	输出
530	...64, finding 1,047 of 1,577 superhost listings while missing 530, and 118 of the 124 superhost listings ...	evidence_model.json → hard_case.fn_total	输出
118	...ng 1,047 of 1,577 superhost listings while missing 530, and 118 of the 124 superhost listings inactive ...	evidence_model.json → false_negative_context.false_negatives_inactive_over_year	输出
124	... 1,577 superhost listings while missing 530, and 118 of the 124 superhost listings inactive for over a ...	evidence_model.json → hard_case.comparison.availability_365.non_superhost_median	输出

二、PPT（The House or the Host?, 19 页）

数字	上下文	来源（输出文件 → 字段 / 推导式）	类型
01 The house or the host?			
66	...Each dot ≈ 66 listings · orange = superhost listing (...)	25,728 ÷ 389 个点（封面设计）	推导
30.65%	...Each dot ≈ 66 listings · orange = superhost listing (30.65%) · COMP20008 · Assignment 2 · W06G02 · ...	evidence_cluster.json → superhost_rate_full (×100)	输出
01 The house or the host? · 备注			
25,728	...on its own scorecard. The house on the right is Melbourne's 25,728 listings, one dot for about 66 of them....	evidence_cluster.json → n_rows_full	输出
66	...the right is Melbourne's 25,728 listings, one dot for about 66 of them. Then the orange dots light up:...	25,728 ÷ 389 个点（封面设计）	推导
30.65%	...ot for about 66 of them. Then the orange dots light up: the 30.65% of listings run by superhosts, scattere...	evidence_model.json → split.test_pos_rate (×100)	输出
03 Airbnb's four criteria, and what we did			
10	...host criterion · Where it lives in the data · Treatment · ≥ 10 completed stays (or 3 totalling 100 nig...	Airbnb 官方超级房东标准（参考文献 Airbnb, n.d.）	推导
100	...the data · Treatment · ≥ 10 completed stays (or 3 totalling 100 nights) · number_of_reviews, ..._ltn, rev...	Airbnb 官方超级房东标准（参考文献 Airbnb, n.d.）	推导
4.8	...ews, ..._ltn, reviews_per_month · Excluded · Overall rating ≥ 4.8 · review_scores_rating and sub-scores · ...	Airbnb 官方超级房东标准（参考文献 Airbnb, n.d.）	推导
90%	... ≥ 4.8 · review_scores_rating and sub-scores · Excluded · ≥ 90% response rate in 24 h · host_response_r...	Airbnb 官方超级房东标准（参考文献 Airbnb, n.d.）	推导
24	...s_rating and sub-scores · Excluded · ≥ 90% response rate in 24 h · host_response_rate — 100% empty · N...	Airbnb 官方超级房东标准（参考文献 Airbnb, n.d.）	推导
100%	...cluded · ≥ 90% response rate in 24 h · host_response_rate — 100% empty · Not observable · Cancellation r...	代码中的设定值（常量 / 参数）	设定
11	...Cancellation rate · no column · Not in data · What remains: 11 features about the listing and the way ...	evidence_model.json → rq[4].n_features	输出
04 One label, many listings			
25,728	...The data · Inside Airbnb, Melbourne · 25,728 · listings · 90 columns · 14,113 · host...	evidence_cluster.json → n_rows_full	输出
90	...The data · Inside Airbnb, Melbourne · 25,728 · listings · 90 columns · 14,113 · hosts · 30.65% · of ...	listings.csv 原始列数（读入后 90 列）	推导
14,113	...Inside Airbnb, Melbourne · 25,728 · listings · 90 columns · 14,113 · hosts · 30.65% · of listings are supe...	evidence_cluster.json → n_hosts_full	输出
30.65%	...lbourne · 25,728 · listings · 90 columns · 14,113 · hosts · 30.65% · of listings are superhost listings · ...	evidence_model.json → split.test_pos_rate (×100)	输出
26.71%	...113 · hosts · 30.65% · of listings are superhost listings · 26.71% · of hosts (3,770) are superhosts · Why...	evidence_preprocess.json → superhost_host_share (×100)	输出
3,770	...% · of listings are superhost listings · 26.71% · of hosts (3,770) are superhosts · Why it matters · The ...	evidence_preprocess.json → superhost_host_count	输出
55.8%	...longs to the host , so all of a host's listings share it. · 55.8% of listings share a host with another l...	evidence_preprocess.json → multi_listing_host_listing_share (×100)	输出
04 One label, many listings · 备注			
30.65%	... That single fact shapes the method. Superhost listings are 30.65% of rows but only 26.71% of hosts, becau...	evidence_cluster.json → superhost_rate_full (×100)	输出

26.71%	... the method. Superhost listings are 30.65% of rows but only 26.71% of hosts, because superhosts tend to ho...	evidence_preprocess.json → superhost_host_share (×100)	输出
05 Three steps, each with a measurable effect			
4,477	...Preprocessing · 3 of 6 candidates · 01 · 4,477 listings (17.4%) never reviewed: only 6...	evidence_preprocess.json → zero_review_count	输出
17.4%	...Preprocessing · 3 of 6 candidates · 01 · 4,477 listings (17.4%) never reviewed: only 6.59% are superho...	4,477 ÷ 25,728	推导
6.59%	...ndidates · 01 · 4,477 listings (17.4%) never reviewed: only 6.59% are superhost listings, vs 35.72%. · Ke...	evidence_preprocess.json → superhost_share_no_review (×100)	输出
35.72%	...7.4%) never reviewed: only 6.59% are superhost listings, vs 35.72%. · Kept: recency = 4,594 plus a has_rev...	evidence_cluster.json → superhost_rate_reviewed (×100)	输出
4,594	... 6.59% are superhost listings, vs 35.72%. · Kept: recency = 4,594 plus a has_review flag. · Remove it · T...	evidence_preprocess.json → days_since_last_review_fill_value	输出
0.575	...cy = 4,594 plus a has_review flag. · Remove it · Tree CV F1 0.575 → 0.526 · 02 · Comma-splitting the amen...	cv_results.csv → DT / cv_mean	输出
0.526	...94 plus a has_review flag. · Remove it · Tree CV F1 0.575 → 0.526 · 02 · Comma-splitting the amenity list...	evidence_model.json → rq[7].cv_mean	输出
3,096	...5 → 0.526 · 02 · Comma-splitting the amenity list miscounts 3,096 rows (12.03%), by up to 21 items. · Par...	evidence_preprocess.json → amenity_naive_diff_count	输出
12.03%	...02 · Comma-splitting the amenity list miscounts 3,096 rows (12.03%), by up to 21 items. · Parsed as JSON i...	evidence_preprocess.json → amenity_naive_diff_pct	输出
21	...ng the amenity list miscounts 3,096 rows (12.03%), by up to 21 items. · Parsed as JSON instead: one ex...	evidence_preprocess.json → amenity_naive_max_diff	输出
0.537	...tead: one exact count per listing. · Remove it · k-NN CV F1 0.537 → 0.475 · 03 · 25.47% of prices missing...	evidence_model.json → rq[4].cv_mean	输出
0.475	...e exact count per listing. · Remove it · k-NN CV F1 0.537 → 0.475 · 03 · 25.47% of prices missing; 99.85%...	evidence_model.json → rq_extra[0].cv_mean	输出
25.47%	... per listing. · Remove it · k-NN CV F1 0.537 → 0.475 · 03 · 25.47% of prices missing; 99.85% of unavailabl...	evidence_preprocess.json → price_missing_pct	输出
99.85%	... · k-NN CV F1 0.537 → 0.475 · 03 · 25.47% of prices missing; 99.85% of unavailable listings have none — MAR...	evidence_preprocess.json → p_price_missing_given_unavailable (×100)	输出
245	...listings have none — MAR, not MCAR. · Training-set median (\$245) plus a price_missing flag. · Delete ro...	evidence_model.json → split_details.price_median_train	输出
30.65%	...price_missing flag. · Delete rows instead · Superhost share 30.65% → 38.06% · Rejected: imputing bedrooms/...	evidence_preprocess.json → superhost_share_overall (×100)	输出
38.06%	...sing flag. · Delete rows instead · Superhost share 30.65% → 38.06% · Rejected: imputing bedrooms/beds · me...	evidence_preprocess.json → superhost_share_if_price_missing_dropped (×100)	输出
41	...0.65% → 38.06% · Rejected: imputing bedrooms/beds · merging 41 rare property types (111 rows) · binnin...	evidence_preprocess.json → property_type_under_10_count	输出
111	...d: imputing bedrooms/beds · merging 41 rare property types (111 rows) · binning minimum nights (1.66% ≥...	evidence_preprocess.json → property_type_rare_row_count	输出
1.66%	...41 rare property types (111 rows) · binning minimum nights (1.66% ≥ 30)...	evidence_preprocess.json → min_nights_ge30_pct	输出
30	...property types (111 rows) · binning minimum nights (1.66% ≥ 30)...	代码中的设定值 (常量 / 参数)	设定
05 Three steps, each with a measurable effect · 备注			
38%	...able, so deleting those rows would have biased the label to 38%....	38.06% 四舍五入 (evidence_preprocess.json → superhost_share_if_price_missing_dropped)	推导
06 Strong on recency, weak on distance			
0.0785	... NMI · NMI with superhost status · days_since_last_review · 0.0785 · availability_365 · 0.0324 · host_list...	correlation_matrix.csv → days_since_last_review ~ y / nmi	输出
0.0324	...atus · days_since_last_review · 0.0785 · availability_365 · 0.0324 · host_listings_count · 0.0097 · dist_c...	correlation_matrix.csv → availability_365 ~ y / nmi	输出
0.0097	... 0.0785 · availability_365 · 0.0324 · host_listings_count · 0.0097 · dist_cbd · the only fundamental · 0.0...	correlation_matrix.csv → host_listings_count ~ y / nmi	输出
0.0047	...listings_count · 0.0097 · dist_cbd · the only fundamental · 0.0047 · With the target · Pearson · Spearman ...	correlation_matrix.csv → dist_cbd ~ y / nmi	输出
-0.337	... · 0.0047 · With the target · Pearson · Spearman · recency · -0.337 · -0.443 · availability · +0.146 · +0.1...	correlation_matrix.csv → days_since_last_review ~ y / pearson	输出
-0.443	... · With the target · Pearson · Spearman · recency · -0.337 · -0.443 · availability · +0.146 · +0.155 · host...	correlation_matrix.csv → days_since_last_review ~ y / spearman	输出

+0.146	...son · Spearman · recency · −0.337 · −0.443 · availability · +0.146 · +0.155 · host listings · −0.069 · +0...	correlation_matrix.csv → availability_365 ~ y / pearson	输出
+0.155	...arman · recency · −0.337 · −0.443 · availability · +0.146 · +0.155 · host listings · −0.069 · +0.070 · dis...	correlation_matrix.csv → availability_365 ~ y / spearman	输出
−0.069	... · −0.443 · availability · +0.146 · +0.155 · host listings · −0.069 · +0.070 · distance · +0.076 · +0.009 · ...	correlation_matrix.csv → host_listings_count ~ y / pearson	输出
+0.070	... · availability · +0.146 · +0.155 · host listings · −0.069 · +0.070 · distance · +0.076 · +0.009 · Pearson ...	correlation_matrix.csv → host_listings_count ~ y / spearman	输出
+0.076	...146 · +0.155 · host listings · −0.069 · +0.070 · distance · +0.076 · +0.009 · Pearson on a 0/1 target is p...	evidence_model.json → feature_ranking.availability_365.filter_MI	输出
+0.009	...155 · host listings · −0.069 · +0.070 · distance · +0.076 · +0.009 · Pearson on a 0/1 target is point-bise...	evidence_model.json → rq_extra[4].cv_std	输出
10	...oint-biserial: it only compares group means. MI and NMI use 10 equal-frequency bins (≈212 listings per...	代码中的设定值 (常量 / 参数)	设定
212	...pares group means. MI and NMI use 10 equal-frequency bins (≈212 listings per cell). · n = 21,251 review...	21,251 ÷ (10 × 10) 个格子	推导
21,251	...use 10 equal-frequency bins (≈212 listings per cell). · n = 21,251 reviewed listings · 10 pairs × 4 method...	evidence_cluster.json → n_rows_reviewed	输出
10	... (≈212 listings per cell). · n = 21,251 reviewed listings · 10 pairs × 4 methods in the report (Table ...	代码中的设定值 (常量 / 参数)	设定
06 Strong on recency, weak on distance · 备注			
0.0047	... to the CBD, the only fundamental here, is the weakest: NMI 0.0047. These are associations, not causes; la...	correlation_matrix.csv → dist_cbd ~ y / nmi	输出
07 Opposite signs, one curve			
−0.069	...Pearson −0.069 vs Spearman +0.070 · 28.5% · 0—1 8,103 ...	correlation_matrix.csv → host_listings_count ~ y / pearson	输出
+0.070	...Pearson −0.069 vs Spearman +0.070 · 28.5% · 0—1 8,103 · 41.5% · 2 2,560 · ...	correlation_matrix.csv → host_listings_count ~ y / spearman	输出
28.5%	...Pearson −0.069 vs Spearman +0.070 · 28.5% · 0—1 8,103 · 41.5% · 2 2,560 · 46.1% · ...	evidence_correlation.json → hlc_band.rate. (−0.001, 1.0] (×100)	输出
8,103	...Pearson −0.069 vs Spearman +0.070 · 28.5% · 0—1 8,103 · 41.5% · 2 2,560 · 46.1% · 3—5 2,832 · ...	evidence_correlation.json → hlc_band.n. (−0.001, 1.0]	输出
41.5%	...Pearson −0.069 vs Spearman +0.070 · 28.5% · 0—1 8,103 · 41.5% · 2 2,560 · 46.1% · 3—5 2,832 · 40.1% · ...	evidence_correlation.json → hlc_band.rate. (1.0, 2.0] (×100)	输出
2,560	...n −0.069 vs Spearman +0.070 · 28.5% · 0—1 8,103 · 41.5% · 2 2,560 · 46.1% · 3—5 2,832 · 40.1% · 6—14 2,481 · ...	evidence_correlation.json → hlc_band.n.(1.0, 2.0]	输出
46.1%	... vs Spearman +0.070 · 28.5% · 0—1 8,103 · 41.5% · 2 2,560 · 46.1% · 3—5 2,832 · 40.1% · 6—14 2,481 · 42.7%...	evidence_correlation.json → hlc_band.rate. (2.0, 5.0] (×100)	输出
2,832	... +0.070 · 28.5% · 0—1 8,103 · 41.5% · 2 2,560 · 46.1% · 3—5 2,832 · 40.1% · 6—14 2,481 · 42.7% · 15—50 2,...	evidence_correlation.json → hlc_band.n.(2.0, 5.0]	输出
40.1%	... · 28.5% · 0—1 8,103 · 41.5% · 2 2,560 · 46.1% · 3—5 2,832 · 40.1% · 6—14 2,481 · 42.7% · 15—50 2,818 · 41...	evidence_correlation.json → hlc_band.rate. (5.0, 14.0] (×100)	输出
14	...0—1 8,103 · 41.5% · 2 2,560 · 46.1% · 3—5 2,832 · 40.1% · 6—14 2,481 · 42.7% · 15—50 2,818 · 41.9% · 5...	代码中的设定值 (常量 / 参数)	设定
2,481	... 8,103 · 41.5% · 2 2,560 · 46.1% · 3—5 2,832 · 40.1% · 6—14 2,481 · 42.7% · 15—50 2,818 · 41.9% · 51—100 ...	evidence_correlation.json → hlc_band.n.(5.0, 14.0]	输出
42.7%	... 41.5% · 2 2,560 · 46.1% · 3—5 2,832 · 40.1% · 6—14 2,481 · 42.7% · 15—50 2,818 · 41.9% · 51—100 1,198 · ...	evidence_correlation.json → hlc_band.rate. (14.0, 50.0] (×100)	输出
15	... 2 2,560 · 46.1% · 3—5 2,832 · 40.1% · 6—14 2,481 · 42.7% · 15—50 2,818 · 41.9% · 51—100 1,198 · 16.9%...	代码中的设定值 (常量 / 参数)	设定
50	...2,560 · 46.1% · 3—5 2,832 · 40.1% · 6—14 2,481 · 42.7% · 15—50 2,818 · 41.9% · 51—100 1,198 · 16.9% · ...	代码中的设定值 (常量 / 参数)	设定
2,818	...60 · 46.1% · 3—5 2,832 · 40.1% · 6—14 2,481 · 42.7% · 15—50 2,818 · 41.9% · 51—100 1,198 · 16.9% · 101+ 1...	evidence_correlation.json → hlc_band.n. (14.0, 50.0]	输出
41.9%	...1% · 3—5 2,832 · 40.1% · 6—14 2,481 · 42.7% · 15—50 2,818 · 41.9% · 51—100 1,198 · 16.9% · 101+ 1,259 · o...	evidence_correlation.json → hlc_band.rate. (50.0, 100.0] (×100)	输出
51	... 2,832 · 40.1% · 6—14 2,481 · 42.7% · 15—50 2,818 · 41.9% · 51—100 1,198 · 16.9% · 101+ 1,259 · overall...	代码中的设定值 (常量 / 参数)	设定
100	...832 · 40.1% · 6—14 2,481 · 42.7% · 15—50 2,818 · 41.9% · 51—100 1,198 · 16.9% · 101+ 1,259 · overall 35...	代码中的设定值 (常量 / 参数)	设定
1,198	... · 40.1% · 6—14 2,481 · 42.7% · 15—50 2,818 · 41.9% · 51—100 1,198 · 16.9% · 101+ 1,259 · overall 35.7% · ...	evidence_correlation.json → hlc_band.n. (50.0, 100.0]	输出

16.9%	... · 6—14 2,481 · 42.7% · 15—50 2,818 · 41.9% · 51—100 1,198 · 16.9% · 101+ 1,259 · overall 35.7% · Spearman...	evidence_correlation.json → hlc_band.rate. (100.0, 700.0] (×100)	输出
101	...,481 · 42.7% · 15—50 2,818 · 41.9% · 51—100 1,198 · 16.9% · 101+ 1,259 · overall 35.7% · Spearman +0.07...	代码中的设定值 (常量 / 参数)	设定
1,259	... · 42.7% · 15—50 2,818 · 41.9% · 51—100 1,198 · 16.9% · 101+ 1,259 · overall 35.7% · Spearman +0.070 · ran...	evidence_correlation.json → hlc_band.n. (100.0, 700.0]	输出
35.7%	...2,818 · 41.9% · 51—100 1,198 · 16.9% · 101+ 1,259 · overall 35.7% · Spearman +0.070 · ranks · 13,495 list...	evidence_cluster.json → superhost_rate_reviewed (×100)	输出
+0.070	...1—100 1,198 · 16.9% · 101+ 1,259 · overall 35.7% · Spearman +0.070 · ranks · 13,495 listings sit in the ri...	correlation_matrix.csv → host_listings_count ~ y / spearman	输出
13,495	...9% · 101+ 1,259 · overall 35.7% · Spearman +0.070 · ranks · 13,495 listings sit in the rising 0—5 range, w...	8,103 + 2,560 + 2,832 (Figure 1 前三档)	推导
28.5%	...listings sit in the rising 0—5 range, where the rate climbs 28.5% → 46.1%. · Pearson −0.069 · raw values ...	evidence_correlation.json → hlc_band.rate. (−0.001, 1.0] (×100)	输出
46.1%	... sit in the rising 0—5 range, where the rate climbs 28.5% → 46.1% . · Pearson −0.069 · raw values · 1,259 ...	evidence_correlation.json → hlc_band.rate. (2.0, 5.0] (×100)	输出
−0.069	...g 0—5 range, where the rate climbs 28.5% → 46.1%. · Pearson −0.069 · raw values · 1,259 listings of hosts ...	correlation_matrix.csv → host_listings_count ~ y / pearson	输出
1,259	... rate climbs 28.5% → 46.1%. · Pearson −0.069 · raw values · 1,259 listings of hosts with 101+ listings (u...	evidence_correlation.json → hlc_band.n. (100.0, 700.0]	输出
101	... Pearson −0.069 · raw values · 1,259 listings of hosts with 101+ listings (up to 667) drop to 16.9% and...	代码中的设定值 (常量 / 参数)	设定
667	... values · 1,259 listings of hosts with 101+ listings (up to 667) drop to 16.9% and pull the line down. ...	evidence_preprocess.json → host_listings_count_max	输出
16.9%	...59 listings of hosts with 101+ listings (up to 667) drop to 16.9% and pull the line down. · Superhost rat...	evidence_correlation.json → hlc_band.rate. (100.0, 700.0] (×100)	输出
07 Opposite signs, one curve · 备注			
40	...: the rate rises up to three to five listings, stays around 40 to 43 percent up to a hundred, then col...	代码中的设定值 (常量 / 参数)	设定
43	...rate rises up to three to five listings, stays around 40 to 43 percent up to a hundred, then collapses...	evidence_correlation.json → dist_band_trend.1.dsl_median	输出
08 The same pattern, now with distance			
−0.081	...Pearson −0.081 vs Spearman +0.061 · Inner city · 0.6—1...	correlation_matrix.csv → days_since_last_review ~ dist_cbd / pearson	输出
+0.061	...Pearson −0.081 vs Spearman +0.061 · Inner city · 0.6—1.4 km · 41—60 days ...	correlation_matrix.csv → days_since_last_review ~ dist_cbd / spearman	输出
0.6	...Pearson −0.081 vs Spearman +0.061 · Inner city · 0.6 —1.4 km · 41—60 days · the most recently...	evidence_preprocess.json → days_since_last_review_vs_reviews_ltm_spearman_recent_pair	输出
1.4	...Pearson −0.081 vs Spearman +0.061 · Inner city · 0.6— 1.4 km · 41—60 days · the most recently rev...	evidence_correlation.json → dist_band_trend.4.dist_km	输出
41	...arson −0.081 vs Spearman +0.061 · Inner city · 0.6—1.4 km · 41 —60 days · the most recently reviewed li...	evidence_preprocess.json → property_type_under_10_count	输出
60	...on −0.081 vs Spearman +0.061 · Inner city · 0.6—1.4 km · 41 —60 days · the most recently reviewed listi...	evidence_cluster.json → crosstab.4.4	输出
1.7	...ys · the most recently reviewed listings · Middle suburbs · 1.7 —5 km · 111 → 122 · days: most listings ...	evidence_correlation.json → dist_band_trend.5.dist_km	输出
111	...st recently reviewed listings · Middle suburbs · 1.7—5 km · 111 → 122 · days: most listings sit on this...	代码中的设定值 (常量 / 参数)	设定
122	...ently reviewed listings · Middle suburbs · 1.7—5 km · 111 → 122 · days: most listings sit on this rise ...	evidence_correlation.json → dist_band_trend.10.dsl_median	输出
34.6	...11 → 122 · days: most listings sit on this rise · Far out · 34.6 —50.6 km · 69 → 47 · days: few listings,...	evidence_correlation.json → dist_band_trend.18.dist_km	输出
50.6	...122 · days: most listings sit on this rise · Far out · 34.6 — 50.6 km · 69 → 47 · days: few listings, rece...	evidence_correlation.json → dist_band_trend.19.dist_km	输出
69	...: most listings sit on this rise · Far out · 34.6 —50.6 km · 69 → 47 · days: few listings, recently rev...	1 − 0.3065 ≈ 0.694 (基线准确率)	推导
47	...t listings sit on this rise · Far out · 34.6 —50.6 km · 69 → 47 · days: few listings, recently reviewed...	evidence_cluster.json → cluster_profiles[1].amenity_count	输出
+0.061	... reviewed · Most listings lie on the rising part → Spearman +0.061 ; the few distant, recently reviewed on...	correlation_matrix.csv → days_since_last_review ~ dist_cbd / spearman	输出
−0.081	...1; the few distant, recently reviewed ones pull Pearson to −0.081 . · 20 equal-frequency distance bands (...)	correlation_matrix.csv → days_since_last_review ~ dist_cbd / pearson	输出

20	... distant, recently reviewed ones pull Pearson to -0.081 . 20 equal-frequency distance bands ($\approx 1,062$...	代码中的设定值 (常量 / 参数)	设定
1,062	...l Pearson to -0.081 . 20 equal-frequency distance bands ($\approx 1,062$ listings each) $\cdot n = 21,251$ $\cdot$ report Se...	21,251 $\div$ 20 个距离分档	推导
21,251	...equal-frequency distance bands ($\approx 1,062$ listings each) $\cdot n = 21,251$ $\cdot$ report Section 4.2...	evidence_correlation.json $\rightarrow$ n_rows	输出
08 The same pattern, now with distance $\cdot$ 备注			
41	...-size bands shows why. Near the CBD, listings were reviewed 41 to 60 days ago. Recency worsens to 111 ...	evidence_preprocess.json $\rightarrow$ property_type_under_10_count	输出
60	...bands shows why. Near the CBD, listings were reviewed 41 to 60 days ago. Recency worsens to 111 days a...	evidence_cluster.json $\rightarrow$ crosstab.4.4	输出
111	...listings were reviewed 41 to 60 days ago. Recency worsens to 111 days at 1.7 kilometres and peaks at 122...	代码中的设定值 (常量 / 参数)	设定
1.7	... reviewed 41 to 60 days ago. Recency worsens to 111 days at 1.7 kilometres and peaks at 122 days around...	correlation_matrix.csv $\rightarrow$ host_listings_count $\sim y / \text{mi_bits} (\times 100)$	输出
122	... Recency worsens to 111 days at 1.7 kilometres and peaks at 122 days around five kilometres. Far out, a...	evidence_correlation.json $\rightarrow$ dist_band_trend.10.dsl_median	输出
35	...s and peaks at 122 days around five kilometres. Far out, at 35 and 50 kilometres, it improves again to...	34.6 $- -0.081$	推导
50	...eaks at 122 days around five kilometres. Far out, at 35 and 50 kilometres, it improves again to 69 and...	evidence_cluster.json $\rightarrow$ cluster_profiles[0].days_since_last_review	输出
69	...res. Far out, at 35 and 50 kilometres, it improves again to 69 and 47 days. [click] Most listings sit ...	1 $- 0.3065 \approx 0.694$ (基线准确率)	推导
47	...r out, at 35 and 50 kilometres, it improves again to 69 and 47 days. [click] Most listings sit on the ...	evidence_cluster.json $\rightarrow$ cluster_profiles[1].amenity_count	输出
09 No host is ever seen twice			
25,728	...Supervised learning $\cdot$ k-NN and decision tree $\cdot$ 25,728 listings $\cdot$ grouped by host $\cdot$ stratified...	evidence_cluster.json $\rightarrow$ n_rows_full	输出
80%	...listings $\cdot$ grouped by host $\cdot$ stratified on the label $\cdot$ Train 80% $\cdot$ 20,582 $\cdot$ 11,292 hosts $\cdot$ 5-fold groupe...	划分比例: 5 折取 1 折作测试集 (设定)	推导
20,582	...s $\cdot$ grouped by host $\cdot$ stratified on the label $\cdot$ Train 80% $\cdot$ 20,582 $\cdot$ 11,292 hosts $\cdot$ 5-fold grouped CV $\cdot$ Te...	evidence_model.json $\rightarrow$ split.train_rows	输出
11,292	...ed by host $\cdot$ stratified on the label $\cdot$ Train 80% $\cdot$ 20,582 $\cdot$ 11,292 hosts $\cdot$ 5-fold grouped CV $\cdot$ Test 20% $\cdot$...	evidence_model.json $\rightarrow$ split.train_hosts	输出
20%	...rain 80% $\cdot$ 20,582 $\cdot$ 11,292 hosts $\cdot$ 5-fold grouped CV $\cdot$ Test 20% $\cdot$ 5,146 $\cdot$ 2,821 hosts $\cdot$ used once $\cdot$ Met...	evidence_model.json $\rightarrow$ rq[1].test_f1 ($\times 100$)	输出
5,146	...0% $\cdot$ 20,582 $\cdot$ 11,292 hosts $\cdot$ 5-fold grouped CV $\cdot$ Test 20% $\cdot$ 5,146 $\cdot$ 2,821 hosts $\cdot$ used once $\cdot$ Metric: sup...	evidence_model.json $\rightarrow$ split.test_rows	输出
2,821	...582 $\cdot$ 11,292 hosts $\cdot$ 5-fold grouped CV $\cdot$ Test 20% $\cdot$ 5,146 $\cdot$ 2,821 hosts $\cdot$ used once $\cdot$ Metric: superhost-c...	evidence_model.json $\rightarrow$ split.test_hosts	输出
0.694	...perhost-class F1 $\cdot$ Always guessing “not a superhost” scores 0.694 accuracy but F1 = 0. $\cdot$ k-NN $\cdot$ CV F1 by ...	evidence_model.json $\rightarrow$ test.Baseline.accuracy	输出
101	...scores 0.694 accuracy but F1 = 0. $\cdot$ k-NN $\cdot$ CV F1 by k $\cdot$ k = 101 $\cdot$ 0.537 $\cdot$ 1 $\cdot$ 5 $\cdot$ 11 $\cdot$ 21 $\cdot$ 51 $\cdot$ 101 $\cdot$...	cv_results.csv $\rightarrow$ KNN / param	输出
0.537	... 0.694 accuracy but F1 = 0. $\cdot$ k-NN $\cdot$ CV F1 by k $\cdot$ k = 101 $\cdot$ 0.537 $\cdot$ 1 $\cdot$ 5 $\cdot$ 11 $\cdot$ 21 $\cdot$ 51 $\cdot$ 101 $\cdot$ 151 $\cdot$ 20...	evidence_model.json $\rightarrow$ rq[4].cv_mean	输出
11	...but F1 = 0. $\cdot$ k-NN $\cdot$ CV F1 by k $\cdot$ k = 101 $\cdot$ 0.537 $\cdot$ 1 $\cdot$ 5 $\cdot$ 11 $\cdot$ 21 $\cdot$ 51 $\cdot$ 101 $\cdot$ 151 $\cdot$ 201 $\cdot$ 301 $\cdot$ Dec...	cv_results.csv $\rightarrow$ KNN / param	输出
21	...1 = 0. $\cdot$ k-NN $\cdot$ CV F1 by k $\cdot$ k = 101 $\cdot$ 0.537 $\cdot$ 1 $\cdot$ 5 $\cdot$ 11 $\cdot$ 21 $\cdot$ 51 $\cdot$ 101 $\cdot$ 151 $\cdot$ 201 $\cdot$ 301 $\cdot$ Decision...	cv_results.csv $\rightarrow$ KNN / param	输出
51	... $\cdot$ k-NN $\cdot$ CV F1 by k $\cdot$ k = 101 $\cdot$ 0.537 $\cdot$ 1 $\cdot$ 5 $\cdot$ 11 $\cdot$ 21 $\cdot$ 51 $\cdot$ 101 $\cdot$ 151 $\cdot$ 201 $\cdot$ 301 $\cdot$ Decision tree...	cv_results.csv $\rightarrow$ KNN / param	输出
101	...-NN $\cdot$ CV F1 by k $\cdot$ k = 101 $\cdot$ 0.537 $\cdot$ 1 $\cdot$ 5 $\cdot$ 11 $\cdot$ 21 $\cdot$ 51 $\cdot$ 101 $\cdot$ 151 $\cdot$ 201 $\cdot$ 301 $\cdot$ Decision tree $\cdot$ CV ...	cv_results.csv $\rightarrow$ KNN / param	输出
151	...CV F1 by k $\cdot$ k = 101 $\cdot$ 0.537 $\cdot$ 1 $\cdot$ 5 $\cdot$ 11 $\cdot$ 21 $\cdot$ 51 $\cdot$ 101 $\cdot$ 151 $\cdot$ 201 $\cdot$ 301 $\cdot$ Decision tree $\cdot$ CV F1 by ...	cv_results.csv $\rightarrow$ KNN / param	输出
201	...by k $\cdot$ k = 101 $\cdot$ 0.537 $\cdot$ 1 $\cdot$ 5 $\cdot$ 11 $\cdot$ 21 $\cdot$ 51 $\cdot$ 101 $\cdot$ 151 $\cdot$ 201 $\cdot$ 301 $\cdot$ Decision tree $\cdot$ CV F1 by max_de...	cv_results.csv $\rightarrow$ KNN / param	输出
301	... k = 101 $\cdot$ 0.537 $\cdot$ 1 $\cdot$ 5 $\cdot$ 11 $\cdot$ 21 $\cdot$ 51 $\cdot$ 101 $\cdot$ 151 $\cdot$ 201 $\cdot$ 301 $\cdot$ Decision tree $\cdot$ CV F1 by max_depth $\cdot$...	cv_results.csv $\rightarrow$ KNN / param	输出
10	...51 $\cdot$ 201 $\cdot$ 301 $\cdot$ Decision tree $\cdot$ CV F1 by max_depth $\cdot$ depth 10 $\cdot$ 0.5745 $\cdot$ 2 $\cdot$ 3 $\cdot$ 4 $\cdot$ 5 $\cdot$ 6 $\cdot$ 8 $\cdot$ 10 $\cdot$...	cv_results.csv $\rightarrow$ DT / param	输出

0.5745	...201 · 301 · Decision tree · CV F1 by max_depth · depth 10 · 0.5745 · 2 · 3 · 4 · 5 · 6 · 8 · 10 · 15 · Non...	cv_results.csv → DT / cv_mean	输出
10	... by max_depth · depth 10 · 0.5745 · 2 · 3 · 4 · 5 · 6 · 8 · 10 · 15 · None · Both curves share one y-s...	cv_results.csv → DT / param	输出
15	...ax_depth · depth 10 · 0.5745 · 2 · 3 · 4 · 5 · 6 · 8 · 10 · 15 · None · Both curves share one y-scale ...	cv_results.csv → DT / param	输出
0.47	...5 · 6 · 8 · 10 · 15 · None · Both curves share one y-scale (0.47–0.59) · depth 4 (blue) is within one SD...	evidence_model.json → rq_extra[0].cv_mean	输出
0.59	... · 8 · 10 · 15 · None · Both curves share one y-scale (0.47–0.59) · depth 4 (blue) is within one SD of d...	evidence_model.json → boot_KNN_vs_DT.b[0]	输出
10	...cale (0.47–0.59) · depth 4 (blue) is within one SD of depth 10: 0.5712 vs 0.5745...	evidence_model.json → best_param.DT	输出
0.5712	... (0.47–0.59) · depth 4 (blue) is within one SD of depth 10: 0.5712 vs 0.5745...	cv_results.csv → DT / cv_mean	输出
0.5745	...9) · depth 4 (blue) is within one SD of depth 10: 0.5712 vs 0.5745...	cv_results.csv → DT / cv_mean	输出
09 No host is ever seen twice · 备注			
30.65%	...t is grouped by host and stratified, so both sides keep the 30.65% superhost share and no host appears on ...	evidence_preprocess.json → superhost_share_overall (×100)	输出
69%	...ading here because a model that always says no already gets 69%, so we optimise F1 on the superhost cla...	$1 - 0.3065 \approx 0.694$ (基线准确率)	推导
101	...se F1 on the superhost class. k-NN improves steadily to k = 101, then flattens. The tree peaks at depth...	代码中的设定值 (常量 / 参数)	设定
10	...steadily to k = 101, then flattens. The tree peaks at depth 10 but depth 4 is within noise, a point we...	evidence_model.json → best_param.DT	输出
0.502	...e come back to in the limitations. Unlimited depth drops to 0.502 : that is the price of memorising. Why f...	cv_results.csv → DT / cv_mean	输出
319	...ising. Why five folds rather than ten: the largest host has 319 listings, which would fill 15.5% of a t...	evidence_model.json → split_details.largest_host_rows	输出
15.5%	...an ten: the largest host has 319 listings, which would fill 15.5% of a ten-fold validation fold against 7...	$319 \div (20,582 \div 10)$	推导
7.75%	...hich would fill 15.5% of a ten-fold validation fold against 7.75% with five....	$319 \div (20,582 \div 5)$	推导
10 Two in three superhost listings found			
5,146	...Test set · 5,146 listings · 1,577 superhost listings · D...	evidence_model.json → split.test_rows	输出
1,577	...Test set · 5,146 listings · 1,577 superhost listings · Decision tree · de...	evidence_model.json → split_details.test_superhost_rows	输出
10	...listings · 1,577 superhost listings · Decision tree · depth 10 · 0.632 · F1 · 95% CI [0.594, 0.676] · ...	evidence_model.json → best_param.DT	输出
0.632	...ngs · 1,577 superhost listings · Decision tree · depth 10 · 0.632 · F1 · 95% CI [0.594, 0.676] · recall 0...	evidence_model.json → test.DT.f1	输出
95%	...uperhost listings · Decision tree · depth 10 · 0.632 · F1 · 95% CI [0.594, 0.676] · recall 0.664 · 1,04...	代码中的设定值 (常量 / 参数)	设定
0.594	... listings · Decision tree · depth 10 · 0.632 · F1 · 95% CI [0.594 , 0.676] · recall 0.664 · 1,047 of 1,577...	evidence_model.json → boot_KNN_vs_DT.b[0]	输出
0.676	...gs · Decision tree · depth 10 · 0.632 · F1 · 95% CI [0.594, 0.676] · recall 0.664 · 1,047 of 1,577 · k-NN...	evidence_model.json → boot_KNN_vs_DT.b[1]	输出
0.664	...ee · depth 10 · 0.632 · F1 · 95% CI [0.594, 0.676] · recall 0.664 · 1,047 of 1,577 · k-NN · k = 101 · 0.5...	evidence_model.json → test.DT.class_1.recall	输出
1,047	...th 10 · 0.632 · F1 · 95% CI [0.594, 0.676] · recall 0.664 · 1,047 of 1,577 · k-NN · k = 101 · 0.577 · F1 ...	evidence_model.json → test_extra.DT.true_positive	输出
1,577	...632 · F1 · 95% CI [0.594, 0.676] · recall 0.664 · 1,047 of 1,577 · k-NN · k = 101 · 0.577 · F1 · CI [0.5...	evidence_model.json → split_details.test_superhost_rows	输出
101	...[0.594, 0.676] · recall 0.664 · 1,047 of 1,577 · k-NN · k = 101 · 0.577 · F1 · CI [0.542, 0.616] · reca...	代码中的设定值 (常量 / 参数)	设定
0.577	..., 0.676] · recall 0.664 · 1,047 of 1,577 · k-NN · k = 101 · 0.577 · F1 · CI [0.542, 0.616] · recall 0.560...	evidence_model.json → test.KNN.f1	输出
0.542	... 0.664 · 1,047 of 1,577 · k-NN · k = 101 · 0.577 · F1 · CI [0.542 , 0.616] · recall 0.560 · 883 found · Ma...	evidence_model.json → boot_KNN_vs_DT.a[0]	输出
0.616	... · 1,047 of 1,577 · k-NN · k = 101 · 0.577 · F1 · CI [0.542, 0.616] · recall 0.560 · 883 found · Majority ...	evidence_model.json → boot_KNN_vs_DT.a[1]	输出
0.560	... · k-NN · k = 101 · 0.577 · F1 · CI [0.542, 0.616] · recall 0.560 · 883 found · Majority baseline · 0.000...	evidence_model.json → test.KNN.class_1.recall	输出

883	... · k = 101 · 0.577 · F1 · CI [0.542, 0.616] · recall 0.560 · 883 found · Majority baseline · 0.000 · F1 ...	evidence_model.json → test_extra.KNN.true_positive	输出
0.000	...42, 0.616] · recall 0.560 · 883 found · Majority baseline · 0.000 · F1 · accuracy 0.694 · finds none of t...	evidence_model.json → test.Baseline.f1 (数值为 0, 多个字段同值)	输出
0.694	...560 · 883 found · Majority baseline · 0.000 · F1 · accuracy 0.694 · finds none of them · Tree — k-NN · +0...	evidence_model.json → test.Baseline.accuracy	输出
+0.055	... · F1 · accuracy 0.694 · finds none of them · Tree — k-NN · +0.055 F1, CI [0.013, 0.099] — excludes zero ...	evidence_model.json → test_extra.KNN.gain_over_baseline.accuracy	输出
0.013	...y 0.694 · finds none of them · Tree — k-NN · +0.055 F1, CI [0.013, 0.099] — excludes zero · Gain over bas...	evidence_model.json → boot_KNN_vs_DT.b_minus_a[0]	输出
0.099	... · finds none of them · Tree — k-NN · +0.055 F1, CI [0.013, 0.099] — excludes zero · Gain over baseline ...	evidence_model.json → boot_KNN_vs_DT.b_minus_a[1]	输出
+0.632	...CI [0.013, 0.099] — excludes zero · Gain over baseline · F1 +0.632 · macro-F1 +0.319 · accuracy +0.069 · l...	evidence_model.json → test_extra.DT.gain_over_baseline.f1	输出
+0.319	... — excludes zero · Gain over baseline · F1 +0.632 · macro-F1 +0.319 · accuracy +0.069 · Intervals: percenti...	evidence_model.json → test_extra.DT.gain_over_baseline.macro_f1	输出
+0.069	...Gain over baseline · F1 +0.632 · macro-F1 +0.319 · accuracy +0.069 · Intervals: percentile bootstrap, 1,00...	evidence_model.json → test_extra.DT.gain_over_baseline.accuracy	输出
1,000	...+0.319 · accuracy +0.069 · Intervals: percentile bootstrap, 1,000 paired resamples of the 2,821 test host...	代码中的设定值 (常量 / 参数)	设定
2,821	...ervals: percentile bootstrap, 1,000 paired resamples of the 2,821 test hosts · report Tables 7–8...	evidence_model.json → split.test_hosts	输出
10 Two in three superhost listings found · 备注			
0.632	...With the criteria removed, the tree reaches F1 0.632 on the superhost class and finds two th...	evidence_model.json → test.DT.class_1.f1-score	输出
0.577	...nd finds two thirds of the superhost listings; k-NN reaches 0.577. The baseline gets 69% accuracy by sayi...	evidence_model.json → test.KNN.class_1.f1-score	输出
69%	...e superhost listings; k-NN reaches 0.577. The baseline gets 69% accuracy by saying no every time, which...	1 - 0.3065 ≈ 0.694 (基线准确率)	推导
11 Where each model goes wrong			
1,047	...Confusion matrices · test set · Superhosts found · 1,047 vs 883 · true positives · tree vs k-NN ...	evidence_model.json → test_extra.DT.true_positive	输出
883	...Confusion matrices · test set · Superhosts found · 1,047 vs 883 · true positives · tree vs k-NN · The p...	evidence_model.json → test_extra.KNN.true_positive	输出
690	... · true positives · tree vs k-NN · The price: false alarms · 690 vs 599 · non-superhosts flagged · tree ...	evidence_model.json → test_extra.DT.false_positive	输出
599	...positives · tree vs k-NN · The price: false alarms · 690 vs 599 · non-superhosts flagged · tree vs k-NN...	evidence_model.json → test_extra.KNN.false_positive	输出
530	...99 · non-superhosts flagged · tree vs k-NN · Still missed · 530 vs 694 · superhost listings not found ...	evidence_model.json → test_extra.DT.false_negative	输出
694	...n-superhosts flagged · tree vs k-NN · Still missed · 530 vs 694 · superhost listings not found · The tr...	evidence_model.json → test_extra.KNN.false_negative	输出
1,737	...530 vs 694 · superhost listings not found · The tree labels 1,737 listings as superhosts against 1,482 fo...	evidence_model.json → test_extra.DT.predicted_superhost	输出
1,482	...ound · The tree labels 1,737 listings as superhosts against 1,482 for k-NN: it finds more, and pays with ...	evidence_model.json → test_extra.KNN.predicted_superhost	输出
5,146	...k-NN: it finds more, and pays with more false alarms. · n = 5,146 test listings · 1,577 superhost listing...	evidence_model.json → split.test_rows	输出
1,577	...nd pays with more false alarms. · n = 5,146 test listings · 1,577 superhost listings · counts from the no...	evidence_model.json → split_details.test_superhost_rows	输出
11 Where each model goes wrong · 备注			
1,047	...ces show the trade-off behind the F1 scores. The tree finds 1,047 superhost listings against 883 for k-NN...	evidence_model.json → test_extra.DT.true_positive	输出
883	... F1 scores. The tree finds 1,047 superhost listings against 883 for k-NN, but it also raises more false...	evidence_model.json → test_extra.KNN.true_positive	输出
690	...against 883 for k-NN, but it also raises more false alarms, 690 against 599. Both models still miss man...	evidence_model.json → test_extra.DT.false_positive	输出
599	...for k-NN, but it also raises more false alarms, 690 against 599. Both models still miss many superhosts...	evidence_model.json → test_extra.KNN.false_positive	输出
530	...s, 690 against 599. Both models still miss many superhosts: 530 for the tree, 694 for k-NN. [click] Ove...	evidence_model.json → test_extra.DT.false_negative	输出

694	 Both models still miss many superhosts: 530 for the tree, 694 for k-NN. [click] Overall the tree call...	evidence_model.json → test_extra.KNN.false_negative	输出
1,737	... for the tree, 694 for k-NN. [click] Overall the tree calls 1,737 listings superhosts against 1,482, whic...	evidence_model.json → test_extra.DT.predicted_superhost	输出
1,482	...k] Overall the tree calls 1,737 listings superhosts against 1,482, which is why its recall is higher and ...	evidence_model.json → test_extra.KNN.predicted_superhost	输出
12 The signal lives in how the listing is run			
0.196	... one family only · Listing fundamentals only · 3 features · 0.196 · Operational strategy only · 8 feature...	rq_experiments.csv → A_fundamentals / test_f1	输出
0.639	...features · 0.196 · Operational strategy only · 8 features · 0.639 · Gap · +0.443 · CI [0.381, 0.505] · No...	rq_experiments.csv → B_operational / test_f1	输出
+0.443	...96 · Operational strategy only · 8 features · 0.639 · Gap · +0.443 · CI [0.381, 0.505] · No review feature...	0.6389 – 0.1955 (B – A 点估计, Table 8 的 0.4434)	推导
0.381	...nal strategy only · 8 features · 0.639 · Gap · +0.443 · CI [0.381, 0.505] · No review features · still 0...	evidence_model.json → boot_A_vs_B.b_minus_a[0]	输出
0.505	...ategy only · 8 features · 0.639 · Gap · +0.443 · CI [0.381, 0.505] · No review features · still 0.566 · +...	evidence_model.json → boot_A_vs_B.b_minus_a[1]	输出
0.566	...p · +0.443 · CI [0.381, 0.505] · No review features · still 0.566 · +0.371 over fundamentals · Price move...	rq_experiments.csv → F_operational_no_review / test_f1	输出
+0.371	...43 · CI [0.381, 0.505] · No review features · still 0.566 · +0.371 over fundamentals · Price moved to fund...	0.5662 – 0.1955 (Table 8: F – A, 0.3707)	推导
0.184	...+0.371 over fundamentals · Price moved to fundamentals · CV 0.184 vs 0.558 · same verdict · Decision tree...	rq_experiments_extra.csv → A2_fundamentals_plus_price / cv_mean	输出
0.558	...er fundamentals · Price moved to fundamentals · CV 0.184 vs 0.558 · same verdict · Decision tree, test-se...	rq_experiments_extra.csv → B2_operational_minus_price / cv_mean	输出
12 The signal lives in how the listing is run · 备注			
0.196	...rained on the three fundamentals alone, the tree reaches F1 0.196. [click] On the eight operational featu...	rq_experiments.csv → A_fundamentals / test_f1	输出
0.639	...eaches F1 0.196. [click] On the eight operational features, 0.639. The gap is 0.443 and its interval stay...	rq_experiments.csv → B_operational / test_f1	输出
0.443	...click] On the eight operational features, 0.639. The gap is 0.443 and its interval stays far from zero. T...	0.6389 – 0.1955 (B – A 点估计, Table 8 的 0.4434)	推导
0.371	...h review features and operations still beat fundamentals by 0.371. Price could be called a fundamental, s...	0.5662 – 0.1955 (Table 8: F – A, 0.3707)	推导
13 Same family, different favourites			
0.227	... test set · Decision tree · k-NN · days_since_last_review · 0.227 · 0.075 · amenity_count · 0.081 · 0.143...	evidence_model.json → permutation_importance.DT.days_since_last_re view	输出
0.075	...t · Decision tree · k-NN · days_since_last_review · 0.227 · 0.075 · amenity_count · 0.081 · 0.143 · price...	evidence_model.json → permutation_importance.KNN.days_since_last_r eview	输出
0.081	... · days_since_last_review · 0.227 · 0.075 · amenity_count · 0.081 · 0.143 · price_missing · 0.022 · 0.062...	evidence_model.json → feature_ranking.amenity_count.DT_perm	输出
0.143	...since_last_review · 0.227 · 0.075 · amenity_count · 0.081 · 0.143 · price_missing · 0.022 · 0.062 · dist_...	evidence_model.json → feature_ranking.amenity_count.KNN_perm	输出
0.022	...7 · 0.075 · amenity_count · 0.081 · 0.143 · price_missing · 0.022 · 0.062 · dist_cbd (fund.) · 0.005 · 0...	evidence_model.json → feature_ranking.price_missing.DT_perm	输出
0.062	...5 · amenity_count · 0.081 · 0.143 · price_missing · 0.022 · 0.062 · dist_cbd (fund.) · 0.005 · 0.032 · ho...	evidence_model.json → feature_ranking.price_missing.KNN_perm	输出
0.005	... 0.143 · price_missing · 0.022 · 0.062 · dist_cbd (fund.) · 0.005 · 0.032 · host_listings_count · 0.020 · ...	evidence_model.json → feature_ranking.dist_cbd.DT_perm	输出
0.032	... price_missing · 0.022 · 0.062 · dist_cbd (fund.) · 0.005 · 0.032 · host_listings_count · 0.020 · 0.004 · ...	evidence_model.json → feature_ranking.dist_cbd.KNN_perm	输出
0.020	... · dist_cbd (fund.) · 0.005 · 0.032 · host_listings_count · 0.020 · 0.004 · availability_365 · 0.013 · 0...	evidence_model.json → feature_ranking.host_listings_count.DT_perm	输出
0.004	...cbd (fund.) · 0.005 · 0.032 · host_listings_count · 0.020 · 0.004 · availability_365 · 0.013 · 0.017 · To...	evidence_model.json → feature_ranking.host_listings_count.KNN_perm	输出
0.013	... · host_listings_count · 0.020 · 0.004 · availability_365 · 0.013 · 0.017 · Top 3 · filter (MI) · recency...	evidence_model.json → feature_ranking.availability_365.DT_perm	输出
0.017	...listings_count · 0.020 · 0.004 · availability_365 · 0.013 · 0.017 · Top 3 · filter (MI) · recency · host ...	evidence_model.json → feature_ranking.availability_365.KNN_perm	输出
4,594	...has_review: MI rank 7, impurity 0 — it is simply “recency = 4,594”. · Drop availability_365 (exp. E) · CV...	evidence_preprocess.json → days_since_last_review_fill_value	输出

0.5846	...“recency = 4,594”. · Drop availability_365 (exp. E) · CV F1 0.5846 vs 0.5745 · test 0.630 vs 0.632 · Mean ...	evidence_model.json → rq[9].cv_mean	输出
0.5745	... 4,594”. · Drop availability_365 (exp. E) · CV F1 0.5846 vs 0.5745 · test 0.630 vs 0.632 · Mean drop in te...	evidence_model.json → rq[5].cv_mean	输出
0.630	...p availability_365 (exp. E) · CV F1 0.5846 vs 0.5745 · test 0.630 vs 0.632 · Mean drop in test F1 when a ...	evidence_model.json → rq[9].test_f1	输出
0.632	...ility_365 (exp. E) · CV F1 0.5846 vs 0.5745 · test 0.630 vs 0.632 · Mean drop in test F1 when a feature i...	evidence_model.json → test.DT.f1	输出
10	...vs 0.632 · Mean drop in test F1 when a feature is shuffled, 10 repeats · full ranking in report Table ...	代码中的设定值（常量 / 参数）	设定
13 Same family, different favourites · 备注			
4,594	...ortance, because it is exactly the same as recency equal to 4,594. [click] Experiment E confirms the over...	evidence_preprocess.json → days_since_last_review_fill_value	输出
14 Listing 6812677: certain, and wrong			
147.5	... · the tree's most confident miss · days_since_last_review ≤ 147.5? · 156 → no · host_listings_count ≤ 1.5...	evidence_model.json → hard_case.tree_path[0].threshold	输出
156	...e's most confident miss · days_since_last_review ≤ 147.5? · 156 → no · host_listings_count ≤ 1.5? · 1 → ...	evidence_model.json → hard_case.comparison.days_since_last_review.this_listing	输出
1.5	...nce_last_review ≤ 147.5? · 156 → no · host_listings_count ≤ 1.5? · 1 → yes · 7 more splits on recency, ...	evidence_model.json → rq_extra[6].cv_std (×100)	输出
24.5	... recency, amenities, price, availability... · amenity_count ≤ 24.5? · 18 → yes · Leaf: 86 training listing...	evidence_model.json → hard_case.tree_path[9].threshold	输出
18	..., amenities, price, availability... · amenity_count ≤ 24.5? · 18 → yes · Leaf: 86 training listings, 0 s...	evidence_model.json → hard_case.comparison.amenity_count.this_listing	输出
86	...e, availability... · amenity_count ≤ 24.5? · 18 → yes · Leaf: 86 training listings, 0 superhosts · P = 0...	evidence_model.json → hard_case.tree_path[10].training_listings	输出
0.000	... · 18 → yes · Leaf: 86 training listings, 0 superhosts · P = 0.000 · What the model could not see · 24 rev...	evidence_model.json → hard_case.proba_superhost（数值为 0，多个字段同值）	输出
24	..., 0 superhosts · P = 0.000 · What the model could not see · 24 reviews in the last 12 months · needs 1...	evidence_model.json → hard_case.number_of_reviews_ltm	输出
12	...000 · What the model could not see · 24 reviews in the last 12 months · needs 10 · rating 4.8 · needs ...	0.0896 ÷ 0.0799 – 1 (NMI 增幅)	推导
10	...el could not see · 24 reviews in the last 12 months · needs 10 · rating 4.8 · needs 4.8 · Meets both c...	Airbnb 官方超级房东标准（参考文献 Airbnb, n.d.）	推导
4.8	... see · 24 reviews in the last 12 months · needs 10 · rating 4.8 · needs 4.8 · Meets both criteria visib...	evidence_model.json → hard_case.review_scores_rating	输出
4.8	...views in the last 12 months · needs 10 · rating 4.8 · needs 4.8 · Meets both criteria visible in the da...	Airbnb 官方超级房东标准（参考文献 Airbnb, n.d.）	推导
156	...luded by design. · An outlier within its class · Recency of 156 days: 54.6th percentile of all listings...	evidence_model.json → hard_case.comparison.days_since_last_review.this_listing	输出
54.6	...esign. · An outlier within its class · Recency of 156 days: 54.6th percentile of all listings, 90.8th am...	evidence_model.json → false_negative_context.hard_case_dsl_percentile_among_all_listings	输出
90.8	...s · Recency of 156 days: 54.6th percentile of all listings, 90.8th among reviewed superhosts. · k–NN mis...	evidence_model.json → false_negative_context.hard_case_dsl_percentile_among_reviewed_superhosts	输出
0.267	...90.8th among reviewed superhosts. · k–NN misses it too: P = 0.267. · Host 35667203 holds one listing · on...	evidence_model.json → hard_case.knn_proba_superhost	输出
35667203	...viewed superhosts. · k–NN misses it too: P = 0.267. · Host 35667203 holds one listing · one of 530 false ne...	evidence_model.json → hard_case.host_id	输出
530	... too: P = 0.267. · Host 35667203 holds one listing · one of 530 false negatives for the tree...	evidence_model.json → test_extra.DT.false_negative	输出
14 Listing 6812677: certain, and wrong · 备注			
8.5	...The path builds from the top: at the very first split it is 8.5 days past the recency threshold, so it ...	156 – 147.5	推导
86	...branch, and a few splits later it lands in a leaf where all 86 training listings are non–superhosts. [...	evidence_model.json → hard_case.tree_path[10].training_listings	输出
24	...] What the model could not see is exactly what Airbnb sees: 24 reviews in the last year and a 4.8 rati...	evidence_model.json → hard_case.number_of_reviews_ltm	输出
4.8	...exactly what Airbnb sees: 24 reviews in the last year and a 4.8 rating. Its recency is ordinary for a l...	evidence_model.json → hard_case.review_scores_rating	输出

15 Variance is not relevance			
32.1%	...host · Superhost · K-Means centroid · PC1 · activity axis · 32.1% · recency +0.563 · amenities −0.504 · a...	pca_table.csv → PC1 / explained_variance (×100)	输出
+0.563	... · K-Means centroid · PC1 · activity axis · 32.1% · recency +0.563 · amenities −0.504 · availability −0.44...	evidence_cluster.json → pca_top_loadings.PC1[0][1]	输出
−0.504	... · PC1 · activity axis · 32.1% · recency +0.563 · amenities −0.504 · availability −0.447 · with superhost ...	evidence_cluster.json → pca_top_loadings.PC1[1][1]	输出
−0.447	... · 32.1% · recency +0.563 · amenities −0.504 · availability −0.447 · with superhost status: r = −0.354 · P...	evidence_cluster.json → pca_top_loadings.PC1[2][1]	输出
−0.354	...s −0.504 · availability −0.447 · with superhost status: r = −0.354 · PC2 · place and scale · 20.45% · dist...	evidence_cluster.json → pc_vs_superhost.PC1.point_biserial_r	输出
20.45%	...with superhost status: r = −0.354 · PC2 · place and scale · 20.45% · distance +0.786 · host listings −0.55...	pca_table.csv → PC2 / explained_variance (×100)	输出
+0.786	...tus: r = −0.354 · PC2 · place and scale · 20.45% · distance +0.786 · host listings −0.556 · capacity +0.20...	evidence_cluster.json → pca_top_loadings.PC2[0][1]	输出
−0.556	... place and scale · 20.45% · distance +0.786 · host listings −0.556 · capacity +0.208 · with superhost stat...	evidence_cluster.json → pca_top_loadings.PC2[1][1]	输出
+0.208	... 20.45% · distance +0.786 · host listings −0.556 · capacity +0.208 · with superhost status: r = +0.050 · O...	evidence_cluster.json → pca_top_loadings.PC2[2][1]	输出
+0.050	...tings −0.556 · capacity +0.208 · with superhost status: r = +0.050 · One fifth of the variance, almost non...	evidence_cluster.json → pc_vs_superhost.PC2.point_biserial_r	输出
21,251	...ne of the signal: the label sits on the operational axis. · 21,251 reviewed listings (5,000 plotted) · PC1...	evidence_cluster.json → n_rows_reviewed	输出
5,000	...l sits on the operational axis. · 21,251 reviewed listings (5,000 plotted) · PC1–PC3 explain 69.4% · NMI ...	notebook 4.5 打印输出	输出
69.4%	... 21,251 reviewed listings (5,000 plotted) · PC1–PC3 explain 69.4% · NMI with label: PC1 0.057, PC2 0.004...	pca_table.csv → PC3 / cumulative_variance (×100)	输出
0.057	...,000 plotted) · PC1–PC3 explain 69.4% · NMI with label: PC1 0.057, PC2 0.004...	evidence_cluster.json → pc_vs_superhost.PC1.nmi_10_bins	输出
0.004	...d) · PC1–PC3 explain 69.4% · NMI with label: PC1 0.057, PC2 0.004...	evidence_cluster.json → pc_vs_superhost.PC2.nmi_10_bins	输出
15 Variance is not relevance · 备注			
0.35	...vity axis, and it correlates with superhost status at minus 0.35: on the scatter, the orange superhost l...	evidence_cluster.json → cluster_profiles[0].superhost_rate	输出
16 Choosing k = 5 and Ward linkage			
12.0%	...ns and Ward · Elbow and silhouette · k = 2 to 8 · SSE falls 12.0% from k = 4 to 5, then only 6.8%; silhou...	evidence_cluster.json → kmeans_sweep[3].sse_drop_pct	输出
6.8%	...e · k = 2 to 8 · SSE falls 12.0% from k = 4 to 5, then only 6.8%; silhouette peaks at k = 5 (0.2232 vs 0...	evidence_cluster.json → kmeans_sweep[4].sse_drop_pct	输出
0.2232	...from k = 4 to 5, then only 6.8%; silhouette peaks at k = 5 (0.2232 vs 0.2227 at k = 4). · Ward dendrogram ...	evidence_cluster.json → kmeans_sweep[3].silhouette_5000_sample	输出
0.2227	... to 5, then only 6.8%; silhouette peaks at k = 5 (0.2232 vs 0.2227 at k = 4). · Ward dendrogram · 2,000–li...	evidence_cluster.json → kmeans_sweep[2].silhouette_5000_sample	输出
2,000	...s at k = 5 (0.2232 vs 0.2227 at k = 4). · Ward dendrogram · 2,000–listing sample · Single linkage leaves ...	树状图抽样数 (代码常量 DENDROGRAM_SAMPLE_SIZE)	推导
21,247	...d dendrogram · 2,000–listing sample · Single linkage leaves 21,247 of 21,251 listings in one cluster; aver...	evidence_cluster.json → linkage_comparison_sizes.single[0]	输出
21,251	...am · 2,000–listing sample · Single linkage leaves 21,247 of 21,251 listings in one cluster; average and co...	evidence_cluster.json → n_rows_reviewed	输出
21,251	...complete also collapse. Ward minimises SSE, like K-Means. · 21,251 reviewed listings · six features: log1p...	evidence_cluster.json → n_rows_reviewed	输出
16 Choosing k = 5 and Ward linkage · 备注			
12	... no minimum; the bend is at five, where the gain drops from 12 percent to under 7, and the silhouette ...	0.0896 ÷ 0.0799 − 1 (NMI 增幅)	推导
21,247	...rom the lectures collapse on this data: single linkage puts 21,247 of 21,251 listings in one cluster. Ward...	evidence_cluster.json → linkage_comparison_sizes.single[0]	输出
21,251	...ctures collapse on this data: single linkage puts 21,247 of 21,251 listings in one cluster. Ward merges th...	evidence_cluster.json → n_rows_reviewed	输出
17 Activity splits the market, not distance			
40	...d, k = 5 · K-Means clusters · superhost rate · 2 · Active · 40 days since review, 47 amenities · 5,340...	evidence_cluster.json → cluster_profiles[1].days_since_last_review	输出

47	...sters · superhost rate · 2 · Active · 40 days since review, 47 amenities · 5,340 · 53.9% · 4 · Larger ...	evidence_cluster.json → cluster_profiles[1].amenity_count	输出
5,340	...st rate · 2 · Active · 40 days since review, 47 amenities · 5,340 · 53.9% · 4 · Larger homes · 20.6 km ou...	evidence_cluster.json → kmeans_sizes.1	输出
53.9%	... · 2 · Active · 40 days since review, 47 amenities · 5,340 · 53.9% · 4 · Larger homes · 20.6 km out · 2,36...	evidence_cluster.json → cluster_profiles[1].superhost_rate (×100)	输出
20.6	...e review, 47 amenities · 5,340 · 53.9% · 4 · Larger homes · 20.6 km out · 2,365 · 43.7% · 5 · Small, ope...	evidence_cluster.json → cluster_profiles[3].dist_cbd	输出
2,365	...menities · 5,340 · 53.9% · 4 · Larger homes · 20.6 km out · 2,365 · 43.7% · 5 · Small, open 310 days · 17...	evidence_cluster.json → kmeans_sizes.3	输出
43.7%	... · 5,340 · 53.9% · 4 · Larger homes · 20.6 km out · 2,365 · 43.7% · 5 · Small, open 310 days · 17.8 km ou...	evidence_cluster.json → cluster_profiles[3].superhost_rate (×100)	输出
310	...arger homes · 20.6 km out · 2,365 · 43.7% · 5 · Small, open 310 days · 17.8 km out · 4,316 · 40.7% · 1 ...	evidence_cluster.json → cluster_profiles[4].availability_365	输出
17.8	... · 20.6 km out · 2,365 · 43.7% · 5 · Small, open 310 days · 17.8 km out · 4,316 · 40.7% · 1 · Large oper...	evidence_cluster.json → cluster_profiles[4].dist_cbd	输出
4,316	... · 2,365 · 43.7% · 5 · Small, open 310 days · 17.8 km out · 4,316 · 40.7% · 1 · Large operators · 50 list...	evidence_cluster.json → kmeans_sizes.4	输出
40.7%	... · 43.7% · 5 · Small, open 310 days · 17.8 km out · 4,316 · 40.7% · 1 · Large operators · 50 listings, 1...	evidence_cluster.json → cluster_profiles[4].superhost_rate (×100)	输出
50	... days · 17.8 km out · 4,316 · 40.7% · 1 · Large operators · 50 listings, 1.4 km · 4,541 · 35.3% · 3 · ...	evidence_cluster.json → cluster_profiles[0].host_listings_count	输出
1.4	...km out · 4,316 · 40.7% · 1 · Large operators · 50 listings, 1.4 km · 4,541 · 35.3% · 3 · Dormant · 1,88...	evidence_correlation.json → dist_band_trend.4.dist_km	输出
4,541	...4,316 · 40.7% · 1 · Large operators · 50 listings, 1.4 km · 4,541 · 35.3% · 3 · Dormant · 1,883 days sinc...	evidence_cluster.json → cluster_profiles[0].listings	输出
35.3%	...40.7% · 1 · Large operators · 50 listings, 1.4 km · 4,541 · 35.3% · 3 · Dormant · 1,883 days since review...	evidence_cluster.json → cluster_profiles[0].superhost_rate (×100)	输出
1,883	...ators · 50 listings, 1.4 km · 4,541 · 35.3% · 3 · Dormant · 1,883 days since review, 0 available · 4,689 ...	evidence_cluster.json → cluster_profiles[2].days_since_last_review	输出
4,689	...5.3% · 3 · Dormant · 1,883 days since review, 0 available · 4,689 · 6.8% · K–Means vs Ward · ARI 0.377 · ...	evidence_cluster.json → kmeans_vs_hierarchical_sizes.kmeans.2	输出
6.8%	... · Dormant · 1,883 days since review, 0 available · 4,689 · 6.8% · K–Means vs Ward · ARI 0.377 · Same st...	evidence_cluster.json → kmeans_sweep[4].sse_drop_pct	输出
0.377	... review, 0 available · 4,689 · 6.8% · K–Means vs Ward · ARI 0.377 · Same structure, different boundaries...	evidence_cluster.json → kmeans_vs_hierarchical_ari	输出
99.2%	...77 · Same structure, different boundaries. · Dormant group: 99.2% of Ward's dormant cluster is K–Means cl...	3,224 ÷ 3,249 (evidence_cluster.json → crosstab)	推导
82.6%	...ard's dormant cluster is K–Means cluster 3. · Active group: 82.6% of K–Means cluster 2 lands in Ward W1. ...	4,410 ÷ 5,340 (evidence_cluster.json → crosstab)	推导
0.075	...undamentals only: superhost–rate range across clusters just 0.075. · Base rate among reviewed listings 35...	evidence_cluster.json → candidate_set_comparison.S2_fundamentals_only.superhost_rate_spread	输出
35.72%	...ss clusters just 0.075. · Base rate among reviewed listings 35.72% · cluster profiles: report Table 11...	evidence_cluster.json → superhost_rate_reviewed (×100)	输出
17 Activity splits the market, not distance · 备注			
54%	...superhost rate differ in activity: the active cluster is at 54%, the dormant one, nearly five years wit...	evidence_cluster.json → cluster_profiles[1].superhost_rate (×100)	输出
7%	... nearly five years without a review and no availability, at 7%. The three in between are separated mai...	evidence_preprocess.json → superhost_share_no_review (×100)	输出
35	...d mainly by location and size, and their rates stay between 35 and 44 percent. [click] Ward agrees on ...	40 – 5	推导
44	...y by location and size, and their rates stay between 35 and 44 percent. [click] Ward agrees on the str...	evidence_correlation.json → dist_band_trend.2.dsl_median	输出
0.38	...ard agrees on the structure, with an adjusted Rand index of 0.38: the dormant group is almost identical ...	evidence_cluster.json → kmeans_vs_hierarchical_ari	输出
18 What we would test next			
14,113	...tations · Rows are not independent. Next: one row per host, 14,113 rows. · $\rho = -0.82$ with reviews in the l...	evidence_preprocess.json → host_count	输出
-0.82	...not independent. Next: one row per host, 14,113 rows. · $\rho = -0.82$ with reviews in the last 12 months. Wit...	evidence_preprocess.json → days_since_last_review_vs_reviews_ltm_spearman	输出
12	...per host, 14,113 rows. · $\rho = -0.82$ with reviews in the last 12 months. Without it, operations still le...	0.0896 ÷ 0.0799 – 1 (NMI 增幅)	推导

0.371	...in the last 12 months. Without it, operations still lead by 0.371. · Depth 10 vs 4: 0.0033 apart, under o...	0.5662 – 0.1955 (Table 8: F – A, 0.3707)	推导
10	...months. Without it, operations still lead by 0.371. · Depth 10 vs 4: 0.0033 apart, under one SD. Train...	evidence_model.json → best_param.DT	输出
0.0033	...ithout it, operations still lead by 0.371. · Depth 10 vs 4: 0.0033 apart, under one SD. Train F1 0.719 vs ...	0.5745 – 0.5712 (cv_results.csv: depth 10 与 depth 4)	推导
0.719	...371. · Depth 10 vs 4: 0.0033 apart, under one SD. Train F1 0.719 vs test 0.632. · 8 operational vs 3 fun...	evidence_model.json → test_extra.DT.train_f1	输出
0.632	...10 vs 4: 0.0033 apart, under one SD. Train F1 0.719 vs test 0.632 . · 8 operational vs 3 fundamental featu...	evidence_model.json → test.DT.f1	输出
0.2232	...tal features; part of the gap may be set size. · Silhouette 0.2232 at k = 5 vs 0.2227 at k = 4. A gap stat...	evidence_cluster.json → candidate_set_comparison.S3_mixed.silhouette_5000_sample	输出
0.2227	...of the gap may be set size. · Silhouette 0.2232 at k = 5 vs 0.2227 at k = 4. A gap statistic would settle ...	evidence_cluster.json → kmeans_sweep[2].silhouette_5000_sample	输出
93.35%	... = 5 vs 0.2227 at k = 4. A gap statistic would settle it. · 93.35% of missing prices belong to unavailable...	evidence_preprocess.json → p_unavailable_given_price_missing (×100)	输出
18 What we would test next · 备注			
10	...ded, so we checked the result without it. Depth 4 and depth 10 cannot be separated. The two families d...	evidence_model.json → best_param.DT	输出
19 Partly — and it is the host, not the house.			
0.639	...The answer · 0.639 · test F1 from operational features alo...	evidence_model.json → rq[3].test_f1	输出
0.196	... · 0.639 · test F1 from operational features alone, against 0.196 for fundamentals · 0.632 · best F1 with...	rq_experiments.csv → A_fundamentals / test_f1	输出
0.632	...perational features alone, against 0.196 for fundamentals · 0.632 · best F1 with the criteria removed; 1,...	evidence_model.json → test.DT.f1	输出
1,047	...r fundamentals · 0.632 · best F1 with the criteria removed; 1,047 of 1,577 found · 118 / 124 · superhost ...	evidence_model.json → test_extra.DT.true_positive	输出
1,577	...ntals · 0.632 · best F1 with the criteria removed; 1,047 of 1,577 found · 118 / 124 · superhost listings ...	evidence_model.json → split_details.test_superhost_rows	输出
118	... · best F1 with the criteria removed; 1,047 of 1,577 found · 118 / 124 · superhost listings inactive for...	evidence_model.json → false_negative_context.false_negatives_inactive_over_year	输出
124	... F1 with the criteria removed; 1,047 of 1,577 found · 118 / 124 · superhost listings inactive for over ...	evidence_model.json → hard_case.comparison.availability_365.non_superhost_median	输出
19 Partly — and it is the host, not the house. · 备注			
0.639	...ng is run, not what it is. Operational features alone reach 0.639 F1 on the test set against 0.196 for fu...	rq_experiments.csv → B_operational / test_f1	输出
0.196	...ional features alone reach 0.639 F1 on the test set against 0.196 for fundamentals. The best model finds ...	rq_experiments.csv → A_fundamentals / test_f1	输出